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Boundary-Value Problems in Rectangular Coordinates

EXERCISES 13.1

Separable Partial Differential Equations

1. Substituting $u(x, y) = X(x)Y(y)$ into the partial differential equation yields $X'Y = XY'$. Separating variables and using the separation constant $-\lambda$, where $\lambda \neq 0$, we obtain

$$\frac{X'}{X} = \frac{Y'}{Y} = -\lambda.$$

When $\lambda \neq 0$

$$X' + \lambda X = 0 \quad \text{and} \quad Y' + \lambda Y = 0$$

so that

$$X = c_1 e^{-\lambda x} \quad \text{and} \quad Y = c_2 e^{-\lambda y}.$$

A particular product solution of the partial differential equation is

$$u = XY = c_3 e^{-\lambda(x+y)}, \quad \lambda \neq 0.$$

When $\lambda = 0$ the differential equations become $X' = 0$ and $Y' = 0$, so in this case $X = c_4$, $Y = c_5$, and $u = XY = c_6$.

2. Substituting $u(x, y) = X(x)Y(y)$ into the partial differential equation yields $X'Y + 3XY' = 0$. Separating variables and using the separation constant $-\lambda$ we obtain

$$\frac{X'}{-3X} = \frac{Y'}{Y} = -\lambda.$$

When $\lambda \neq 0$

$$X' - 3\lambda X = 0 \quad \text{and} \quad Y' + \lambda Y = 0$$

so that

$$X = c_1 e^{3\lambda x} \quad \text{and} \quad Y = c_2 e^{-\lambda y}.$$

A particular product solution of the partial differential equation is

$$u = XY = c_3 e^{\lambda(3x-y)}.$$

When $\lambda = 0$ the differential equations become $X' = 0$ and $Y' = 0$, so in this case $X = c_4$, $Y = c_5$, and $u = XY = c_6$.

3. Substituting $u(x, y) = X(x)Y(y)$ into the partial differential equation yields $X'Y + XY' = XY$. Separating variables and using the separation constant $-\lambda$ we obtain

$$\frac{X'}{X} = \frac{Y - Y'}{Y} = -\lambda.$$

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Then

$$X' + \lambda X = 0 \quad \text{and} \quad Y' - (1 + \lambda)Y = 0$$

so that

$$X = c_1 e^{-\lambda x} \quad \text{and} \quad Y = c_2 e^{(1+\lambda)y}.$$

A particular product solution of the partial differential equation is

$$u = XY = c_3 e^{y+\lambda(y-x)}.$$

4. Substituting $u(x, y) = X(x)Y(y)$ into the partial differential equation yields $X'Y = XY' + XY$. Separating variables and using the separation constant $-\lambda$ we obtain

$$\frac{X'}{X} = \frac{Y + Y'}{Y} = -\lambda.$$

Then

$$X' + \lambda X = 0 \quad \text{and} \quad y' + (1 + \lambda)Y = 0$$

so that

$$X = c_1 e^{-\lambda x} \quad \text{and} \quad Y = c_2 e^{-(1+\lambda)y} = 0.$$

A particular product solution of the partial differential equation is

$$u = XY = c_3 e^{-y-\lambda(x+y)}.$$

5. Substituting $u(x, y) = X(x)Y(y)$ into the partial differential equation yields $xX'Y = yXY'$. Separating variables and using the separation constant $-\lambda$ we obtain

$$\frac{xX'}{X} = \frac{yY'}{Y} = -\lambda.$$

When $\lambda \neq 0$

$$xX' + \lambda X = 0 \quad \text{and} \quad yY' + \lambda Y = 0$$

so that

$$X = c_1 x^{-\lambda} \quad \text{and} \quad Y = c_2 y^{-\lambda}.$$

A particular product solution of the partial differential equation is

$$u = XY = c_3 (xy)^{-\lambda}.$$

When $\lambda = 0$ the differential equations become $X' = 0$ and $Y' = 0$, so in this case $X = c_4$, $Y = c_5$, and $u = XY = c_6$.

6. Substituting $u(x, y) = X(x)Y(y)$ into the partial differential equation yields $yX'Y + xXY' = 0$. Separating variables and using the separation constant $-\lambda$ we obtain

$$\frac{X'}{xX} = -\frac{Y'}{yY} = -\lambda.$$

When $\lambda \neq 0$

$$X' + \lambda xX = 0 \quad \text{and} \quad Y' - \lambda yY = 0$$

so that

$$X = c_1 e^{\lambda x^2/2} \quad \text{and} \quad Y = c_2 e^{-\lambda y^2/2}.$$

A particular product solution of the partial differential equation is

$$u = XY = c_3 e^{\lambda(x^2-y^2)/2}.$$

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When $\lambda = 0$ the differential equations become $X' = 0$ and $Y' = 0$, so in this case $X = c_4$, $Y = c_5$, and $u = XY = c_6$.

7. Substituting $u(x, y) = X(x)Y(y)$ into the partial differential equation yields $X''Y + X'Y' + XY'' = 0'$, which is not separable.
8. Substituting $u(x, y) = X(x)Y(y)$ into the partial differential equation yields $yX'Y' + XY = 0$. Separating variables and using the separation constant $-\lambda$ we obtain

$$\frac{X'}{X} = -\frac{Y}{yY'} = -\lambda.$$

When $\lambda \neq 0$

$$X' + \lambda X = 0 \quad \text{and} \quad \lambda yY' - Y = 0$$

so that

$$X = c_1 e^{-\lambda x} \quad \text{and} \quad Y = c_2 y^{1/\lambda}.$$

A particular product solution of the partial differential equation is

$$u = XY = c_3 e^{-\lambda x} y^{1/\lambda}.$$

In this case $\lambda = 0$ yields no solution.

9. Substituting $u(x, t) = X(x)T(t)$ into the partial differential equation yields $kX''T - XT = XT'$. Separating variables and using the separation constant $-\lambda$ we obtain

$$\frac{kX'' - X}{X} = \frac{T'}{T} = -\lambda.$$

Then

$$X'' + \frac{\lambda - 1}{k}X = 0 \quad \text{and} \quad T' + \lambda T = 0.$$

The second differential equation implies $T(t) = c_1 e^{-\lambda t}$. For the first differential equation we consider three cases:

- I. If $(\lambda - 1)/k = 0$ then $\lambda = 1$, $X'' = 0$, and $X(x) = c_2 x + c_3$, so

$$u = XT = e^{-t}(A_1 x + A_2).$$

- II. If $(\lambda - 1)/k = -\alpha^2 < 0$, then $\lambda = 1 - k\alpha^2$, $X'' - \alpha^2 X = 0$, and $X(x) = c_4 \cosh \alpha x + c_5 \sinh \alpha x$, so

$$u = XT = (A_3 \cosh \alpha x + A_4 \sinh \alpha x)e^{-(1-k\alpha^2)t}.$$

- III. If $(\lambda - 1)/k = \alpha^2 > 0$, then $\lambda = 1 + k\alpha^2$, $X'' + \alpha^2 X = 0$, and $X(x) = c_6 \cos \alpha x + c_7 \sin \alpha x$, so

$$u = XT = (A_5 \cos \alpha x + A_6 \sin \alpha x)e^{-(1+k\alpha^2)t}.$$

10. Substituting $u(x, t) = X(x)T(t)$ into the partial differential equation yields $kX''T = XT'$. Separating variables and using the separation constant $-\lambda$ we obtain

$$\frac{X''}{X} = \frac{T'}{kT} = -\lambda.$$

Then

$$X'' + \lambda X = 0 \quad \text{and} \quad T' + \lambda kT = 0.$$

The second differential equation implies $T(t) = c_1 e^{-\lambda kt}$. For the first differential equation we consider three cases:

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I. If $\lambda = 0$ then $X'' = 0$ and $X(x) = c_2x + c_3$, so

$$u = XT = A_1x + A_2.$$

II. If $\lambda = -\alpha^2 < 0$, then $X'' - \alpha^2X = 0$, and $X(x) = c_4 \cosh \alpha x + c_5 \sinh \alpha x$, so

$$u = XT = (A_3 \cosh \alpha x + A_4 \sinh \alpha x)e^{k\alpha^2 t}.$$

III. If $\lambda = \alpha^2 > 0$, then $X'' + \alpha^2X = 0$, and $X(x) = c_6 \cos \alpha x + c_7 \sin \alpha x$, so

$$u = XT = (A_5 \cos \alpha x + A_6 \sin \alpha x)e^{-k\alpha^2 t}.$$

- 11.** Substituting $u(x, t) = X(x)T(t)$ into the partial differential equation yields $a^2X''T = XT''$. Separating variables and using the separation constant $-\lambda$ we obtain

$$\frac{X''}{X} = \frac{T''}{a^2T} = -\lambda.$$

Then

$$X'' + \lambda X = 0 \quad \text{and} \quad T'' + a^2\lambda T = 0.$$

We consider three cases:

I. If $\lambda = 0$ then $X'' = 0$ and $X(x) = c_1x + c_2$. Also, $T'' = 0$ and $T(t) = c_3t + c_4$, so

$$u = XT = (c_1x + c_2)(c_3t + c_4).$$

II. If $\lambda = -\alpha^2 < 0$, then $X'' - \alpha^2X = 0$, and $X(x) = c_5 \cosh \alpha x + c_6 \sinh \alpha x$. Also, $T'' - \alpha^2a^2T = 0$ and $T(t) = c_7 \cosh \alpha at + c_8 \sinh \alpha at$, so

$$u = XT = (c_5 \cosh \alpha x + c_6 \sinh \alpha x)(c_7 \cosh \alpha at + c_8 \sinh \alpha at).$$

III. If $\lambda = \alpha^2 > 0$, then $X'' + \alpha^2X = 0$, and $X(x) = c_9 \cos \alpha x + c_{10} \sin \alpha x$. Also, $T'' + \alpha^2a^2T = 0$ and $T(t) = c_{11} \cos \alpha at + c_{12} \sin \alpha at$, so

$$u = XT = (c_9 \cos \alpha x + c_{10} \sin \alpha x)(c_{11} \cos \alpha at + c_{12} \sin \alpha at).$$

- 12.** Substituting $u(x, t) = X(x)T(t)$ into the partial differential equation yields $a^2X''T = XT'' + 2kXT'$. Separating variables and using the separation constant $-\lambda$ we obtain

$$\frac{X''}{X} = \frac{T'' + 2kT'}{a^2T} = -\lambda.$$

Then

$$X'' + \lambda X = 0 \quad \text{and} \quad T'' + 2kT' + a^2\lambda T = 0.$$

We consider three cases:

I. If $\lambda = 0$ then $X'' = 0$ and $X(x) = c_1x + c_2$. Also, $T'' + 2kT' = 0$ and $T(t) = c_3 + c_4e^{-2kt}$, so

$$u = XT = (c_1x + c_2)(c_3 + c_4e^{-2kt}).$$

II. If $\lambda = -\alpha^2 < 0$, then $X'' - \alpha^2X = 0$, and $X(x) = c_5 \cosh \alpha x + c_6 \sinh \alpha x$. The auxiliary equation of $T'' + 2kT' - \alpha^2a^2T = 0$ is $m^2 + 2km - \alpha^2a^2 = 0$. Solving for m we obtain $m = -k \pm \sqrt{k^2 + \alpha^2a^2}$, so $T(t) = c_7e^{(-k+\sqrt{k^2+\alpha^2a^2})t} + c_8e^{(-k-\sqrt{k^2+\alpha^2a^2})t}$. Then

$$u = XT = (c_5 \cosh \alpha x + c_6 \sinh \alpha x) \left(c_7e^{(-k+\sqrt{k^2+\alpha^2a^2})t} + c_8e^{(-k-\sqrt{k^2+\alpha^2a^2})t} \right).$$

III. If $\lambda = \alpha^2 > 0$, then $X'' + \alpha^2X = 0$, and $X(x) = c_9 \cos \alpha x + c_{10} \sin \alpha x$. The auxiliary equation

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of $T'' + 2kT' + \alpha^2 a^2 T = 0$ is $m^2 + 2km + \alpha^2 a^2 = 0$. Solving for m we obtain $m = -k \pm \sqrt{k^2 - \alpha^2 a^2}$. We consider three possibilities for the discriminant $k^2 - \alpha^2 a^2$:

(i) If $k^2 - \alpha^2 a^2 = 0$ then $T(t) = c_{11}e^{-kt} + c_{12}te^{-kt}$ and

$$u = XT = (c_9 \cos \alpha x + c_{10} \sin \alpha x)(c_{11}e^{-kt} + c_{12}te^{-kt}).$$

From $k^2 - \alpha^2 a^2 = 0$ we have $\alpha = k/a$ so the solution can be written

$$u = XT = (c_9 \cos kx/a + c_{10} \sin kx/a)(c_{11}e^{-kt} + c_{12}te^{-kt}).$$

(ii) If $k^2 - \alpha^2 a^2 < 0$ then $T(t) = e^{-kt} (c_{13} \cos \sqrt{\alpha^2 a^2 - k^2} t + c_{14} \sin \sqrt{\alpha^2 a^2 - k^2} t)$ and

$$u = XT = (c_9 \cos \alpha x + c_{10} \sin \alpha x)e^{-kt} (c_{13} \cos \sqrt{\alpha^2 a^2 - k^2} t + c_{14} \sin \sqrt{\alpha^2 a^2 - k^2} t).$$

(iii) If $k^2 - \alpha^2 a^2 > 0$ then $T(t) = c_{15}e^{(-k+\sqrt{k^2-\alpha^2 a^2})t} + c_{16}e^{(-k-\sqrt{k^2-\alpha^2 a^2})t}$ and

$$u = XT = (c_9 \cos \alpha x + c_{10} \sin \alpha x) (c_{15}e^{(-k+\sqrt{k^2-\alpha^2 a^2})t} + c_{16}e^{(-k-\sqrt{k^2-\alpha^2 a^2})t}).$$

13. Substituting $u(x, y) = X(x)Y(y)$ into the partial differential equation yields $X''Y + XY'' = 0$. Separating variables and using the separation constant $-\lambda$ we obtain

$$-\frac{X''}{X} = \frac{Y''}{Y} = -\lambda.$$

Then

$$X'' - \lambda X = 0 \quad \text{and} \quad Y'' + \lambda Y = 0.$$

We consider three cases:

I. If $\lambda = 0$ then $X'' = 0$ and $X(x) = c_1x + c_2$. Also, $Y'' = 0$ and $Y(y) = c_3y + c_4$ so

$$u = XY = (c_1x + c_2)(c_3y + c_4).$$

II. If $\lambda = -\alpha^2 < 0$ then $X'' + \alpha^2 X = 0$ and $X(x) = c_5 \cos \alpha x + c_6 \sin \alpha x$. Also, $Y'' - \alpha^2 Y = 0$ and $Y(y) = c_7 \cosh \alpha y + c_8 \sinh \alpha y$ so

$$u = XY = (c_5 \cos \alpha x + c_6 \sin \alpha x)(c_7 \cosh \alpha y + c_8 \sinh \alpha y).$$

III. If $\lambda = \alpha^2 > 0$ then $X'' - \alpha^2 X = 0$ and $X(x) = c_9 \cosh \alpha x + c_{10} \sinh \alpha x$. Also, $Y'' + \alpha^2 Y = 0$ and $Y(y) = c_{11} \cos \alpha y + c_{12} \sin \alpha y$ so

$$u = XY = (c_9 \cosh \alpha x + c_{10} \sinh \alpha x)(c_{11} \cos \alpha y + c_{12} \sin \alpha y).$$

14. Substituting $u(x, y) = X(x)Y(y)$ into the partial differential equation yields $x^2 X''Y + XY'' = 0$. Separating variables and using the separation constant $-\lambda$ we obtain

$$-\frac{x^2 X''}{X} = \frac{Y''}{Y} = -\lambda.$$

Then

$$x^2 X'' - \lambda X = 0 \quad \text{and} \quad Y'' + \lambda Y = 0.$$

We consider three cases:

I. If $\lambda = 0$ then $x^2 X'' = 0$ and $X(x) = c_1x + c_2$. Also, $Y'' = 0$ and $Y(y) = c_3y + c_4$ so

$$u = XY = (c_1x + c_2)(c_3y + c_4).$$

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II. If $\lambda = -\alpha^2 < 0$ then $x^2 X'' + \alpha^2 X = 0$ and $Y'' - \alpha^2 Y = 0$. The solution of the second differential equation is $Y(y) = c_5 \cosh \alpha y + c_6 \sinh \alpha y$. The first equation is Cauchy-Euler with auxiliary equation $m^2 - m + \alpha^2 = 0$. Solving for m we obtain $m = \frac{1}{2} \pm \frac{1}{2}\sqrt{1-4\alpha^2}$. We consider three possibilities for the discriminant $1-4\alpha^2$.

(i) If $1-4\alpha^2 = 0$ then $X(x) = c_7 x^{1/2} + c_8 x^{1/2} \ln x$ and

$$u = XY = x^{1/2}(c_7 + c_8 \ln x)(c_5 \cosh \alpha y + c_6 \sinh \alpha y).$$

(ii) If $1-4\alpha^2 < 0$ then $X(x) = x^{1/2} [c_9 \cos(\sqrt{4\alpha^2-1} \ln x) + c_{10} \sin(\sqrt{4\alpha^2-1} \ln x)]$ and

$$u = XY = x^{1/2} [c_9 \cos(\sqrt{4\alpha^2-1} \ln x) + c_{10} \sin(\sqrt{4\alpha^2-1} \ln x)] (c_5 \cosh \alpha y + c_6 \sinh \alpha y).$$

(iii) If $1-4\alpha^2 > 0$ then $X(x) = x^{1/2} (c_{11} x^{\sqrt{1-4\alpha^2}/2} + c_{12} x^{-\sqrt{1-4\alpha^2}/2})$ and

$$u = XY = x^{1/2} (c_{11} x^{\sqrt{1-4\alpha^2}/2} + c_{12} x^{-\sqrt{1-4\alpha^2}/2}) (c_5 \cosh \alpha y + c_6 \sinh \alpha y).$$

III. If $\lambda = \alpha^2 > 0$ then $x^2 X'' - \alpha^2 X = 0$ and $Y'' + \alpha^2 Y = 0$. The solution of the second differential equation is $Y(y) = c_{13} \cos \alpha y + c_{14} \sin \alpha y$. The first equation is Cauchy-Euler with auxiliary equation $m^2 - m - \alpha^2 = 0$. Solving for m we obtain $m = \frac{1}{2} \pm \frac{1}{2}\sqrt{1+4\alpha^2}$. In this case the discriminant is always positive so the solution of the differential equation is $X(x) = x^{1/2} (c_{15} x^{\sqrt{1+4\alpha^2}/2} + c_{16} x^{-\sqrt{1+4\alpha^2}/2})$ and

$$u = XY = x^{1/2} (c_{15} x^{\sqrt{1+4\alpha^2}/2} + c_{16} x^{-\sqrt{1+4\alpha^2}/2}) (c_{13} \cos \alpha y + c_{14} \sin \alpha y).$$

15. Substituting $u(x, y) = X(x)Y(y)$ into the partial differential equation yields $X''Y + XY'' = XY$. Separating variables and using the separation constant $-\lambda$ we obtain

$$\frac{X''}{X} = \frac{Y - Y''}{Y} = -\lambda.$$

Then

$$X'' + \lambda X = 0 \quad \text{and} \quad Y'' - (1 + \lambda)Y = 0.$$

We consider three cases:

I. If $\lambda = 0$ then $X'' = 0$ and $X(x) = c_1 x + c_2$. Also $Y'' - Y = 0$ and $Y(y) = c_3 \cosh y + c_4 \sinh y$ so

$$u = XY = (c_1 x + c_2)(c_3 \cosh y + c_4 \sinh y).$$

II. If $\lambda = -\alpha^2 < 0$ then $X'' - \alpha^2 X = 0$ and $Y'' + (\alpha^2 - 1)Y = 0$. The solution of the first differential equation is $X(x) = c_5 \cosh \alpha x + c_6 \sinh \alpha x$. The solution of the second differential equation depends on the nature of $\alpha^2 - 1$. We consider three cases:

(i) If $\alpha^2 - 1 = 0$, or $\alpha^2 = 1$, then $Y(y) = c_7 y + c_8$ and

$$u = XY = (c_5 \cosh \alpha x + c_6 \sinh \alpha x)(c_7 y + c_8).$$

(ii) If $\alpha^2 - 1 < 0$, or $0 < \alpha^2 < 1$, then $Y(y) = c_9 \cosh \sqrt{1-\alpha^2} y + c_{10} \sinh \sqrt{1-\alpha^2} y$ and

$$u = XY = (c_5 \cosh \alpha x + c_6 \sinh \alpha x) (c_9 \cosh \sqrt{1-\alpha^2} y + c_{10} \sinh \sqrt{1-\alpha^2} y).$$

(iii) If $\alpha^2 - 1 > 0$, or $\alpha^2 > 1$, then $Y(y) = c_{11} \cos \sqrt{\alpha^2-1} y + c_{12} \sin \sqrt{\alpha^2-1} y$ and

$$u = XY = (c_5 \cosh \alpha x + c_6 \sinh \alpha x) (c_{11} \cos \sqrt{\alpha^2-1} y + c_{12} \sin \sqrt{\alpha^2-1} y).$$

III. If $\lambda = \alpha^2 > 0$, then $X'' + \alpha^2 X = 0$ and $X(x) = c_{13} \cos \alpha x + c_{14} \sin \alpha x$. Also,

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$Y'' - (1 + \alpha^2)Y = 0$ and $Y(y) = c_{15} \cosh \sqrt{1 + \alpha^2} y + c_{16} \sinh \sqrt{1 + \alpha^2} y$ so

$$u = XY = (c_{13} \cos \alpha x + c_{14} \sin \alpha x) \left(c_{15} \cosh \sqrt{1 + \alpha^2} y + c_{16} \sinh \sqrt{1 + \alpha^2} y \right).$$

16. Substituting $u(x, t) = X(x)T(t)$ into the partial differential equation yields $a^2 X''T - g = XT''$, which is not separable.
17. Identifying $A = B = C = 1$, we compute $B^2 - 4AC = -3 < 0$. The equation is elliptic.
18. Identifying $A = 3$, $B = 5$, and $C = 1$, we compute $B^2 - 4AC = 13 > 0$. The equation is hyperbolic.
19. Identifying $A = 1$, $B = 6$, and $C = 9$, we compute $B^2 - 4AC = 0$. The equation is parabolic.
20. Identifying $A = 1$, $B = -1$, and $C = -3$, we compute $B^2 - 4AC = 13 > 0$. The equation is hyperbolic.
21. Identifying $A = 1$, $B = -9$, and $C = 0$, we compute $B^2 - 4AC = 81 > 0$. The equation is hyperbolic.
22. Identifying $A = 0$, $B = 1$, and $C = 0$, we compute $B^2 - 4AC = 1 > 0$. The equation is hyperbolic.
23. Identifying $A = 1$, $B = 2$, and $C = 1$, we compute $B^2 - 4AC = 0$. The equation is parabolic.
24. Identifying $A = 1$, $B = 0$, and $C = 1$, we compute $B^2 - 4AC = -4 < 0$. The equation is elliptic.
25. Identifying $A = a^2$, $B = 0$, and $C = -1$, we compute $B^2 - 4AC = 4a^2 > 0$. The equation is hyperbolic.
26. Identifying $A = k > 0$, $B = 0$, and $C = 0$, we compute $B^2 - 4AC = -4k < 0$. The equation is elliptic.
27. Substituting $u(r, t) = R(r)T(t)$ into the partial differential equation yields

$$k \left(R''T + \frac{1}{r} R'T \right) = RT'.$$

Separating variables and using the separation constant $-\lambda$ we obtain

$$\frac{rR'' + R'}{rR} = \frac{T'}{kT} = -\lambda.$$

Then

$$rR'' + R' + \lambda rR = 0 \quad \text{and} \quad T' + \lambda kT = 0.$$

Letting $\lambda = \alpha^2$ and writing the first equation as $r^2 R'' + rR' = \alpha^2 r^2 R = 0$ we see that it is a parametric Bessel equation of order 0. As discussed in Chapter 5 of the text, it has solution $R(r) = c_1 J_0(\alpha r) + c_2 Y_0(\alpha r)$. Since a solution of $T' + \alpha^2 kT$ is $T(t) = e^{-k\alpha^2 t}$, we see that a solution of the partial differential equation is

$$u = RT = e^{-k\alpha^2 t} [c_1 J_0(\alpha r) + c_2 Y_0(\alpha r)].$$

28. Substituting $u(r, \theta) = R(r)\Theta(\theta)$ into the partial differential equation yields

$$R''\Theta + \frac{1}{r} R'\Theta + \frac{1}{r^2} R\Theta'' = 0.$$

Separating variables and using the separation constant $-\lambda$ we obtain

$$\frac{r^2 R'' + rR'}{R} = -\frac{\Theta''}{\Theta} = -\lambda.$$

Then

$$r^2 R'' + rR' + \lambda R = 0 \quad \text{and} \quad \Theta'' - \lambda \Theta = 0.$$

Letting $\lambda = -\alpha^2$ we have the Cauchy-Euler equation $r^2 R'' + rR' - \alpha^2 R = 0$ whose solution is $R(r) = c_3 r^\alpha + c_4 r^{-\alpha}$. Since the solution of $\Theta'' + \alpha^2 \Theta = 0$ is $\Theta(\theta) = c_1 \cos \alpha \theta + c_2 \sin \alpha \theta$ we see that a solution of the partial differential equation is

$$u = R\Theta = (c_1 \cos \alpha \theta + c_2 \sin \alpha \theta)(c_3 r^\alpha + c_4 r^{-\alpha}).$$

29. For $u = A_1 + B_1x$ we compute $\partial^2 u / \partial x^2 = 0 = \partial u / \partial y$. Then $\partial^2 u / \partial x^2 = 4 \partial u / \partial y$.

For $u = A_2 e^{\alpha^2 y} \cos 2\alpha x + B_2 e^{\alpha^2 y} \sin 2\alpha x$ we compute

$$\begin{aligned}\frac{\partial u}{\partial x} &= 2\alpha A_2 e^{\alpha^2 y} \sinh 2\alpha x + 2\alpha B_2 e^{\alpha^2 y} \cosh 2\alpha x \\ \frac{\partial^2 u}{\partial x^2} &= 4\alpha^2 A_2 e^{\alpha^2 y} \cosh 2\alpha x + 4\alpha^2 B_2 e^{\alpha^2 y} \sinh 2\alpha x\end{aligned}$$

and

$$\frac{\partial u}{\partial y} = \alpha^2 A_2 e^{\alpha^2 y} \cosh 2\alpha x + \alpha^2 B_2 e^{\alpha^2 y} \sinh 2\alpha x.$$

Then $\partial^2 u / \partial x^2 = 4 \partial u / \partial y$.

For $u = A_3 e^{-\alpha^2 y} \cosh 2\alpha x + B_3 e^{-\alpha^2 y} \sinh 2\alpha x$ we compute

$$\begin{aligned}\frac{\partial u}{\partial x} &= -2\alpha A_3 e^{-\alpha^2 y} \sin 2\alpha x + 2\alpha B_3 e^{-\alpha^2 y} \cos 2\alpha x \\ \frac{\partial^2 u}{\partial x^2} &= -4\alpha^2 A_3 e^{-\alpha^2 y} \cos 2\alpha x - 4\alpha^2 B_3 e^{-\alpha^2 y} \sin 2\alpha x\end{aligned}$$

and

$$\frac{\partial u}{\partial y} = -\alpha^2 A_3 e^{-\alpha^2 y} \cosh 2\alpha x - \alpha^2 B_3 e^{-\alpha^2 y} \sinh 2\alpha x.$$

Then $\partial^2 u / \partial x^2 = 4 \partial u / \partial y$.

30. We identify $A = xy + 1$, $B = x + 2y$, and $C = 1$. Then $B^2 - 4AC = x^2 + 4y^2 - 4$. The equation $x^2 + 4y^2 = 4$ defines an ellipse. The partial differential equation is hyperbolic outside the ellipse, parabolic on the ellipse, and elliptic inside the ellipse.
31. Assuming $u(x, y) = X(x)Y(y)$ and substituting into $\partial^2 u / \partial x^2 - u = 0$ we get $X''Y - XY = 0$ or $Y(X'' - X) = 0$. This implies $X(x) = c_1 e^x$ or $X(x) = c_2 e^{-x}$. For these choices of X , Y can be any function of y . Two solutions of the partial differential equation are then

$$u_1(x, y) = A(y)e^x \quad \text{and} \quad u_2(x, y) = B(y)e^{-x}.$$

Since the partial differential equation is linear and homogeneous the superposition principle indicates that another solution is

$$u(x, y) = u_1(x, y) + u_2(x, y) = A(y)e^x + B(y)e^{-x}.$$

32. Assuming $u(x, y) = X(x)Y(y)$ and substituting into $\partial^2 u / \partial x \partial y + \partial u / \partial x = 0$ we get $X'Y' + X'Y = 0$ or $X'(Y' + Y) = 0$. This implies $Y(y) = c_1 e^{-y}$. For this choice of Y , X can be any function of x . A solution of the partial differential equation is then $u(x, y) = A(x)e^{-y}$. In addition, noting that the partial differential equation can be written

$$\frac{\partial}{\partial x} \left[\frac{\partial u}{\partial y} + u \right] = 0,$$

any function $u_2(x, y) = B(y)$ will satisfy the partial differential equation since, in this case, $\partial u_2 / \partial y + u_2 = B'(y) + B(y)$ and the x -partial of $B'(y) + B(y)$ is 0. Thus, using the superposition principle, a solution of the partial differential equation is

$$u(x, y) = u_1(x, y) + u_2(x, y) = A(x)e^{-y} + B(y).$$

EXERCISES 13.2**Classical Equations and
Boundary-Value Problems**

1. $k \frac{\partial^2 u}{\partial x^2} = \frac{\partial u}{\partial t}, \quad 0 < x < L, \quad t > 0$
 $u(0, t) = 0, \quad \left. \frac{\partial u}{\partial x} \right|_{x=L} = 0, \quad t > 0$
 $u(x, 0) = f(x), \quad 0 < x < L$
2. $k \frac{\partial^2 u}{\partial x^2} = \frac{\partial u}{\partial t}, \quad 0 < x < L, \quad t > 0$
 $u(0, t) = u_0, \quad u(L, t) = u_1, \quad t > 0$
 $u(x, 0) = 0, \quad 0 < x < L$
3. $k \frac{\partial^2 u}{\partial x^2} = \frac{\partial u}{\partial t}, \quad 0 < x < L, \quad t > 0$
 $u(0, t) = 100, \quad \left. \frac{\partial u}{\partial x} \right|_{x=L} = -hu(L, t), \quad t > 0$
 $u(x, 0) = f(x), \quad 0 < x < L$
4. $k \frac{\partial^2 u}{\partial x^2} = \frac{\partial u}{\partial t}, \quad 0 < x < L, \quad t > 0$
 $\left. \frac{\partial u}{\partial x} \right|_{x=0} = h[u(0, t) - 20], \quad \left. \frac{\partial u}{\partial x} \right|_{x=L} = 0, \quad t > 0$
 $u(x, 0) = f(x), \quad 0 < x < L$
5. $k \frac{\partial^2 u}{\partial x^2} - hu = \frac{\partial u}{\partial t}, \quad 0 < x < L, \quad t > 0, \quad h \text{ a constant}$
 $u(0, t) = \sin \frac{\pi t}{L}, \quad u(L, t) = 0, \quad t > 0$
 $u(x, 0) = f(x), \quad 0 < x < L$
6. $k \frac{\partial^2 u}{\partial x^2} + h(u - 50) = \frac{\partial u}{\partial t}, \quad 0 < x < L, \quad t > 0$
 $\left. \frac{\partial u}{\partial x} \right|_{x=0} = 0, \quad \left. \frac{\partial u}{\partial x} \right|_{x=L} = 0, \quad t > 0$
 $u(x, 0) = 100, \quad 0 < x < L$
7. $a^2 \frac{\partial^2 u}{\partial x^2} = \frac{\partial^2 u}{\partial t^2}, \quad 0 < x < L, \quad t > 0$
 $u(0, t) = 0, \quad u(L, t) = 0, \quad t > 0$
 $u(x, 0) = x(L - x), \quad \left. \frac{\partial u}{\partial t} \right|_{t=0} = 0, \quad 0 < x < L$

8. $a^2 \frac{\partial^2 u}{\partial x^2} = \frac{\partial^2 u}{\partial t^2}, \quad 0 < x < L, \quad t > 0$
 $u(0, t) = 0, \quad u(L, t) = 0, \quad t > 0$
 $u(x, 0) = 0, \quad \left. \frac{\partial u}{\partial t} \right|_{t=0} = \sin \frac{\pi x}{L}, \quad 0 < x < L$
9. $a^2 \frac{\partial^2 u}{\partial x^2} - 2\beta \frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial t^2}, \quad 0 < x < L, \quad t > 0$
 $u(0, t) = 0, \quad u(L, t) = \sin \pi t, \quad t > 0$
 $u(x, 0) = f(x), \quad \left. \frac{\partial u}{\partial t} \right|_{t=0} = 0, \quad 0 < x < L$
10. $a^2 \frac{\partial^2 u}{\partial x^2} + Ax = \frac{\partial^2 u}{\partial t^2}, \quad 0 < x < L, \quad t > 0, \quad A \text{ a constant}$
 $u(0, t) = 0, \quad u(L, t) = 0, \quad t > 0$
 $u(x, 0) = 0, \quad \left. \frac{\partial u}{\partial t} \right|_{t=0} = 0, \quad 0 < x < L$
11. $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0, \quad 0 < x < 4, \quad 0 < y < 2$
 $\left. \frac{\partial u}{\partial x} \right|_{x=0} = 0, \quad u(4, y) = f(y), \quad 0 < y < 2$
 $\left. \frac{\partial u}{\partial y} \right|_{y=0} = 0, \quad u(x, 2) = 0, \quad 0 < x < 4$
12. $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0, \quad 0 < x < \pi, \quad y > 0$
 $u(0, y) = e^{-y}, \quad u(\pi, y) = \begin{cases} 100, & 0 < y \leq 1 \\ 0, & y > 1 \end{cases}$
 $u(x, 0) = f(x), \quad 0 < x < \pi$

EXERCISES 13.3

Heat Equation

1. Using $u = XT$ and $-\lambda$ as a separation constant we obtain

$$X'' + \lambda X = 0,$$

$$X(0) = 0,$$

$$X(L) = 0,$$

and

$$T' + k\lambda T = 0.$$

This leads to

$$X = c_1 \sin \frac{n\pi}{L}x \quad \text{and} \quad T = c_2 e^{-kn^2 \pi^2 t/L^2}$$

13.3 Heat Equation

for $n = 1, 2, 3, \dots$ so that

$$u = \sum_{n=1}^{\infty} A_n e^{-kn^2\pi^2 t/L^2} \sin \frac{n\pi}{L} x.$$

Imposing

$$u(x, 0) = \sum_{n=1}^{\infty} A_n \sin \frac{n\pi}{L} x$$

gives

$$A_n = \frac{2}{L} \int_0^{L/2} \sin \frac{n\pi}{L} x \, dx = \frac{2}{n\pi} \left(1 - \cos \frac{n\pi}{2}\right)$$

for $n = 1, 2, 3, \dots$ so that

$$u(x, t) = \frac{2}{\pi} \sum_{n=1}^{\infty} \frac{1 - \cos \frac{n\pi}{2}}{n} e^{-kn^2\pi^2 t/L^2} \sin \frac{n\pi}{L} x.$$

2. Using $u = XT$ and $-\lambda$ as a separation constant we obtain

$$X'' + \lambda X = 0,$$

$$X(0) = 0,$$

$$X(L) = 0,$$

and

$$T' + k\lambda T = 0.$$

This leads to

$$X = c_1 \sin \frac{n\pi}{L} x \quad \text{and} \quad T = c_2 e^{-kn^2\pi^2 t/L^2}$$

for $n = 1, 2, 3, \dots$ so that

$$u = \sum_{n=1}^{\infty} A_n e^{-kn^2\pi^2 t/L^2} \sin \frac{n\pi}{L} x.$$

Imposing

$$u(x, 0) = \sum_{n=1}^{\infty} A_n \sin \frac{n\pi}{L} x$$

gives

$$A_n = \frac{2}{L} \int_0^L x(L-x) \sin \frac{n\pi}{L} x \, dx = \frac{4L^2}{n^3\pi^3} [1 - (-1)^n]$$

for $n = 1, 2, 3, \dots$ so that

$$u(x, t) = \frac{4L^2}{\pi^3} \sum_{n=1}^{\infty} \frac{1 - (-1)^n}{n^3} e^{-kn^2\pi^2 t/L^2} \sin \frac{n\pi}{L} x.$$

3. Using $u = XT$ and $-\lambda$ as a separation constant we obtain

$$X'' + \lambda X = 0,$$

$$X'(0) = 0,$$

$$X'(L) = 0,$$

and

$$T' + k\lambda T = 0.$$

This leads to

$$X = c_1 \cos \frac{n\pi}{L} x \quad \text{and} \quad T = c_2 e^{-kn^2\pi^2 t/L^2}$$

for $n = 0, 1, 2, \dots$ ($\lambda = 0$ is an eigenvalue in this case) so that

$$u = \sum_{n=0}^{\infty} A_n e^{-kn^2\pi^2 t/L^2} \cos \frac{n\pi}{L} x.$$

Imposing

$$u(x, 0) = f(x) = A_0 + \sum_{n=1}^{\infty} A_n \cos \frac{n\pi}{L} x$$

gives

$$u(x, t) = \frac{1}{L} \int_0^L f(x) dx + \frac{2}{L} \sum_{n=1}^{\infty} \left(\int_0^L f(x) \cos \frac{n\pi}{L} x dx \right) e^{-kn^2\pi^2 t/L^2} \cos \frac{n\pi}{L} x.$$

4. If $L = 2$ and $f(x)$ is x for $0 < x < 1$ and $f(x)$ is 0 for $1 < x < 2$ then

$$u(x, t) = \frac{1}{4} + 4 \sum_{n=1}^{\infty} \left[\frac{1}{2n\pi} \sin \frac{n\pi}{2} + \frac{1}{n^2\pi^2} \left(\cos \frac{n\pi}{2} - 1 \right) \right] e^{-kn^2\pi^2 t/4} \cos \frac{n\pi}{2} x.$$

5. Using $u = XT$ and $-\lambda$ as a separation constant leads to

$$X'' + \lambda X = 0,$$

$$X'(0) = 0,$$

$$X'(L) = 0,$$

and

$$T' + (h + k\lambda)T = 0.$$

Then

$$X = c_1 \cos \frac{n\pi}{L} x \quad \text{and} \quad T = c_2 e^{-ht - kn^2\pi^2 t/L^2}$$

for $n = 0, 1, 2, \dots$ ($\lambda = 0$ is an eigenvalue in this case) so that

$$u = A_0 e^{-ht} + e^{-ht} \sum_{n=1}^{\infty} A_n e^{-kn^2\pi^2 t/L^2} \cos \frac{n\pi}{L} x.$$

Imposing

$$u(x, 0) = f(x) = \sum_{n=0}^{\infty} A_n \cos \frac{n\pi}{L} x$$

gives

$$u(x, t) = \frac{e^{-ht}}{L} \int_0^L f(x) dx + \frac{2e^{-ht}}{L} \sum_{n=1}^{\infty} \left(\int_0^L f(x) \cos \frac{n\pi}{L} x dx \right) e^{-kn^2\pi^2 t/L^2} \cos \frac{n\pi}{L} x.$$

6. In Problem 5 we instead find that $X(0) = 0$ and $X(L) = 0$ so that

$$X = c_1 \sin \frac{n\pi}{L} x$$

and

$$u = \frac{2e^{-ht}}{L} \sum_{n=1}^{\infty} \left(\int_0^L f(x) \sin \frac{n\pi}{L} x dx \right) e^{-kn^2\pi^2 t/L^2} \sin \frac{n\pi}{L} x.$$

7. (a) The solution is

$$u(x, t) = \sum_{n=1}^{\infty} A_n e^{-kn^2\pi^2 t/100^2} \sin \frac{n\pi}{100} x,$$

13.3 Heat Equation

where

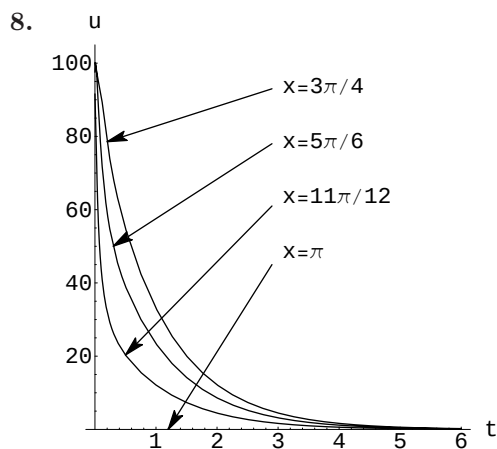
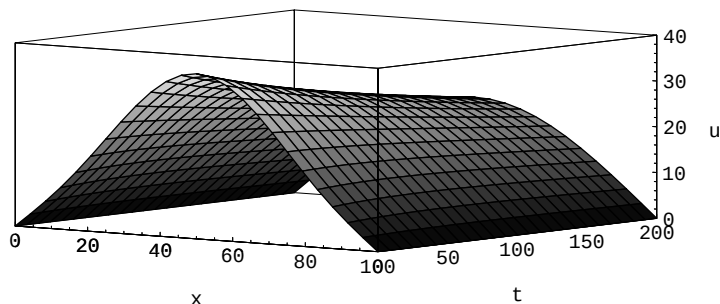
$$A_n = \frac{2}{100} \left[\int_0^{50} 0.8x \sin \frac{n\pi}{100} x \, dx + \int_{50}^{100} 0.8(100-x) \sin \frac{n\pi}{100} x \, dx \right] = \frac{320}{n^2 \pi^2} \sin \frac{n\pi}{2}.$$

Thus,

$$u(x, t) = \frac{320}{\pi^2} \sum_{n=1}^{\infty} \frac{1}{n^2} \left(\sin \frac{n\pi}{2} \right) e^{-kn^2 \pi^2 t / 100^2} \sin \frac{n\pi}{100} x.$$

- (b) Since $A_n = 0$ for n even, the first five nonzero terms correspond to $n = 1, 3, 5, 7, 9$. In this case $\sin(n\pi/2) = \sin(2p-1)/2 = (-1)^{p+1}$ for $p = 1, 2, 3, 4, 5$, and

$$u(x, t) = \frac{320}{\pi^2} \sum_{p=1}^{\infty} \frac{(-1)^{p+1}}{(2p-1)^2} e^{(-1.6352(2p-1)^2 \pi^2 / 100^2)t} \sin \frac{(2p-1)\pi}{100} x.$$



EXERCISES 13.4

Wave Equation

1. Using $u = XT$ and $-\lambda$ as a separation constant we obtain

$$X'' + \lambda X = 0,$$

$$X(0) = 0,$$

$$X(L) = 0,$$

and

$$T'' + \lambda a^2 T = 0,$$

$$T'(0) = 0.$$

Solving the differential equations we get

$$X = c_1 \sin \frac{n\pi}{L}x + c_2 \cos \frac{n\pi}{L}x \quad \text{and} \quad T = c_3 \cos \frac{n\pi a}{L}t + c_4 \sin \frac{n\pi a}{L}t$$

for $n = 1, 2, 3, \dots$. The boundary and initial conditions give

$$u = \sum_{n=1}^{\infty} A_n \cos \frac{n\pi a}{L}t \sin \frac{n\pi}{L}x.$$

Imposing

$$u(x, 0) = \frac{1}{4}x(L - x) = \sum_{n=1}^{\infty} A_n \sin \frac{n\pi}{L}x$$

gives

$$A_n = \frac{L^2}{n^3\pi^3}[1 - (-1)^n]$$

for $n = 1, 2, 3, \dots$ so that

$$u(x, t) = \frac{L^2}{\pi^3} \sum_{n=1}^{\infty} \frac{1 - (-1)^n}{n^3} \cos \frac{n\pi a}{L}t \sin \frac{n\pi}{L}x.$$

2. Using $u = XT$ and $-\lambda$ as a separation constant we obtain

$$X'' + \lambda X = 0,$$

$$X(0) = 0,$$

$$X(L) = 0,$$

and

$$T'' + \lambda a^2 T = 0,$$

$$T(0) = 0.$$

Solving the differential equations we get

$$X = c_1 \sin \frac{n\pi}{L}x + c_2 \cos \frac{n\pi}{L}x \quad \text{and} \quad T = c_3 \cos \frac{n\pi a}{L}t + c_4 \sin \frac{n\pi a}{L}t$$

13.4 Wave Equation

for $n = 1, 2, 3, \dots$. The boundary and initial conditions give

$$u = \sum_{n=1}^{\infty} B_n \sin \frac{n\pi a}{L} t \sin \frac{n\pi}{L} x.$$

Imposing

$$u_t(x, 0) = x(L - x) = \sum_{n=1}^{\infty} B_n \frac{n\pi a}{L} \sin \frac{n\pi}{L} x$$

gives

$$B_n \frac{n\pi a}{L} = \frac{4L^2}{n^3\pi^3} [1 - (-1)^n]$$

for $n = 1, 2, 3, \dots$ so that

$$u(x, t) = \frac{4L^3}{a\pi^4} \sum_{n=1}^{\infty} \frac{1 - (-1)^n}{n^4} \sin \frac{n\pi a}{L} t \sin \frac{n\pi}{L} x.$$

3. Using $u = XT$ and $-\lambda$ as a separation constant we obtain

$$X'' + \lambda X = 0,$$

$$X(0) = 0,$$

$$X(L) = 0,$$

and

$$T'' + \lambda a^2 T = 0,$$

$$T'(0) = 0.$$

Solving the differential equations we get

$$X = c_1 \sin \frac{n\pi}{L} x + c_2 \cos \frac{n\pi}{L} x \quad \text{and} \quad T = c_3 \cos \frac{n\pi a}{L} t + c_4 \sin \frac{n\pi a}{L} t$$

for $n = 1, 2, 3, \dots$. The boundary and initial conditions give

$$u = \sum_{n=1}^{\infty} A_n \cos \frac{n\pi a}{L} t \sin \frac{n\pi}{L} x.$$

Imposing

$$u(x, 0) = \sum_{n=1}^{\infty} A_n \sin \frac{n\pi}{L} x$$

gives

$$A_n = \frac{2}{L} \left(\int_0^{L/3} \frac{3}{L} x \sin \frac{n\pi}{L} x dx + \int_{L/3}^{2L/3} \sin \frac{n\pi}{L} x dx + \int_{2L/3}^L \left(3 - \frac{3}{L} x \right) \sin \frac{n\pi}{L} x dx \right)$$

so that

$$A_1 = \frac{6\sqrt{3}}{\pi^2},$$

$$A_2 = A_3 = A_4 = 0,$$

$$A_5 = -\frac{6\sqrt{3}}{5^2\pi^2},$$

$$A_6 = 0,$$

$$A_7 = \frac{6\sqrt{3}}{7^2\pi^2}$$

$\vdots$

and

$$u(x, t) = \frac{6\sqrt{3}}{\pi^2} \left(\cos \frac{\pi a}{L} t \sin \frac{\pi}{L} x - \frac{1}{5^2} \cos \frac{5\pi a}{L} t \sin \frac{5\pi}{L} x + \frac{1}{7^2} \cos \frac{7\pi a}{L} t \sin \frac{7\pi}{L} x - \cdots \right).$$

4. Using $u = XT$ and $-\lambda$ as a separation constant we obtain

$$X'' + \lambda X = 0,$$

$$X(0) = 0,$$

$$X(\pi) = 0,$$

and

$$T'' + \lambda a^2 T = 0,$$

$$T'(0) = 0.$$

Solving the differential equations we get

$$X = c_1 \sin nx + c_2 \cos nx \quad \text{and} \quad T = c_3 \cos nat + c_4 \sin nat$$

for $n = 1, 2, 3, \dots$. The boundary and initial conditions give

$$u = \sum_{n=1}^{\infty} A_n \cos nt \sin nx.$$

Imposing

$$u(x, 0) = \frac{1}{6}x(\pi^2 - x^2) = \sum_{n=1}^{\infty} A_n \sin nx \quad \text{and} \quad u_t(x, 0) = 0$$

gives

$$A_n = \frac{2}{n^3}(-1)^{n+1}$$

for $n = 1, 2, 3, \dots$ so that

$$u(x, t) = 2 \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n^3} \cos nat \sin nx.$$

5. Using $u = XT$ and $-\lambda$ as a separation constant we obtain

$$X'' + \lambda X = 0,$$

$$X(0) = 0,$$

$$X(\pi) = 0,$$

and

$$T'' + \lambda a^2 T = 0,$$

$$T'(0) = 0.$$

Solving the differential equations we get

$$X = c_1 \sin nx + c_2 \cos nx \quad \text{and} \quad T = c_3 \cos nat + c_4 \sin nat$$

for $n = 1, 2, 3, \dots$. The boundary and initial conditions give

$$u = \sum_{n=1}^{\infty} A_n \cos nt \sin nx.$$

Imposing

$$u_t(x, 0) = \sin x = \sum_{n=1}^{\infty} B_n na \sin nx$$

13.4 Wave Equation

gives

$$B_1 = \frac{1}{a^2}, \quad \text{and} \quad B_n = 0$$

for $n = 2, 3, 4, \dots$ so that

$$u(x, t) = \frac{1}{a} \sin at \sin x.$$

6. Using $u = XT$ and $-\lambda$ as a separation constant we obtain

$$X'' + \lambda X = 0,$$

$$X(0) = 0,$$

$$X(1) = 0,$$

and

$$T'' + \lambda a^2 T = 0,$$

$$T'(0) = 0.$$

Solving the differential equations we get

$$X = c_1 \sin n\pi x + c_2 \cos n\pi x \quad \text{and} \quad T = c_3 \cos n\pi at + c_4 \sin n\pi at$$

for $n = 1, 2, 3, \dots$. The boundary and initial conditions give

$$u = \sum_{n=1}^{\infty} A_n \cos nt \sin nx.$$

Imposing

$$u(x, 0) = 0.01 \sin 3\pi x = \sum_{n=1}^{\infty} A_n \sin n\pi x$$

gives $A_3 = 0.01$, and $A_n = 0$ for $n = 1, 2, 4, 5, 6, \dots$ so that

$$u(x, t) = 0.01 \sin 3\pi x \cos 3\pi at.$$

7. Using $u = XT$ and $-\lambda$ as a separation constant we obtain

$$X'' + \lambda X = 0,$$

$$X(0) = 0,$$

$$X(L) = 0,$$

and

$$T'' + \lambda a^2 T = 0,$$

$$T'(0) = 0.$$

Solving the differential equations we get

$$X = c_1 \sin \frac{n\pi}{L} x + c_2 \cos \frac{n\pi}{L} x \quad \text{and} \quad T = c_3 \cos \frac{n\pi a}{L} t + c_4 \sin \frac{n\pi a}{L} t$$

for $n = 1, 2, 3, \dots$. The boundary and initial conditions give

$$u = \sum_{n=1}^{\infty} A_n \cos \frac{n\pi a}{L} t \sin \frac{n\pi}{L} x.$$

Imposing

$$u(x, 0) = \sum_{n=1}^{\infty} A_n \sin \frac{n\pi}{L} x$$

gives

$$A_n = \frac{8h}{n^2\pi^2} \sin \frac{n\pi}{2}$$

for $n = 1, 2, 3, \dots$ so that

$$u(x, t) = \frac{8h}{\pi^2} \sum_{n=1}^{\infty} \frac{1}{n^2} \sin \frac{n\pi}{2} \cos \frac{n\pi a}{L} t \sin \frac{n\pi}{L} x.$$

8. Using $u = XT$ and $-\lambda$ as a separation constant we obtain

$$X'' + \lambda X = 0,$$

$$X'(0) = 0,$$

$$X'(L) = 0,$$

and

$$T'' + \lambda a^2 T = 0,$$

$$T'(0) = 0.$$

Solving the differential equations we get

$$X = c_1 \sin \frac{n\pi}{L} x + c_2 \cos \frac{n\pi}{L} x \quad \text{and} \quad T = c_3 \cos \frac{n\pi a}{L} t + c_4 \sin \frac{n\pi a}{L} t$$

for $n = 1, 2, 3, \dots$. The boundary and initial conditions, together with the fact that $\lambda = 0$ is an eigenvalue with eigenfunction $X(x) = 1$, give

$$u = A_0 + \sum_{n=1}^{\infty} A_n \cos \frac{n\pi a}{L} t \sin \frac{n\pi}{L} x.$$

Imposing

$$u(x, 0) = x = A_0 + \sum_{n=1}^{\infty} A_n \cos \frac{n\pi}{L} x$$

gives

$$A_0 = \frac{1}{L} \int_0^L x \, dx = \frac{L}{2}$$

and

$$A_n = \frac{2}{L} \int_0^L x \cos \frac{n\pi}{L} x \, dx = \frac{2L}{n^2\pi^2} [(-1)^n - 1]$$

for $n = 1, 2, 3, \dots$, so that

$$u(x, t) = \frac{L}{2} + \frac{2L}{\pi^2} \sum_{n=1}^{\infty} \frac{(-1)^n - 1}{n^2} \cos \frac{n\pi a}{L} t \cos \frac{n\pi}{L} x.$$

9. Using $u = XT$ and $-\lambda$ as a separation constant we obtain

$$X'' + \lambda X = 0,$$

$$X(0) = 0,$$

$$X(\pi) = 0,$$

and

$$T'' + 2\beta T' + \lambda T = 0,$$

$$T'(0) = 0.$$

13.4 Wave Equation

Solving the differential equations we get

$$X = c_1 \sin nx + c_2 \cos nx \quad \text{and} \quad T = e^{-\beta t} \left(c_3 \cos \sqrt{n^2 - \beta^2} t + c_4 \sin \sqrt{n^2 - \beta^2} t \right)$$

The boundary conditions on X imply $c_2 = 0$ so

$$X = c_1 \sin nx \quad \text{and} \quad T = e^{-\beta t} \left(c_3 \cos \sqrt{n^2 - \beta^2} t + c_4 \sin \sqrt{n^2 - \beta^2} t \right)$$

and

$$u = \sum_{n=1}^{\infty} e^{-\beta t} \left(A_n \cos \sqrt{n^2 - \beta^2} t + B_n \sin \sqrt{n^2 - \beta^2} t \right) \sin nx.$$

Imposing

$$u(x, 0) = f(x) = \sum_{n=1}^{\infty} A_n \sin nx$$

and

$$u_t(x, 0) = 0 = \sum_{n=1}^{\infty} \left(B_n \sqrt{n^2 - \beta^2} - \beta A_n \right) \sin nx$$

gives

$$u(x, t) = e^{-\beta t} \sum_{n=1}^{\infty} A_n \left(\cos \sqrt{n^2 - \beta^2} t + \frac{\beta}{\sqrt{n^2 - \beta^2}} \sin \sqrt{n^2 - \beta^2} t \right) \sin nx,$$

where

$$A_n = \frac{2}{\pi} \int_0^{\pi} f(x) \sin nx \, dx.$$

10. Using $u = XT$ and $-\lambda = \alpha^2$ as a separation constant leads to $X'' + \alpha^2 X = 0$, $X(0) = 0$, $X(\pi) = 0$ and $T'' + (1 + \alpha^2)T = 0$, $T'(0) = 0$. Then $X = c_2 \sin nx$ and $T = c_3 \cos \sqrt{n^2 + 1} t$ for $n = 1, 2, 3, \dots$ so that

$$u = \sum_{n=1}^{\infty} B_n \cos \sqrt{n^2 + 1} t \sin nx.$$

Imposing $u(x, 0) = \sum_{n=1}^{\infty} B_n \sin nx$ gives

$$\begin{aligned} B_n &= \frac{2}{\pi} \int_0^{\pi/2} x \sin nx \, dx + \frac{2}{\pi} \int_{\pi/2}^{\pi} (\pi - x) \sin nx \, dx = \frac{4}{\pi n^2} \sin \frac{n\pi}{2} \\ &= \begin{cases} 0, & n \text{ even,} \\ \frac{4}{\pi n^2} (-1)^{(n+3)/2}, & n = 2k - 1, k = 1, 2, 3, \dots \end{cases} \end{aligned}$$

Thus with $n = 2k - 1$,

$$u(x, t) = \frac{4}{\pi} \sum_{n=1}^{\infty} \frac{\sin \frac{n\pi}{2}}{n^2} \cos \sqrt{n^2 + 1} t \sin nx = \frac{4}{\pi} \sum_{k=1}^{\infty} \frac{(-1)^{k+1}}{(2k-1)^2} \cos \sqrt{(2k-1)^2 + 1} t \sin(2k-1)x.$$

11. From (8) in the text we have

$$u(x, t) = \sum_{n=1}^{\infty} \left(A_n \cos \frac{n\pi a}{L} t + B_n \sin \frac{n\pi a}{L} t \right) \sin \frac{n\pi}{L} x.$$

Since $u_t(x, 0) = g(x) = 0$ we have $B_n = 0$ and

$$\begin{aligned} u(x, t) &= \sum_{n=1}^{\infty} A_n \cos \frac{n\pi a}{L} t \sin \frac{n\pi}{L} x \\ &= \sum_{n=1}^{\infty} A_n \frac{1}{2} \left[\sin \left(\frac{n\pi}{L} x + \frac{n\pi a}{L} t \right) + \sin \left(\frac{n\pi}{L} x - \frac{n\pi a}{L} t \right) \right] \\ &= \frac{1}{2} \sum_{n=1}^{\infty} A_n \left[\sin \frac{n\pi}{L} (x + at) + \sin \frac{n\pi}{L} (x - at) \right]. \end{aligned}$$

From

$$u(x, 0) = f(x) = \sum_{n=1}^{\infty} A_n \sin \frac{n\pi}{L} x$$

we identify

$$f(x + at) = \sum_{n=1}^{\infty} A_n \sin \frac{n\pi}{L} (x + at)$$

and

$$f(x - at) = \sum_{n=1}^{\infty} A_n \sin \frac{n\pi}{L} (x - at),$$

so that

$$u(x, t) = \frac{1}{2} [f(x + at) + f(x - at)].$$

12. (a) We note that $\xi_x = \eta_x = 1$, $\xi_t = a$, and $\eta_t = -a$. Then

$$\frac{\partial u}{\partial x} = \frac{\partial u}{\partial \xi} \frac{\partial \xi}{\partial x} + \frac{\partial u}{\partial \eta} \frac{\partial \eta}{\partial x} = u_\xi + u_\eta$$

and

$$\begin{aligned} \frac{\partial^2 u}{\partial x^2} &= \frac{\partial}{\partial x} (u_\xi + u_\eta) = \frac{\partial u_\xi}{\partial \xi} \frac{\partial \xi}{\partial x} + \frac{\partial u_\xi}{\partial \eta} \frac{\partial \eta}{\partial x} + \frac{\partial u_\eta}{\partial \xi} \frac{\partial \xi}{\partial x} + \frac{\partial u_\eta}{\partial \eta} \frac{\partial \eta}{\partial x} \\ &= u_{\xi\xi} + 2u_{\xi\eta} + u_{\eta\eta}. \end{aligned}$$

Similarly

$$\frac{\partial^2 u}{\partial t^2} = a^2 (u_{\xi\xi} - 2u_{\xi\eta} + u_{\eta\eta}).$$

Thus

$$a^2 \frac{\partial^2 u}{\partial x^2} = \frac{\partial^2 u}{\partial t^2} \quad \text{becomes} \quad \frac{\partial^2 u}{\partial \xi \partial \eta} = 0.$$

(b) Integrating

$$\frac{\partial^2 u}{\partial \xi \partial \eta} = \frac{\partial}{\partial \eta} u_\xi = 0$$

we obtain

$$\begin{aligned} \int \frac{\partial}{\partial \eta} u_\xi d\eta &= \int 0 d\eta \\ u_\xi &= f(\xi). \end{aligned}$$

Integrating this result with respect to ξ we obtain

$$\begin{aligned} \int \frac{\partial u}{\partial \xi} d\xi &= \int f(\xi) d\xi \\ u &= F(\xi) + G(\eta). \end{aligned}$$

13.4 Wave Equation

Since $\xi = x + at$ and $\eta = x - at$, we then have

$$u = F(\xi) + G(\eta) = F(x + at) + G(x - at).$$

Next, we have

$$\begin{aligned} u(x, t) &= F(x + at) + G(x - at) \\ u(x, 0) &= F(x) + G(x) = f(x) \\ u_t(x, 0) &= aF'(x) - aG'(x) = g(x) \end{aligned}$$

Integrating the last equation with respect to x gives

$$F(x) - G(x) = \frac{1}{a} \int_{x_0}^x g(s) ds + c_1.$$

Substituting $G(x) = f(x) - F(x)$ we obtain

$$F(x) = \frac{1}{2} f(x) + \frac{1}{2a} \int_{x_0}^x g(s) ds + c$$

where $c = c_1/2$. Thus

$$G(x) = \frac{1}{2} f(x) - \frac{1}{2a} \int_{x_0}^x g(s) ds - c.$$

(c) From the expressions for F and G ,

$$\begin{aligned} F(x + at) &= \frac{1}{2} f(x + at) + \frac{1}{2a} \int_{x_0}^{x+at} g(s) ds + c \\ G(x - at) &= \frac{1}{2} f(x - at) - \frac{1}{2a} \int_{x_0}^{x-at} g(s) ds - c. \end{aligned}$$

Thus,

$$u(x, t) = F(x + at) + G(x - at) = \frac{1}{2} [f(x + at) + f(x - at)] + \frac{1}{2a} \int_{x-at}^{x+at} g(s) ds.$$

Here we have used $-\int_{x_0}^{x-at} g(s) ds = \int_{x-at}^{x_0} g(s) ds$.

$$\begin{aligned} \text{13. } u(x, t) &= \frac{1}{2} [\sin(x + at) + \sin(x - at)] + \frac{1}{2a} \int_{x-at}^{x+at} ds \\ &= \frac{1}{2} [\sin x \cos at + \cos x \sin at + \sin x \cos at - \cos x \sin at] + \frac{1}{2a} s \Big|_{x-at}^{x+at} = \sin x \cos at + t \end{aligned}$$

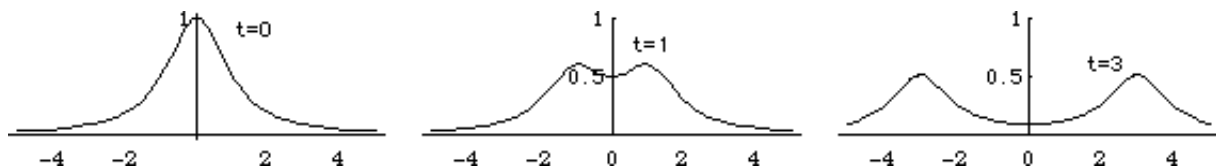
$$\begin{aligned} \text{14. } u(x, t) &= \frac{1}{2} [\sin(x + at) + \sin(x - at)] + \frac{1}{2a} \int_{x-at}^{x+at} \cos s ds \\ &= \sin x \cos at + \frac{1}{2a} [\sin(x + at) - \sin(x - at)] = \sin x \cos at + \frac{1}{a} \cos x \sin at \end{aligned}$$

$$\begin{aligned} \text{15. } u(x, t) &= 0 + \frac{1}{2a} \int_{x-at}^{x+at} \sin 2s ds = \frac{1}{2a} \left[\frac{-\cos(2x + 2at) + \cos(2x - 2at)}{2} \right] \\ &= \frac{1}{4a} [-\cos 2x \cos 2at + \sin 2x \sin 2at + \cos 2x \cos 2at + \sin 2x \sin 2at] = \frac{1}{2a} \sin 2x \sin 2at \end{aligned}$$

16. As noted in Problem 12 of this exercise set, when the initial velocity is $g(x) = 0$, d'Alembert's solution is $u(x, t) = \frac{1}{2} [f(x + at) + f(x - at)]$, $-\infty < x < \infty$. with $a = 1$ and $f(x) = 1/(1 + x^2)$ this becomes

$$u(x, t) = \frac{1}{2} \left[\frac{1}{1 + (x + t)^2} + \frac{1}{1 + (x - t)^2} \right].$$

The graphs of this function for $t = 0$, $t = 1$, and $t = 3$ are shown below.



17. Separating variables in the partial differential equation and using the separation constant $-\lambda = \alpha^4$ gives

$$\frac{X^{(4)}}{X} = -\frac{T''}{a^2 T} = \alpha^4$$

so that

$$X^{(4)} - \alpha^4 X = 0$$

$$T'' + a^2 \alpha^4 T = 0$$

and

$$X = c_1 \cosh \alpha x + c_2 \sinh \alpha x + c_3 \cos \alpha x + c_4 \sin \alpha x$$

$$T = c_5 \cos a \alpha^2 t + c_6 \sin a \alpha^2 t.$$

The boundary conditions translate into $X(0) = X(L) = 0$ and $X''(0) = X''(L) = 0$. From $X(0) = X''(0) = 0$ we find $c_1 = c_3 = 0$. From

$$X(L) = c_2 \sinh \alpha L + c_4 \sin \alpha L = 0$$

$$X''(L) = \alpha^2 c_2 \sinh \alpha L - \alpha^2 c_4 \sin \alpha L = 0$$

we see by subtraction that $c_4 \sin \alpha L = 0$. This equation yields the eigenvalues $\alpha = n\pi/L$ for $n = 1, 2, 3, \dots$.

The corresponding eigenfunctions are

$$X = c_4 \sin \frac{n\pi}{L} x.$$

Thus

$$u(x, t) = \sum_{n=1}^{\infty} \left(A_n \cos \frac{n^2 \pi^2}{L^2} at + B_n \sin \frac{n^2 \pi^2}{L^2} at \right) \sin \frac{n\pi}{L} x.$$

From

$$u(x, 0) = f(x) = \sum_{n=1}^{\infty} A_n \sin \frac{n\pi}{L} x$$

we obtain

$$A_n = \frac{2}{L} \int_0^L f(x) \sin \frac{n\pi}{L} x dx.$$

From

$$\frac{\partial u}{\partial t} = \sum_{n=1}^{\infty} \left(-A_n \frac{n^2 \pi^2 a}{L^2} \sin \frac{n^2 \pi^2}{L^2} at + B_n \frac{n^2 \pi^2 a}{L^2} \cos \frac{n^2 \pi^2}{L^2} at \right) \sin \frac{n\pi}{L} x$$

and

$$\left. \frac{\partial u}{\partial t} \right|_{t=0} = g(x) = \sum_{n=1}^{\infty} B_n \frac{n^2 \pi^2 a}{L^2} \sin \frac{n\pi}{L} x$$

we obtain

$$B_n \frac{n^2 \pi^2 a}{L^2} = \frac{2}{L} \int_0^L g(x) \sin \frac{n\pi}{L} x dx$$

and

$$B_n = \frac{2L}{n^2 \pi^2 a} \int_0^L g(x) \sin \frac{n\pi}{L} x dx.$$

13.4 Wave Equation

18. (a) Write the differential equation in X from Problem 17 as $X^{(4)} - \alpha^4 X = 0$ where the eigenvalues are $\lambda = \alpha^2$. Then

$$X = c_1 \cosh \alpha x + c_2 \sinh \alpha x + c_3 \cos \alpha x + c_4 \sin \alpha x$$

and using $X(0) = 0$ and $X'(0) = 0$ we find $c_3 = -c_1$ and $c_4 = -c_2$. The conditions $X(L) = 0$ and $X'(L) = 0$ yield the system of equations

$$c_1(\cosh \alpha L - \cos \alpha L) + c_2(\sinh \alpha L - \sin \alpha L) = 0$$

$$c_1(\alpha \sinh \alpha L + \alpha \sin \alpha L) + c_2(\alpha \cosh \alpha L - \alpha \cos \alpha L) = 0.$$

In order for this system to have nontrivial solutions the determinant of the coefficients must be zero. That is,

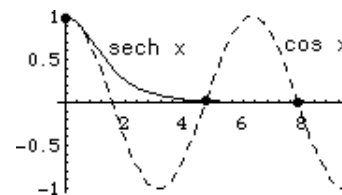
$$\alpha(\cosh \alpha L - \cos \alpha L)^2 - \alpha(\sinh^2 \alpha L - \sin^2 \alpha L) = 0.$$

Since $\alpha = 0$ leads to $X = 0$, $\lambda = \alpha^2 = 0^2 = 0$ is not an eigenvalue. Then, dividing the above equation by α , we have

$$\begin{aligned} & (\cosh \alpha L - \cos \alpha L)^2 - (\sinh^2 \alpha L - \sin^2 \alpha L) \\ &= \cosh^2 \alpha L - 2 \cosh \alpha L \cos \alpha L + \cos^2 \alpha L - \sinh^2 \alpha L + \sin^2 \alpha L \\ &= -2 \cosh \alpha L \cos \alpha L + 2 = 0 \end{aligned}$$

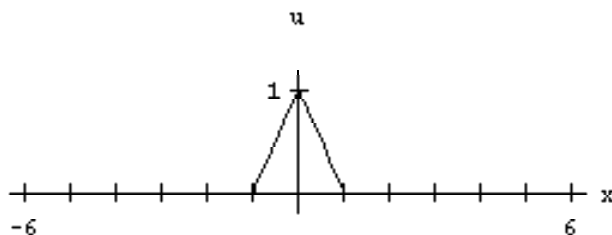
or $\cosh \alpha L \cos \alpha L = 1$. Letting $x = \alpha L$ we see that the eigenvalues are $\lambda_n = \alpha_n^2 = x_n^2/L^2$ where x_n , $n = 1, 2, 3, \dots$, are the positive roots of the equation $\cosh x \cos x = 1$.

- (b) The equation $\cosh x \cos x = 1$ is the same as $\cos x = \operatorname{sech} x$. The figure indicates that the equation has an infinite number of roots.



- (c) Using a CAS we find the first four positive roots of $\cosh x \cos x = 1$ to be $x_1 = 4.7300$, $x_2 = 7.8532$, $x_3 = 10.9956$, and $x_4 = 14.1372$. Thus the first four eigenvalues are $\lambda_1 = x_1^2/L = 22.3733/L$, $\lambda_2 = x_2^2/L = 61.6728/L$, $\lambda_3 = x_3^2/L = 120.9034/L$, and $\lambda_4 = x_4^2/L = 199.8594/L$.

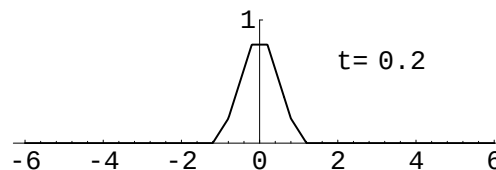
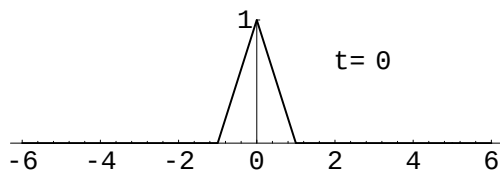
19. (a)

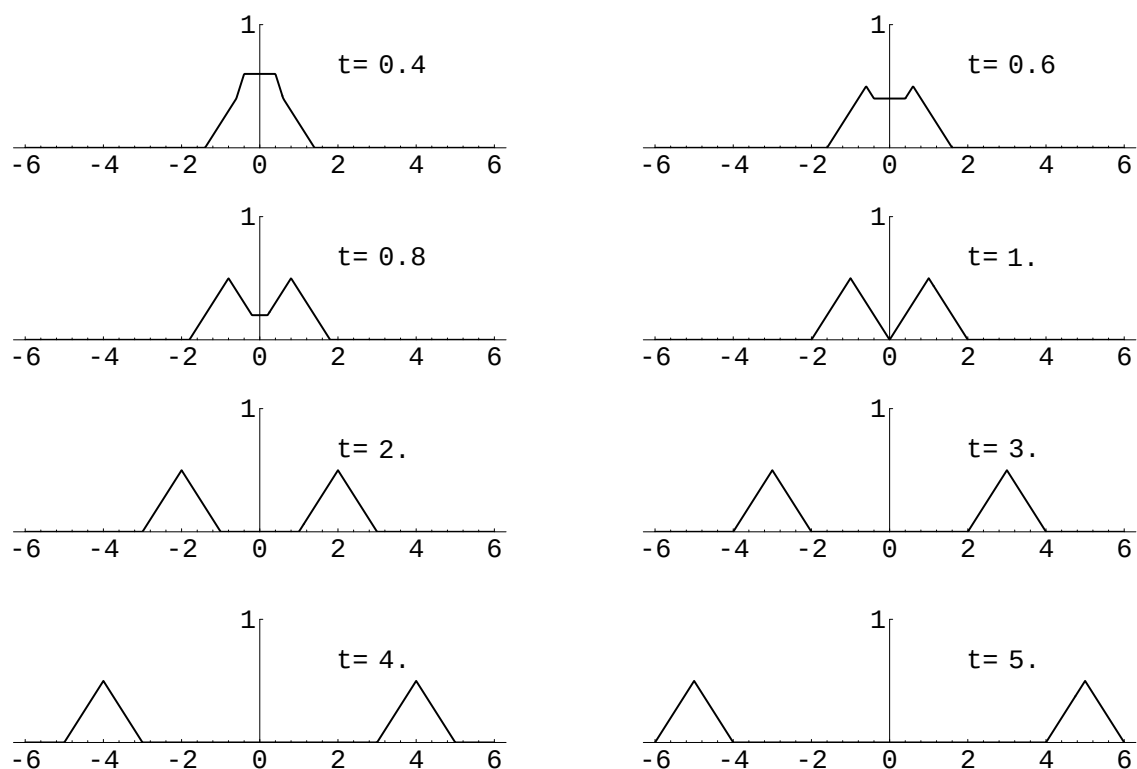


- (b) Since $g(x) = 0$, d'Alembert's solution with $a = 1$ is

$$u(x, t) = \frac{1}{2}[f(x+t) + f(x-t)].$$

Sample plots are shown below.



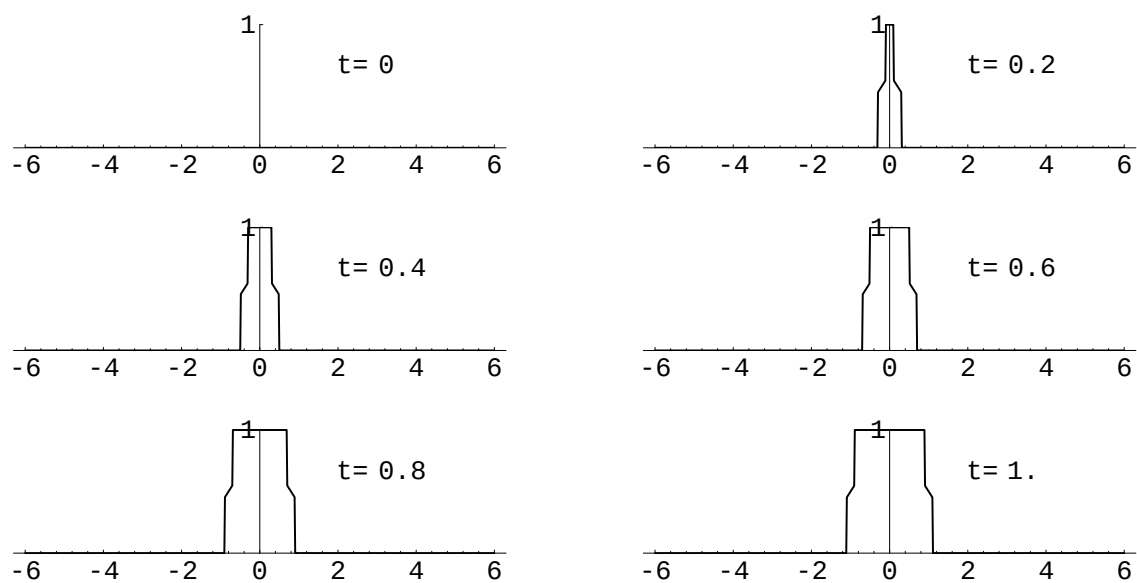


(c) The single peaked wave dissolves into two peaks moving outward.

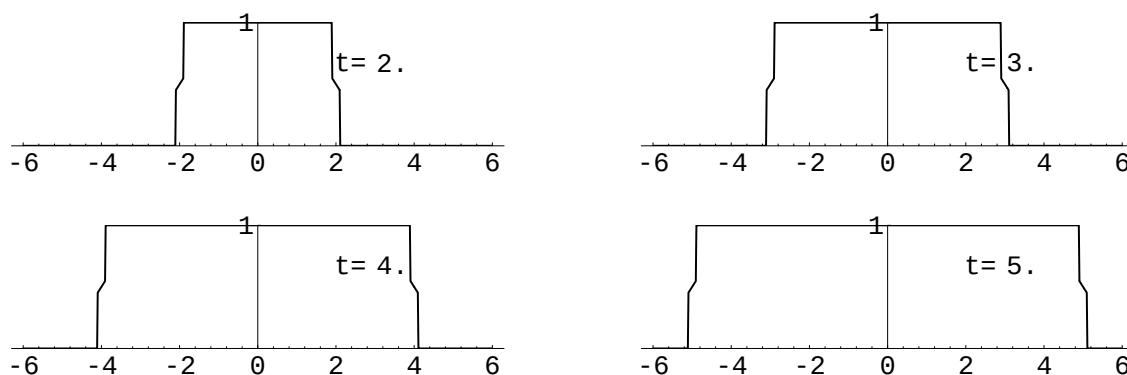
20. (a) With $a = 1$, d'Alembert's solution is

$$u(x, t) = \frac{1}{2} \int_{x-t}^{x+t} g(s) ds \quad \text{where} \quad g(s) = \begin{cases} 1, & |s| \leq 0.1 \\ 0, & |s| > 0.1. \end{cases}$$

Sample plots are shown below.



13.4 Wave Equation



Some frames of the movie are shown in part (a). The string has a roughly rectangular shape with the base on the x -axis increasing in length.

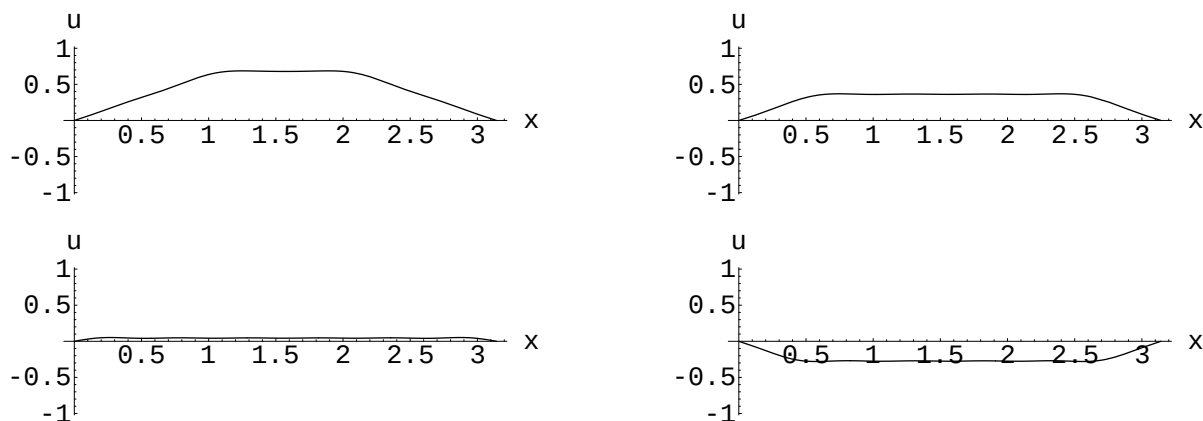
21. (a) and (b) With the given parameters, the solution is

$$u(x, t) = \frac{8}{\pi^2} \sum_{n=1}^{\infty} \frac{1}{n^2} \sin \frac{n\pi}{2} \cos nt \sin nx.$$

For n even, $\sin(n\pi/2) = 0$, so the first six nonzero terms correspond to $n = 1, 3, 5, 7, 9, 11$. In this case $\sin(n\pi/2) = \sin(2p-1)/2 = (-1)^{p+1}$ for $p = 1, 2, 3, 4, 5, 6$, and

$$u(x, t) = \frac{8}{\pi^2} \sum_{p=1}^{\infty} \frac{(-1)^{p+1}}{(2p-1)^2} \cos(2p-1)t \sin(2p-1)x.$$

Frames of the movie corresponding to $t = 0.5, 1, 1.5$, and 2 are shown.



EXERCISES 13.5

Laplace's Equation

1. Using $u = XY$ and $-\lambda$ as a separation constant we obtain

$$X'' + \lambda X = 0,$$

$$X(0) = 0,$$

$$X(a) = 0,$$

and

$$Y'' - \lambda Y = 0,$$

$$Y(0) = 0.$$

With $\lambda = \alpha^2 > 0$ the solutions of the differential equations are

$$X = c_1 \cos \alpha x + c_2 \sin \alpha x \quad \text{and} \quad Y = c_3 \cosh \alpha y + c_4 \sinh \alpha y$$

The boundary and initial conditions imply

$$X = c_2 \sin \frac{n\pi}{a} x \quad \text{and} \quad Y = c_4 \sinh \frac{n\pi}{a} y$$

for $n = 1, 2, 3, \dots$ so that

$$u = \sum_{n=1}^{\infty} A_n \sin \frac{n\pi}{a} x \sinh \frac{n\pi}{a} y.$$

Imposing

$$u(x, b) = f(x) = \sum_{n=1}^{\infty} A_n \sinh \frac{n\pi b}{a} \sin \frac{n\pi}{a} x$$

gives

$$A_n \sinh \frac{n\pi b}{a} = \frac{2}{a} \int_0^a f(x) \sin \frac{n\pi}{a} x \, dx$$

so that

$$u(x, y) = \sum_{n=1}^{\infty} A_n \sin \frac{n\pi}{a} x \sinh \frac{n\pi}{a} y$$

where

$$A_n = \frac{2}{a} \operatorname{csch} \frac{n\pi b}{a} \int_0^a f(x) \sin \frac{n\pi}{a} x \, dx.$$

2. Using $u = XY$ and $-\lambda$ as a separation constant we obtain

$$X'' + \lambda X = 0,$$

$$X(0) = 0,$$

$$X(a) = 0,$$

and

$$Y'' - \lambda Y = 0,$$

$$Y'(0) = 0.$$

13.5 Laplace's Equation

With $\lambda = \alpha^2 > 0$ the solutions of the differential equations are

$$X = c_1 \cos \alpha x + c_2 \sin \alpha x \quad \text{and} \quad Y = c_3 \cosh \alpha y + c_4 \sinh \alpha y$$

The boundary and initial conditions imply

$$X = c_2 \sin \frac{n\pi}{a} x \quad \text{and} \quad Y = c_3 \cosh \frac{n\pi}{a} y$$

for $n = 1, 2, 3, \dots$ so that

$$u = \sum_{n=1}^{\infty} A_n \sin \frac{n\pi}{a} x \cosh \frac{n\pi}{a} y.$$

Imposing

$$u(x, b) = f(x) = \sum_{n=1}^{\infty} A_n \cosh \frac{n\pi b}{a} \sin \frac{n\pi}{a} x$$

gives

$$A_n \cosh \frac{n\pi b}{a} = \frac{2}{a} \int_0^a f(x) \sin \frac{n\pi}{a} x dx$$

so that

$$u(x, y) = \sum_{n=1}^{\infty} A_n \sin \frac{n\pi}{a} x \cosh \frac{n\pi}{a} y$$

where

$$A_n = \frac{2}{a} \operatorname{sech} \frac{n\pi b}{a} \int_0^a f(x) \sin \frac{n\pi}{a} x dx.$$

3. Using $u = XY$ and $-\lambda$ as a separation constant we obtain

$$X'' + \lambda X = 0,$$

$$X(0) = 0,$$

$$X(a) = 0,$$

and

$$Y'' - \lambda Y = 0,$$

$$Y(b) = 0.$$

With $\lambda = \alpha^2 > 0$ the solutions of the differential equations are

$$X = c_1 \cos \alpha x + c_2 \sin \alpha x \quad \text{and} \quad Y = c_3 \cosh \alpha y + c_4 \sinh \alpha y$$

The boundary and initial conditions imply

$$X = c_2 \sin \frac{n\pi}{a} x \quad \text{and} \quad Y = c_2 \cosh \frac{n\pi}{a} y - c_2 \frac{\cosh \frac{n\pi b}{a}}{\sinh \frac{n\pi b}{a}} \sinh \frac{n\pi}{a} y$$

for $n = 1, 2, 3, \dots$ so that

$$u = \sum_{n=1}^{\infty} A_n \left(\cosh \frac{n\pi}{a} y - \frac{\cosh \frac{n\pi b}{a}}{\sinh \frac{n\pi b}{a}} \sinh \frac{n\pi}{a} y \right) \sin \frac{n\pi}{a} x.$$

Imposing

$$u(x, 0) = f(x) = \sum_{n=1}^{\infty} A_n \sin \frac{n\pi}{a} x$$

gives

$$A_n = \frac{2}{a} \int_0^a f(x) \sin \frac{n\pi}{a} x dx$$

so that

$$u(x, y) = \frac{2}{a} \sum_{n=1}^{\infty} \left(\int_0^a f(x) \sin \frac{n\pi}{a} x dx \right) \left(\cosh \frac{n\pi}{a} y - \frac{\cosh \frac{n\pi b}{a}}{\sinh \frac{n\pi b}{a}} \sinh \frac{n\pi}{a} y \right) \sin \frac{n\pi}{a} x.$$

4. Using $u = XY$ and $-\lambda$ as a separation constant we obtain

$$X'' + \lambda X = 0,$$

$$X'(0) = 0,$$

$$X'(a) = 0,$$

and

$$Y'' - \lambda Y = 0,$$

$$Y(b) = 0.$$

With $\lambda = \alpha^2 > 0$ the solutions of the differential equations are

$$X = c_1 \cos \alpha x + c_2 \sin \alpha x \quad \text{and} \quad Y = c_3 \cosh \alpha y + c_4 \sinh \alpha y$$

The boundary and initial conditions imply

$$X = c_1 \cos \frac{n\pi}{a} x \quad \text{and} \quad Y = c_3 \cosh \frac{n\pi}{a} y - c_3 \frac{\cosh \frac{n\pi b}{a}}{\sinh \frac{n\pi b}{a}} \sinh \frac{n\pi}{a} y$$

for $n = 1, 2, 3, \dots$. Since $\lambda = 0$ is an eigenvalue for both differential equations with corresponding eigenfunctions 1 and $y - b$, respectively we have

$$u = A_0(y - b) + \sum_{n=1}^{\infty} A_n \cos \frac{n\pi}{a} x \left(\cosh \frac{n\pi}{a} y - \frac{\cosh \frac{n\pi b}{a}}{\sinh \frac{n\pi b}{a}} \sinh \frac{n\pi}{a} y \right).$$

Imposing

$$u(x, 0) = x = -A_0 b + \sum_{n=1}^{\infty} A_n \cos \frac{n\pi}{a} x$$

gives

$$-A_0 b = \frac{1}{a} \int_0^a x dx = \frac{1}{2} a$$

and

$$A_n = \frac{2}{a} \int_0^a x \cos \frac{n\pi}{a} x dx = \frac{2a}{n^2 \pi^2} [(-1)^n - 1]$$

so that

$$u(x, y) = \frac{a}{2b}(b - y) + \frac{2a}{\pi^2} \sum_{n=1}^{\infty} \frac{(-1)^n - 1}{n^2} \cos \frac{n\pi}{a} x \left(\cosh \frac{n\pi}{a} y - \frac{\cosh \frac{n\pi b}{a}}{\sinh \frac{n\pi b}{a}} \sinh \frac{n\pi}{a} y \right).$$

5. Using $u = XY$ and $-\lambda$ as a separation constant we obtain

$$X'' + \lambda X = 0,$$

$$X'(0) = 0,$$

$$X'(a) = 0,$$

and

$$Y'' - \lambda Y = 0,$$

$$Y(b) = 0.$$

13.5 Laplace's Equation

With $\lambda = -\alpha^2 < 0$ the solutions of the differential equations are

$$X = c_1 \cosh \alpha x + c_2 \sinh \alpha x \quad \text{and} \quad Y = c_3 \cos \alpha y + c_4 \sin \alpha y$$

for $n = 1, 2, 3, \dots$. The boundary and initial conditions imply

$$X = c_2 \sinh n\pi x \quad \text{and} \quad Y = c_3 \cos n\pi y$$

for $n = 1, 2, 3, \dots$. Since $\lambda = 0$ is an eigenvalue for the differential equation in X with corresponding eigenfunction x we have

$$u = A_0 x + \sum_{n=1}^{\infty} A_n \sinh n\pi x \cos n\pi y.$$

Imposing

$$u(1, y) = 1 - y = A_0 + \sum_{n=1}^{\infty} A_n \sinh n\pi \cos n\pi y$$

gives

$$A_0 = \int_0^1 (1 - y) dy$$

and

$$A_n \sinh n\pi = 2 \int_0^1 (1 - y) \cos n\pi y dy = \frac{2[1 - (-1)^n]}{n^2 \pi^2}$$

for $n = 1, 2, 3, \dots$ so that

$$u(x, y) = \frac{1}{2}x + \frac{2}{\pi^2} \sum_{n=1}^{\infty} \frac{1 - (-1)^n}{n^2 \sinh n\pi} \sinh n\pi x \cos n\pi y.$$

6. Using $u = XY$ and $-\lambda$ as a separation constant we obtain

$$X'' + \lambda X = 0,$$

$$X'(1) = 0$$

and

$$Y'' - \lambda Y = 0,$$

$$Y'(0) = 0,$$

$$Y'(\pi) = 0.$$

With $\lambda = \alpha^2 < 0$ the solutions of the differential equations are

$$X = c_1 \cosh \alpha x + c_2 \sinh \alpha x \quad \text{and} \quad Y = c_3 \cos \alpha y + c_4 \sin \alpha y$$

The boundary and initial conditions imply

$$X = c_1 \cosh nx - c_1 \frac{\sinh n}{\cosh n} \sinh nx \quad \text{and} \quad Y = c_3 \cos ny$$

for $n = 1, 2, 3, \dots$. Since $\lambda = 0$ is an eigenvalue for both differential equations with corresponding eigenfunctions 1 and 1 we have

$$u = A_0 + \sum_{n=1}^{\infty} A_n \left(\cosh nx - \frac{\sinh n}{\cosh n} \sinh nx \right) \cos ny.$$

Imposing

$$u(0, y) = g(y) = A_0 + \sum_{n=1}^{\infty} A_n \cos ny$$

gives

$$A_0 = \frac{1}{\pi} \int_0^\pi g(y) dy \quad \text{and} \quad A_n = \frac{2}{\pi} \int_0^\pi g(y) \cos ny dy$$

for $n = 1, 2, 3, \dots$ so that

$$u(x, y) = \frac{1}{\pi} \int_0^\pi g(y) dy + \sum_{n=1}^{\infty} \left(\frac{2}{\pi} \int_0^\pi g(y) \cos ny dy \right) \left(\cosh nx - \frac{\sinh n}{\cosh n} \sinh nx \right) \cos ny.$$

7. Using $u = XY$ and $-\lambda$ as a separation constant we obtain

$$X'' + \lambda X = 0,$$

$$X'(0) = X(0)$$

and

$$Y'' - \lambda Y = 0,$$

$$Y(0) = 0,$$

$$Y(\pi) = 0.$$

With $\lambda = \alpha^2 < 0$ the solutions of the differential equations are

$$X = c_1 \cosh \alpha x + c_2 \sinh \alpha x \quad \text{and} \quad Y = c_3 \cos \alpha y + c_4 \sin \alpha y$$

The boundary and initial conditions imply

$$Y = c_4 \sin ny \quad \text{and} \quad X = c_2(n \cosh nx + \sinh nx)$$

for $n = 1, 2, 3, \dots$ so that

$$u = \sum_{n=1}^{\infty} A_n(n \cosh nx + \sinh nx) \sin ny.$$

Imposing

$$u(\pi, y) = 1 = \sum_{n=1}^{\infty} A_n(n \cosh n\pi + \sinh n\pi) \sin ny$$

gives

$$A_n(n \cosh n\pi + \sinh n\pi) = \frac{2}{\pi} \int_0^\pi \sin ny dy = \frac{2[1 - (-1)^n]}{n\pi}$$

for $n = 1, 2, 3, \dots$ so that

$$u(x, y) = \frac{2}{\pi} \sum_{n=1}^{\infty} \frac{1 - (-1)^n}{n} \frac{n \cosh nx + \sinh nx}{n \cosh n\pi + \sinh n\pi} \sin ny.$$

8. Using $u = XY$ and $-\lambda$ as a separation constant we obtain

$$X'' + \lambda X = 0,$$

$$X(0) = 0,$$

$$X(1) = 0,$$

and

$$Y'' - \lambda Y = 0,$$

$$Y'(0) = Y(0).$$

With $\lambda = \alpha^2 > 0$ the solutions of the differential equations are

$$X = c_1 \cos \alpha x + c_2 \sin \alpha x \quad \text{and} \quad Y = c_3 \cosh \alpha y + c_4 \sinh \alpha y$$

13.5 Laplace's Equation

The boundary and initial conditions imply

$$X = c_2 \sin n\pi x \quad \text{and} \quad Y = c_4(n \cosh n\pi y + \sinh n\pi y)$$

for $n = 1, 2, 3, \dots$ so that

$$u = \sum_{n=1}^{\infty} A_n(n \cosh n\pi y + \sinh n\pi y) \sin n\pi x.$$

Imposing

$$u(x, 1) = f(x) = \sum_{n=1}^{\infty} A_n(n \cosh n\pi + \sinh n\pi) \sin n\pi x$$

gives

$$A_n(n \cosh n\pi + \sinh n\pi) = \frac{2}{\pi} \int_0^{\pi} f(x) \sin n\pi x \, dx$$

for $n = 1, 2, 3, \dots$ so that

$$u(x, y) = \sum_{n=1}^{\infty} A_n(n \cosh n\pi y + \sinh n\pi y) \sin n\pi x$$

where

$$A_n = \frac{2}{n\pi \cosh n\pi + \pi \sinh n\pi} \int_0^1 f(x) \sin n\pi x \, dx.$$

9. This boundary-value problem has the form of Problem 1 in this section, with $a = b = 1$, $f(x) = 100$, and $g(x) = 200$. The solution, then, is

$$u(x, y) = \sum_{n=1}^{\infty} (A_n \cosh n\pi y + B_n \sinh n\pi y) \sin n\pi x,$$

where

$$A_n = 2 \int_0^1 100 \sin n\pi x \, dx = 200 \left(\frac{1 - (-1)^n}{n\pi} \right)$$

and

$$\begin{aligned} B_n &= \frac{1}{\sinh n\pi} \left[2 \int_0^1 200 \sin n\pi x \, dx - A_n \cosh n\pi \right] \\ &= \frac{1}{\sinh n\pi} \left[400 \left(\frac{1 - (-1)^n}{n\pi} \right) - 200 \left(\frac{1 - (-1)^n}{n\pi} \right) \cosh n\pi \right] \\ &= 200 \left[\frac{1 - (-1)^n}{n\pi} \right] [2 \operatorname{csch} n\pi - \coth n\pi]. \end{aligned}$$

10. This boundary-value problem has the form of Problem 2 in this section, with $a = 1$ and $b = 1$. Thus, the solution has the form

$$u(x, y) = \sum_{n=1}^{\infty} (A_n \cosh n\pi x + B_n \sinh n\pi x) \sin n\pi y.$$

The boundary condition $u(0, y) = 10y$ implies

$$10y = \sum_{n=1}^{\infty} A_n \sin n\pi y$$

and

$$A_n = \frac{2}{1} \int_0^1 10y \sin n\pi y \, dy = \frac{20}{n\pi} (-1)^{n+1}.$$

The boundary condition $u_x(1, y) = -1$ implies

$$-1 = \sum_{n=1}^{\infty} (n\pi A_n \sinh n\pi + n\pi B_n \cosh n\pi) \sin n\pi y$$

and

$$n\pi A_n \sinh n\pi + n\pi B_n \cosh n\pi = \frac{2}{1} \int_0^1 (-\sin n\pi y) dy$$

$$A_n \sinh n\pi + B_n \cosh n\pi = -\frac{2}{n\pi} [1 - (-1)^n]$$

$$B_n = \frac{2}{n\pi} [(-1)^n - 1] \operatorname{sech} n\pi - \frac{20}{n\pi} (-1)^{n+1} \tanh n\pi.$$

11. Using $u = XY$ and $-\lambda$ as a separation constant we obtain

$$X'' + \lambda X = 0,$$

$$X(0) = 0,$$

$$X(\pi) = 0,$$

and

$$Y'' - \lambda Y = 0.$$

With $\lambda = \alpha^2 > 0$ the solutions of the differential equations are

$$X = c_1 \cos \alpha x + c_2 \sin \alpha x \quad \text{and} \quad Y = c_3 e^{\alpha y} + c_4 e^{-\alpha y}$$

Then the boundedness of u as $y \rightarrow \infty$ implies $c_3 = 0$, so $Y = c_4 e^{-ny}$. The boundary conditions at $x = 0$ and $x = \pi$ imply $c_1 = 0$ so $X = c_2 \sin nx$ for $n = 1, 2, 3, \dots$ and

$$u = \sum_{n=1}^{\infty} A_n e^{-ny} \sin nx.$$

Imposing

$$u(x, 0) = f(x) = \sum_{n=1}^{\infty} A_n \sin nx$$

gives

$$A_n = \frac{2}{\pi} \int_0^{\pi} f(x) \sin nx \, dx$$

so that

$$u(x, y) = \sum_{n=1}^{\infty} \left(\frac{2}{\pi} \int_0^{\pi} f(x) \sin nx \, dx \right) e^{-ny} \sin nx.$$

12. Using $u = XY$ and $-\lambda$ as a separation constant we obtain

$$X'' + \lambda X = 0,$$

$$X'(0) = 0,$$

$$X'(\pi) = 0,$$

and

$$Y'' - \lambda Y = 0.$$

With $\lambda = \alpha^2 > 0$ the solutions of the differential equations are

$$X = c_1 \cos \alpha x + c_2 \sin \alpha x \quad \text{and} \quad Y = c_3 e^{\alpha y} + c_4 e^{-\alpha y}$$

13.5 Laplace's Equation

The boundary conditions at $x = 0$ and $x = \pi$ imply $c_2 = 0$ so $X = c_1 \cos nx$ for $n = 1, 2, 3, \dots$. Now the boundedness of u as $y \rightarrow \infty$ implies $c_3 = 0$, so $Y = c_4 e^{-ny}$. In this problem $\lambda = 0$ is also an eigenvalue with corresponding eigenfunction 1 so that

$$u = A_0 + \sum_{n=1}^{\infty} A_n e^{-ny} \cos nx.$$

Imposing

$$u(x, 0) = f(x) = A_0 + \sum_{n=1}^{\infty} A_n \cos nx$$

gives

$$A_0 = \frac{1}{\pi} \int_0^{\pi} f(x) dx \quad \text{and} \quad A_n = \frac{2}{\pi} \int_0^{\pi} f(x) \cos nx dx$$

so that

$$u(x, y) = \frac{1}{\pi} \int_0^{\pi} f(x) dx + \sum_{n=1}^{\infty} \left(\frac{2}{\pi} \int_0^{\pi} f(x) \cos nx dx \right) e^{-ny} \cos nx.$$

13. Since the boundary conditions at $y = 0$ and $y = b$ are functions of x we choose to separate Laplace's equation as

$$\frac{X''}{X} = -\frac{Y''}{Y} = -\lambda$$

so that

$$X'' + \lambda X = 0$$

$$Y'' - \lambda Y = 0.$$

Then with $\lambda = \alpha^2$ we have

$$X(x) = c_1 \cos \alpha x + c_2 \sin \alpha x$$

$$Y(y) = c_3 \cosh \alpha y + c_4 \sinh \alpha y.$$

Now $X(0) = 0$ gives $c_1 = 0$ and $X(a) = 0$ implies $\sin \alpha a = 0$ or $\alpha = n\pi/a$ for $n = 1, 2, 3, \dots$. Thus

$$u_n(x, y) = XY = \left(A_n \cosh \frac{n\pi}{a} y + B_n \sinh \frac{n\pi}{a} y \right) \sin \frac{n\pi}{a} x$$

and

$$u(x, y) = \sum_{n=1}^{\infty} \left(A_n \cosh \frac{n\pi}{a} y + B_n \sinh \frac{n\pi}{a} y \right) \sin \frac{n\pi}{a} x. \quad (1)$$

At $y = 0$ we then have

$$f(x) = \sum_{n=1}^{\infty} A_n \sin \frac{n\pi}{a} x$$

and consequently

$$A_n = \frac{2}{a} \int_0^a f(x) \sin \frac{n\pi}{a} x dx. \quad (2)$$

At $y = b$,

$$g(y) = \sum_{n=1}^{\infty} \left(A_n \cosh \frac{n\pi}{a} b + B_n \sinh \frac{n\pi}{a} b \right) \sin \frac{n\pi}{a} x$$

indicates that the entire expression in the parentheses is given by

$$A_n \cosh \frac{n\pi}{a} b + B_n \sinh \frac{n\pi}{a} b = \frac{2}{a} \int_0^a g(x) \sin \frac{n\pi}{a} x dx.$$

We can now solve for B_n :

$$B_n \sinh \frac{n\pi}{a} b = \frac{2}{a} \int_0^a g(x) \sin \frac{n\pi}{a} x \, dx - A_n \cosh \frac{n\pi}{a} b$$

$$B_n = \frac{1}{\sinh \frac{n\pi}{a} b} \left(\frac{2}{a} \int_0^a g(x) \sin \frac{n\pi}{a} x \, dx - A_n \cosh \frac{n\pi}{a} b \right). \quad (3)$$

A solution to the given boundary-value problem consists of the series (1) with coefficients A_n and B_n given in (2) and (3), respectively.

14. Since the boundary conditions at $x = 0$ and $x = a$ are functions of y we choose to separate Laplace's equation as

$$\frac{X''}{X} = -\frac{Y''}{Y} = -\lambda$$

so that

$$X'' + \lambda X = 0$$

$$Y'' - \lambda Y = 0.$$

Then with $\lambda = -\alpha^2$ we have

$$X(x) = c_1 \cosh \alpha x + c_2 \sinh \alpha x$$

$$Y(y) = c_3 \cos \alpha y + c_4 \sin \alpha y.$$

Now $Y(0) = 0$ gives $c_3 = 0$ and $Y(b) = 0$ implies $\sin \alpha b = 0$ or $\alpha = n\pi/b$ for $n = 1, 2, 3, \dots$. Thus

$$u_n(x, y) = XY = \left(A_n \cosh \frac{n\pi}{b} x + B_n \sinh \frac{n\pi}{b} x \right) \sin \frac{n\pi}{b} y$$

and

$$u(x, y) = \sum_{n=1}^{\infty} \left(A_n \cosh \frac{n\pi}{b} x + B_n \sinh \frac{n\pi}{b} x \right) \sin \frac{n\pi}{b} y. \quad (4)$$

At $x = 0$ we then have

$$F(y) = \sum_{n=1}^{\infty} A_n \sin \frac{n\pi}{b} y$$

and consequently

$$A_n = \frac{2}{b} \int_0^b F(y) \sin \frac{n\pi}{b} y \, dy. \quad (5)$$

At $x = a$,

$$G(y) = \sum_{n=1}^{\infty} \left(A_n \cosh \frac{n\pi}{b} a + B_n \sinh \frac{n\pi}{b} a \right) \sin \frac{n\pi}{b} y$$

indicates that the entire expression in the parentheses is given by

$$A_n \cosh \frac{n\pi}{b} a + B_n \sinh \frac{n\pi}{b} a = \frac{2}{b} \int_0^b G(y) \sin \frac{n\pi}{b} y \, dy.$$

We can now solve for B_n :

$$B_n \sinh \frac{n\pi}{b} a = \frac{2}{b} \int_0^b G(y) \sin \frac{n\pi}{b} y \, dy - A_n \cosh \frac{n\pi}{b} a$$

$$B_n = \frac{1}{\sinh \frac{n\pi}{b} a} \left(\frac{2}{b} \int_0^b G(y) \sin \frac{n\pi}{b} y \, dy - A_n \cosh \frac{n\pi}{b} a \right). \quad (6)$$

A solution to the given boundary-value problem consists of the series (4) with coefficients A_n and B_n given in (5) and (6), respectively.

13.5 Laplace's Equation

15. Referring to the discussion in this section of the text we identify $a = b = \pi$, $f(x) = 0$, $g(x) = 1$, $F(y) = 1$, and $G(y) = 1$. Then $A_n = 0$ and

$$u_1(x, y) = \sum_{n=1}^{\infty} B_n \sinh ny \sin nx$$

where

$$B_n = \frac{2}{\pi \sinh n\pi} \int_0^{\pi} \sin nx \, dx = \frac{2[1 - (-1)^n]}{n\pi \sinh n\pi}.$$

Next

$$u_2(x, y) = \sum_{n=1}^{\infty} (A_n \cosh nx + B_n \sinh nx) \sin ny$$

where

$$A_n = \frac{2}{\pi} \int_0^{\pi} \sin ny \, dy = \frac{2[1 - (-1)^n]}{n\pi}$$

and

$$\begin{aligned} B_n &= \frac{1}{\sinh n\pi} \left(\frac{2}{\pi} \int_0^{\pi} \sin ny \, dy - A_n \cosh n\pi \right) \\ &= \frac{1}{\sinh n\pi} \left(\frac{2[1 - (-1)^n]}{n\pi} - \frac{2[1 - (-1)^n]}{n\pi} \cosh n\pi \right) \\ &= \frac{2[1 - (-1)^n]}{n\pi \sinh n\pi} (1 - \cosh n\pi). \end{aligned}$$

Now

$$\begin{aligned} A_n \cosh nx + B_n \sinh nx &= \frac{2[1 - (-1)^n]}{n\pi} \left[\cosh nx + \frac{\sinh nx}{\sinh n\pi} (1 - \cosh n\pi) \right] \\ &= \frac{2[1 - (-1)^n]}{n\pi \sinh n\pi} [\cosh nx \sinh n\pi + \sinh nx - \sinh nx \cosh n\pi] \\ &= \frac{2[1 - (-1)^n]}{n\pi \sinh n\pi} [\sinh nx + \sinh n(\pi - x)] \end{aligned}$$

and

$$\begin{aligned} u(x, y) = u_1 + u_2 &= \frac{2}{\pi} \sum_{n=1}^{\infty} \frac{1 - (-1)^n}{n \sinh n\pi} \sinh ny \sin nx \\ &\quad + \frac{2}{\pi} \sum_{n=1}^{\infty} \frac{[1 - (-1)^n][\sinh nx + \sinh n(\pi - x)]}{n \sinh n\pi} \sin ny. \end{aligned}$$

16. Referring to the discussion in this section of the text we identify $a = b = 2$, $f(x) = 0$,

$$g(x) = \begin{cases} x, & 0 < x < 1 \\ 2 - x, & 1 < x < 2, \end{cases}$$

$F(y) = 0$, and $G(y) = y(2 - y)$. Then $A_n = 0$ and

$$u_1(x, y) = \sum_{n=1}^{\infty} B_n \sinh \frac{n\pi}{2} y \sin \frac{n\pi}{2} x$$

where

$$\begin{aligned} B_n &= \frac{1}{\sinh n\pi} \int_0^2 g(x) \sin \frac{n\pi}{2} x dx \\ &= \frac{1}{\sinh n\pi} \left(\int_0^1 x \sin \frac{n\pi}{2} x dx + \int_1^2 (2-x) \sin \frac{n\pi}{2} x dx \right) \\ &= \frac{8 \sin \frac{n\pi}{2}}{n^2 \pi^2 \sinh n\pi}. \end{aligned}$$

Next, since $A_n = 0$ in u_2 , we have

$$u_2(x, y) = \sum_{n=1}^{\infty} B_n \sinh \frac{n\pi}{2} x \sin \frac{n\pi}{2} y$$

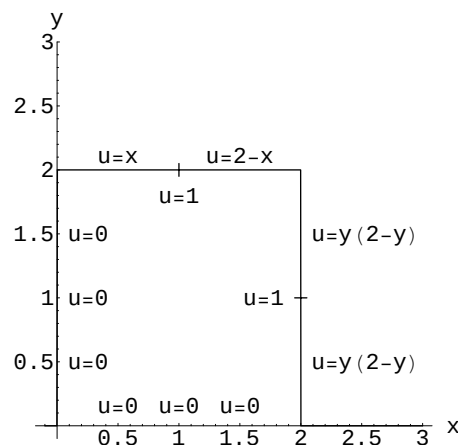
where

$$B_n = \frac{1}{\sinh n\pi} \int_0^b y(2-y) \sin \frac{n\pi}{2} y dy = \frac{16[1 - (-1)^n]}{n^3 \pi^3 \sinh n\pi}.$$

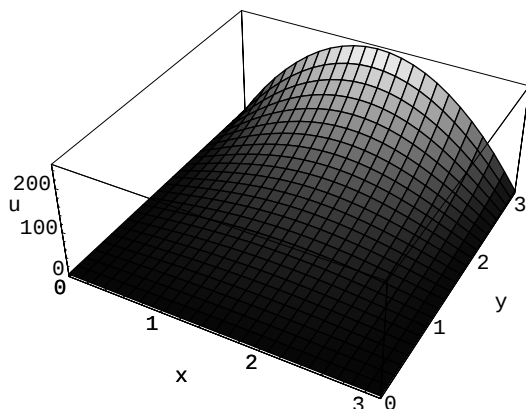
Thus

$$\begin{aligned} u(x, y) = u_1 + u_2 &= \frac{8}{\pi^2} \sum_{n=1}^{\infty} \frac{\sin \frac{n\pi}{2}}{n^2 \sinh n\pi} \sinh \frac{n\pi}{2} x \sin \frac{n\pi}{2} y \\ &\quad + \frac{16}{\pi^3} \sum_{n=1}^{\infty} \frac{[1 - (-1)^n]}{n^3 \sinh n\pi} \sinh \frac{n\pi}{2} x \sin \frac{n\pi}{2} y. \end{aligned}$$

17. From the figure showing the boundary conditions we see that the maximum value of the temperature is 1 at (1, 2) and (2, 1).



18. (a)



13.5 Laplace's Equation

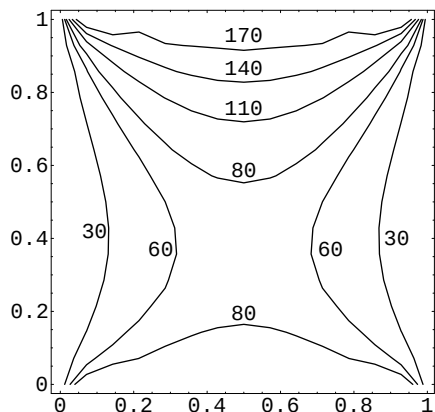
(b) The maximum value occurs at $(\pi/2, \pi)$ and is $f(\pi/2) = 25\pi^2$.

(c) The coefficients are

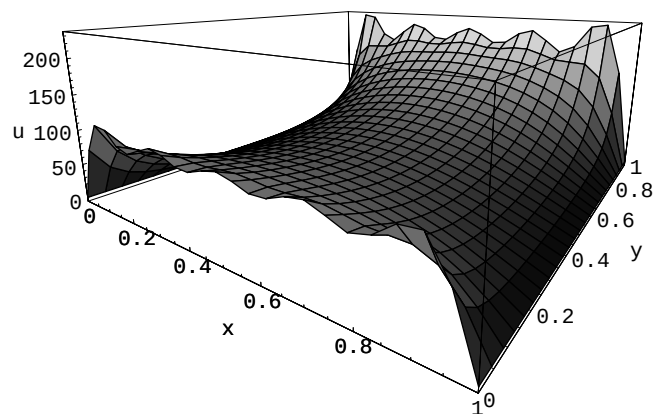
$$\begin{aligned} A_n &= \frac{2}{\pi} \operatorname{csch} n\pi \int_0^\pi 100x(\pi - x) \sin nx \, dx \\ &= \frac{200 \operatorname{csch} n\pi}{\pi} \left[\frac{200}{n^3} (1 - (-1)^n) \right] = \frac{400}{n^3 \pi} [1 - (-1)^n] \operatorname{csch} n\pi. \end{aligned}$$

See part (a) for the graph.

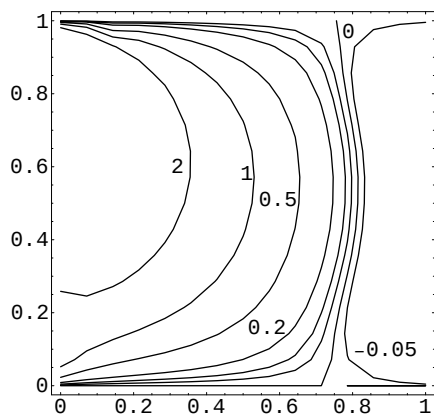
19. (a)



(b)



20.



21. Assuming $u(x, y) = X(x)Y(y)$ and substituting into the partial differential equation we get $X''Y + XY'' = 0$. Separating variables and using $\lambda = \alpha^2$ we get

$$X'' - \alpha^2 X = 0, \quad X'(0) = 0,$$

which implies $X(x) = c_3 \cosh \alpha x$. From

$$Y'' + \alpha^2 Y = 0, \quad Y'(0) = 0, \quad Y'(b) = 0$$

we get $Y(y) = c_1 \cos \alpha y + c_2 \sin \alpha y$ and eigenvalues $\lambda_n = n^2 \pi^2 / b^2$, $n = 1, 2, 3, \dots$. The corresponding eigenfunctions are $Y(y) = c_1 \cos(n\pi y/b)$. For $\lambda = 0$ the boundary conditions applied to $X(x) = c_3 + c_4 x$ and $Y(y) = c_1 + c_2 y$ imply $X = c_3$ and $Y = c_1$. Forming products and using the superposition principle then gives

$$u(x, y) = A_0 + \sum_{n=1}^{\infty} A_n \cosh \frac{n\pi}{b} x \cos \frac{n\pi}{b} y.$$

The remaining boundary condition, $u_x(a) = g(y)$ implies

$$g(y) = \frac{\partial u}{\partial x} \Big|_{x=a} = \sum_{n=1}^{\infty} A_n \frac{n\pi}{b} \sinh \frac{n\pi}{b} a \cos \frac{n\pi}{b} y$$

and so A_0 remains arbitrary. In order that the series expression for $g(y)$ be a cosine series, the constant term in the series, $a_0/2$, must be 0. Thus, from Section 12.3 in the text,

$$a_0 = \frac{2}{b} \int_0^b g(y) dy = 0 \quad \text{so} \quad \int_0^b g(y) dy = 0.$$

Also,

$$A_n \frac{n\pi}{b} \sinh \frac{n\pi}{b} a = \frac{2}{b} \int_0^b g(y) \cos \frac{n\pi}{b} y dy$$

and

$$A_n = \frac{2}{n\pi \sinh n\pi a/b} \int_0^b g(y) \cos \frac{n\pi}{b} y dy.$$

The solution is then

$$u(x, y) = A_0 + \sum_{n=1}^{\infty} A_n \cosh \frac{n\pi}{b} x \cos \frac{n\pi}{b} y,$$

where the A_n are defined above and A_0 is arbitrary. In general, Neumann problems do not have unique solutions.

For a physical interpretation of the compatibility condition $\int_0^b g(y) dy = 0$ see the texts *Elementary Partial Differential Equations* by Paul Berg and James McGregor (Holden-Day) and *Partial Differential Equations of Mathematical Physics* by Tyn Myint-U (North Holland).

EXERCISES 13.6**Nonhomogeneous Equations
and Boundary Conditions**

1. Using $v(x, t) = u(x, t) - 100$ we wish to solve $kv_{xx} = v_t$ subject to $v(0, t) = 0$, $v(1, t) = 0$, and $v(x, 0) = -100$. Let $v = XT$ and use $-\lambda$ as a separation constant so that

$$X'' + \lambda X = 0,$$

$$X(0) = 0,$$

$$X(1) = 0,$$

and

$$T' + \lambda kT = 0.$$

This leads to

$$X = c_2 \sin(n\pi x) \quad \text{and} \quad T = c_3 e^{-kn^2\pi^2 t}$$

for $n = 1, 2, 3, \dots$ so that

$$v = \sum_{n=1}^{\infty} A_n e^{-kn^2\pi^2 t} \sin n\pi x.$$

Imposing

$$v(x, 0) = -100 = \sum_{n=1}^{\infty} A_n \sin n\pi x$$

gives

$$A_n = 2 \int_0^1 (-100) \sin n\pi x \, dx = \frac{-200}{n\pi} [1 - (-1)^n]$$

so that

$$u(x, t) = v(x, t) + 100 = 100 + \frac{200}{\pi} \sum_{n=1}^{\infty} \frac{(-1)^n - 1}{n} e^{-kn^2\pi^2 t} \sin n\pi x.$$

2. Letting $u(x, t) = v(x, t) + \psi(x)$ and proceeding as in Example 1 in the text we find $\psi(x) = u_0 - u_0 x$. Then $v(x, t) = u(x, t) + u_0 x - u_0$ and we wish to solve $kv_{xx} = v_t$ subject to $v(0, t) = 0$, $v(1, t) = 0$, and $v(x, 0) = f(x) + u_0 x - u_0$. Let $v = XT$ and use $-\lambda$ as a separation constant so that

$$X'' + \lambda X = 0,$$

$$X(0) = 0,$$

$$X(1) = 0,$$

and

$$T' + \lambda kT = 0.$$

Then

$$X = c_2 \sin n\pi x \quad \text{and} \quad T = c_3 e^{-kn^2\pi^2 t}$$

for $n = 1, 2, 3, \dots$ so that

$$v = \sum_{n=1}^{\infty} A_n e^{-kn^2\pi^2 t} \sin n\pi x.$$

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Imposing

$$v(x, 0) = f(x) + u_0x - u_0 = \sum_{n=1}^{\infty} A_n \sin n\pi x$$

gives

$$A_n = 2 \int_0^1 (f(x) + u_0x - u_0) \sin n\pi x \, dx$$

so that

$$u(x, t) = v(x, t) + u_0 - u_0x = u_0 - u_0x + \sum_{n=1}^{\infty} A_n e^{-kn^2\pi^2 t} \sin n\pi x.$$

3. If we let $u(x, t) = v(x, t) + \psi(x)$, then we obtain as in Example 1 in the text

$$k\psi'' + r = 0$$

or

$$\psi(x) = -\frac{r}{2k}x^2 + c_1x + c_2.$$

The boundary conditions become

$$u(0, t) = v(0, t) + \psi(0) = u_0$$

$$u(1, t) = v(1, t) + \psi(1) = u_0.$$

Letting $\psi(0) = \psi(1) = u_0$ we obtain homogeneous boundary conditions in v :

$$v(0, t) = 0 \quad \text{and} \quad v(1, t) = 0.$$

Now $\psi(0) = \psi(1) = u_0$ implies $c_2 = u_0$ and $c_1 = r/2k$. Thus

$$\psi(x) = -\frac{r}{2k}x^2 + \frac{r}{2k}x + u_0 = u_0 - \frac{r}{2k}x(x-1).$$

To determine $v(x, t)$ we solve

$$k \frac{\partial^2 v}{\partial x^2} = \frac{\partial v}{\partial t}, \quad 0 < x < 1, \quad t > 0$$

$$v(0, t) = 0, \quad v(1, t) = 0,$$

$$v(x, 0) = \frac{r}{2k}x(x-1) - u_0.$$

Separating variables, we find

$$v(x, t) = \sum_{n=1}^{\infty} A_n e^{-kn^2\pi^2 t} \sin n\pi x,$$

where

$$A_n = 2 \int_0^1 \left[\frac{r}{2k}x(x-1) - u_0 \right] \sin n\pi x \, dx = 2 \left[\frac{u_0}{n\pi} + \frac{r}{kn^3\pi^3} \right] [(-1)^n - 1]. \quad (1)$$

Hence, a solution of the original problem is

$$u(x, t) = \psi(x) + v(x, t) = u_0 - \frac{r}{2k}x(x-1) + \sum_{n=1}^{\infty} A_n e^{-kn^2\pi^2 t} \sin n\pi x,$$

where A_n is defined in (1).

4. If we let $u(x, t) = v(x, t) + \psi(x)$, then we obtain as in Example 1 in the text

$$k\psi'' + r = 0.$$

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Integrating gives

$$\psi(x) = -\frac{r}{2k}x^2 + c_1x + c_2.$$

The boundary conditions become

$$u(0, t) = v(0, t) + \psi(0) = u_0$$

$$u(1, t) = v(1, t) + \psi(1) = u_1.$$

Letting $\psi(0) = u_0$ and $\psi(1) = u_1$ we obtain homogeneous boundary conditions in v :

$$v(0, t) = 0 \quad \text{and} \quad v(1, t) = 0.$$

Now $\psi(0) = u_0$ and $\psi(1) = u_1$ imply $c_2 = u_0$ and $c_1 = u_1 - u_0 + r/2k$. Thus

$$\psi(x) = -\frac{r}{2k}x^2 + \left(u_1 - u_0 + \frac{r}{2k}\right)x + u_0.$$

To determine $v(x, t)$ we solve

$$k \frac{\partial^2 v}{\partial x^2} = \frac{\partial v}{\partial t}, \quad 0 < x < 1, \quad t > 0$$

$$v(0, t) = 0, \quad v(1, t) = 0,$$

$$v(x, 0) = f(x) - \psi(x).$$

Separating variables, we find

$$v(x, t) = \sum_{n=1}^{\infty} A_n e^{-kn^2\pi^2 t} \sin n\pi x,$$

where

$$A_n = 2 \int_0^1 [f(x) - \psi(x)] \sin n\pi x \, dx. \quad (2)$$

Hence, a solution of the original problem is

$$u(x, t) = \psi(x) + v(x, t) = -\frac{r}{2k}x^2 + \left(u_1 - u_0 + \frac{r}{2k}\right)x + u_0 + \sum_{n=1}^{\infty} A_n e^{-kn^2\pi^2 t} \sin n\pi x,$$

where A_n is defined in (2).

5. Substituting $u(x, t) = v(x, t) + \psi(x)$ into the partial differential equation gives

$$k \frac{\partial^2 v}{\partial x^2} + k\psi'' + Ae^{-\beta x} = \frac{\partial v}{\partial t}.$$

This equation will be homogeneous provided ψ satisfies

$$k\psi'' + Ae^{-\beta x} = 0.$$

The solution of this differential equation is obtained by successive integrations:

$$\psi(x) = -\frac{A}{\beta^2 k} e^{-\beta x} + c_1 x + c_2.$$

From $\psi(0) = 0$ and $\psi(1) = 0$ we find

$$c_1 = \frac{A}{\beta^2 k} (e^{-\beta} - 1) \quad \text{and} \quad c_2 = \frac{A}{\beta^2 k}.$$

Hence

$$\begin{aligned} \psi(x) &= -\frac{A}{\beta^2 k} e^{-\beta x} + \frac{A}{\beta^2 k} (e^{-\beta} - 1)x + \frac{A}{\beta^2 k} \\ &= \frac{A}{\beta^2 k} [1 - e^{-\beta x} + (e^{-\beta} - 1)x]. \end{aligned}$$

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Now the new problem is

$$\begin{aligned} k \frac{\partial^2 v}{\partial x^2} &= \frac{\partial v}{\partial t}, \quad 0 < x < 1, \quad t > 0, \\ v(0, t) &= 0, \quad v(1, t) = 0, \quad t > 0, \\ v(x, 0) &= f(x) - \psi(x), \quad 0 < x < 1. \end{aligned}$$

Identifying this as the heat equation solved in Section 13.3 in the text with $L = 1$ we obtain

$$v(x, t) = \sum_{n=1}^{\infty} A_n e^{-kn^2\pi^2 t} \sin n\pi x$$

where

$$A_n = 2 \int_0^1 [f(x) - \psi(x)] \sin n\pi x \, dx.$$

Thus

$$u(x, t) = \frac{A}{\beta^2 k} [1 - e^{-\beta x} + (e^{-\beta} - 1)x] + \sum_{n=1}^{\infty} A_n e^{-kn^2\pi^2 t} \sin n\pi x.$$

6. Substituting $u(x, t) = v(x, t) + \psi(x)$ into the partial differential equation gives

$$k \frac{\partial^2 v}{\partial x^2} + k\psi'' - hv - h\psi = \frac{\partial v}{\partial t}.$$

This equation will be homogeneous provided ψ satisfies

$$k\psi'' - h\psi = 0.$$

Since k and h are positive, the general solution of this latter linear second-order equation is

$$\psi(x) = c_1 \cosh \sqrt{\frac{h}{k}} x + c_2 \sinh \sqrt{\frac{h}{k}} x.$$

From $\psi(0) = 0$ and $\psi(\pi) = u_0$ we find $c_1 = 0$ and $c_2 = u_0 / \sinh \sqrt{h/k} \pi$. Hence

$$\psi(x) = u_0 \frac{\sinh \sqrt{h/k} x}{\sinh \sqrt{h/k} \pi}.$$

Now the new problem is

$$\begin{aligned} k \frac{\partial^2 v}{\partial x^2} - hv &= \frac{\partial v}{\partial t}, \quad 0 < x < \pi, \quad t > 0 \\ v(0, t) &= 0, \quad v(\pi, t) = 0, \quad t > 0 \\ v(x, 0) &= -\psi(x), \quad 0 < x < \pi. \end{aligned}$$

If we let $v = XT$ then

$$\frac{X''}{X} = \frac{T' + hT}{kT} = -\lambda.$$

With $\lambda = \alpha^2 > 0$, the separated differential equations

$$X'' + \alpha^2 X = 0 \quad \text{and} \quad T' + (h + k\alpha^2) T = 0.$$

have the respective solutions

$$\begin{aligned} X(x) &= c_3 \cos \alpha x + c_4 \sin \alpha x \\ T(t) &= c_5 e^{-(h+k\alpha^2)t}. \end{aligned}$$

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From $X(0) = 0$ we get $c_3 = 0$ and from $X(\pi) = 0$ we find $\alpha = n$ for $n = 1, 2, 3, \dots$. Consequently, it follows that

$$v(x, t) = \sum_{n=1}^{\infty} A_n e^{-(h+kn^2)t} \sin nx$$

where

$$A_n = -\frac{2}{\pi} \int_0^{\pi} \psi(x) \sin nx \, dx.$$

Hence a solution of the original problem is

$$u(x, t) = u_0 \frac{\sinh \sqrt{h/k} x}{\sinh \sqrt{h/k} \pi} + e^{-ht} \sum_{n=1}^{\infty} A_n e^{-kn^2 t} \sin nx$$

where

$$A_n = -\frac{2}{\pi} \int_0^{\pi} u_0 \frac{\sinh \sqrt{h/k} x}{\sinh \sqrt{h/k} \pi} \sin nx \, dx.$$

Using the exponential definition of the hyperbolic sine and integration by parts we find

$$A_n = \frac{2u_0 nk(-1)^n}{\pi(h + kn^2)}.$$

7. Substituting $u(x, t) = v(x, t) + \psi(x)$ into the partial differential equation gives

$$k \frac{\partial^2 v}{\partial x^2} + k\psi'' - hv - h\psi + hu_0 = \frac{\partial v}{\partial t}.$$

This equation will be homogeneous provided ψ satisfies

$$k\psi'' - h\psi + hu_0 = 0 \quad \text{or} \quad k\psi'' - h\psi = -hu_0.$$

This non-homogeneous, linear, second-order, differential equation has solution

$$\psi(x) = c_1 \cosh \sqrt{\frac{h}{k}} x + c_2 \sinh \sqrt{\frac{h}{k}} x + u_0,$$

where we assume $h > 0$ and $k > 0$. From $\psi(0) = u_0$ and $\psi(1) = 0$ we find $c_1 = 0$ and $c_2 = -u_0 / \sinh \sqrt{h/k}$. Thus, the steady-state solution is

$$\psi(x) = -\frac{u_0}{\sinh \sqrt{\frac{h}{k}}} \sinh \sqrt{\frac{h}{k}} x + u_0 = u_0 \left(1 - \frac{\sinh \sqrt{\frac{h}{k}} x}{\sinh \sqrt{\frac{h}{k}}} \right).$$

8. The partial differential equation is

$$k \frac{\partial^2 u}{\partial x^2} - hu = \frac{\partial u}{\partial t}.$$

Substituting $u(x, t) = v(x, t) + \psi(x)$ gives

$$k \frac{\partial^2 v}{\partial x^2} + k\psi'' - hv - h\psi = \frac{\partial v}{\partial t}.$$

This equation will be homogeneous provided ψ satisfies

$$k\psi'' - h\psi = 0.$$

Assuming $h > 0$ and $k > 0$, we have

$$\psi = c_1 e^{\sqrt{h/k} x} + c_2 e^{-\sqrt{h/k} x},$$

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where we have used the exponential form of the solution since the rod is infinite. Now, in order that the steady-state temperature $\psi(x)$ be bounded as $x \rightarrow \infty$, we require $c_1 = 0$. Then

$$\psi(x) = c_2 e^{-\sqrt{h/k} x}$$

and $\psi(0) = u_0$ implies $c_2 = u_0$. Thus

$$\psi(x) = u_0 e^{-\sqrt{h/k} x}.$$

9. Substituting $u(x, t) = v(x, t) + \psi(x)$ into the partial differential equation gives

$$a^2 \frac{\partial^2 v}{\partial x^2} + a^2 \psi'' + Ax = \frac{\partial^2 v}{\partial t^2}.$$

This equation will be homogeneous provided ψ satisfies

$$a^2 \psi'' + Ax = 0.$$

The solution of this differential equation is

$$\psi(x) = -\frac{A}{6a^2} x^3 + c_1 x + c_2.$$

From $\psi(0) = 0$ we obtain $c_2 = 0$, and from $\psi(1) = 0$ we obtain $c_1 = A/6a^2$. Hence

$$\psi(x) = \frac{A}{6a^2} (x - x^3).$$

Now the new problem is

$$\begin{aligned} a^2 \frac{\partial^2 v}{\partial x^2} &= \frac{\partial^2 v}{\partial t^2} \\ v(0, t) &= 0, \quad v(1, t) = 0, \quad t > 0, \\ v(x, 0) &= -\psi(x), \quad v_t(x, 0) = 0, \quad 0 < x < 1. \end{aligned}$$

Identifying this as the wave equation solved in Section 13.4 in the text with $L = 1$, $f(x) = -\psi(x)$, and $g(x) = 0$ we obtain

$$v(x, t) = \sum_{n=1}^{\infty} A_n \cos n\pi at \sin n\pi x$$

where

$$A_n = 2 \int_0^1 [-\psi(x)] \sin n\pi x \, dx = \frac{A}{3a^2} \int_0^1 (x^3 - x) \sin n\pi x \, dx = \frac{2A(-1)^n}{a^2 \pi^3 n^3}.$$

Thus

$$u(x, t) = \frac{A}{6a^2} (x - x^3) + \frac{2A}{a^2 \pi^3} \sum_{n=1}^{\infty} \frac{(-1)^n}{n^3} \cos n\pi at \sin n\pi x.$$

10. We solve

$$\begin{aligned} a^2 \frac{\partial^2 u}{\partial x^2} - g &= \frac{\partial^2 u}{\partial t^2}, \quad 0 < x < 1, \quad t > 0 \\ u(0, t) &= 0, \quad u(1, t) = 0, \quad t > 0 \\ u(x, 0) &= 0, \quad \left. \frac{\partial u}{\partial t} \right|_{t=0} = 0, \quad 0 < x < 1. \end{aligned}$$

The partial differential equation is nonhomogeneous. The substitution $u(x, t) = v(x, t) + \psi(x)$ yields a homogeneous partial differential equation provided ψ satisfies

$$a^2 \psi'' - g = 0.$$

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By integrating twice we find

$$\psi(x) = \frac{g}{2a^2}x^2 + c_1x + c_2.$$

The imposed conditions $\psi(0) = 0$ and $\psi(1) = 0$ then lead to $c_2 = 0$ and $c_1 = -g/2a^2$. Hence

$$\psi(x) = \frac{g}{2a^2}(x^2 - x).$$

The new problem is now

$$\begin{aligned} a^2 \frac{\partial^2 v}{\partial x^2} &= \frac{\partial^2 v}{\partial t^2}, \quad 0 < x < 1, \quad t > 0 \\ v(0, t) &= 0, \quad v(1, t) = 0 \\ v(x, 0) &= \frac{g}{2a^2}(x - x^2), \quad \left. \frac{\partial v}{\partial t} \right|_{t=0} = 0. \end{aligned}$$

Substituting $v = XT$ we find in the usual manner

$$\begin{aligned} X'' + \alpha^2 X &= 0 \\ T'' + a^2 \alpha^2 T &= 0 \end{aligned}$$

with solutions

$$\begin{aligned} X(x) &= c_3 \cos \alpha x + c_4 \sin \alpha x \\ T(t) &= c_5 \cos a\alpha t + c_6 \sin a\alpha t. \end{aligned}$$

The conditions $X(0) = 0$ and $X(1) = 0$ imply in turn that $c_3 = 0$ and $\alpha = n\pi$ for $n = 1, 2, 3, \dots$. The condition $T'(0) = 0$ implies $c_6 = 0$. Hence, by the superposition principle

$$v(x, t) = \sum_{n=1}^{\infty} A_n \cos(an\pi t) \sin(n\pi x).$$

At $t = 0$,

$$\frac{g}{2a^2}(x - x^2) = \sum_{n=1}^{\infty} A_n \sin(n\pi x)$$

and so

$$A_n = \frac{g}{a^2} \int_0^1 (x - x^2) \sin(n\pi x) dx = \frac{2g}{a^2 n^3 \pi^3} [1 - (-1)^n].$$

Thus the solution to the original problem is

$$u(x, t) = \psi(x) + v(x, t) = \frac{g}{2a^2}(x^2 - x) + \frac{2g}{a^2 \pi^3} \sum_{n=1}^{\infty} \frac{1 - (-1)^n}{n^3} \cos(an\pi t) \sin(n\pi x).$$

11. Substituting $u(x, y) = v(x, y) + \psi(y)$ into Laplace's equation we obtain

$$\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} + \psi''(y) = 0.$$

This equation will be homogeneous provided ψ satisfies $\psi(y) = c_1 y + c_2$. Considering

$$\begin{aligned} u(x, 0) &= v(x, 0) + \psi(0) = u_1 \\ u(x, 1) &= v(x, 1) + \psi(1) = u_0 \\ u(0, y) &= v(0, y) + \psi(y) = 0 \end{aligned}$$

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we require that $\psi(0) = u_1$, $\psi_1 = u_0$ and $v(0, y) = -\psi(y)$. Then $c_1 = u_0 - u_1$ and $c_2 = u_1$. The new boundary-value problem is

$$\begin{aligned}\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} &= 0 \\ v(x, 0) &= 0, \quad v(x, 1) = 0, \\ v(0, y) &= -\psi(y), \quad 0 < y < 1,\end{aligned}$$

where $v(x, y)$ is bounded at $x \rightarrow \infty$. This problem is similar to Problem 11 in Section 13.5. The solution is

$$\begin{aligned}v(x, y) &= \sum_{n=1}^{\infty} \left(2 \int_0^1 [-\psi(y) \sin n\pi y] dy \right) e^{-n\pi x} \sin n\pi y \\ &= 2 \sum_{n=1}^{\infty} \left[(u_1 - u_0) \int_0^1 y \sin n\pi y dy - u_1 \int_0^1 \sin n\pi y dy \right] e^{-n\pi x} \sin n\pi y \\ &= \frac{2}{\pi} \sum_{n=1}^{\infty} \frac{u_0(-1)^n - u_1}{n} e^{-n\pi x} \sin n\pi y.\end{aligned}$$

Thus

$$\begin{aligned}u(x, y) &= v(x, y) + \psi(y) \\ &= (u_0 - u_1)y + u_1 + \frac{2}{\pi} \sum_{n=1}^{\infty} \frac{u_0(-1)^n - u_1}{n} e^{-n\pi x} \sin n\pi y.\end{aligned}$$

12. Substituting $u(x, y) = v(x, y) + \psi(x)$ into Poisson's equation we obtain

$$\frac{\partial^2 v}{\partial x^2} + \psi''(x) + h + \frac{\partial^2 v}{\partial y^2} = 0.$$

The equation will be homogeneous provided ψ satisfies $\psi''(x) + h = 0$ or $\psi(x) = -\frac{h}{2}x^2 + c_1x + c_2$. From $\psi(0) = 0$ we obtain $c_2 = 0$. From $\psi(\pi) = 1$ we obtain

$$c_1 = \frac{1}{\pi} + \frac{h\pi}{2}.$$

Then

$$\psi(x) = \left(\frac{1}{\pi} + \frac{h\pi}{2} \right) x - \frac{h}{2}x^2.$$

The new boundary-value problem is

$$\begin{aligned}\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} &= 0 \\ v(0, y) &= 0, \quad v(\pi, y) = 0, \\ v(x, 0) &= -\psi(x), \quad 0 < x < \pi.\end{aligned}$$

This is Problem 11 in Section 13.5. The solution is

$$v(x, y) = \sum_{n=1}^{\infty} A_n e^{-ny} \sin nx$$

where

$$\begin{aligned}A_n &= \frac{2}{\pi} \int_0^{\pi} [-\psi(x) \sin nx] dx \\ &= \frac{2(-1)^n}{n} \left(\frac{1}{\pi} + \frac{h\pi}{2} \right) - h(-1)^n \left(\frac{\pi}{n} + \frac{2}{n^2} \right).\end{aligned}$$

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Thus

$$u(x, y) = v(x, y) + \psi(x) = \left(\frac{1}{\pi} + \frac{h\pi}{2} \right) x - \frac{h}{2} x^2 + \sum_{n=1}^{\infty} A_n e^{-ny} \sin nx.$$

13. Identifying $k = 1$ and $L = \pi$ we see that the eigenfunctions of $X'' + \lambda X = 0$, $X(0) = 0$, $X(\pi) = 0$ are $\sin nx$, $n = 1, 2, 3, \dots$. Assuming that $u(x, t) = \sum_{n=1}^{\infty} u_n(t) \sin nx$, the formal partial derivatives of u are

$$\frac{\partial^2 u}{\partial x^2} = \sum_{n=1}^{\infty} u_n(t)(-n^2) \sin nx \quad \text{and} \quad \frac{\partial u}{\partial t} = \sum_{n=1}^{\infty} u'_n(t) \sin nx.$$

Assuming that $xe^{-3t} = \sum_{n=1}^{\infty} F_n(t) \sin nx$ we have

$$F_n(t) = \frac{2}{\pi} \int_0^{\pi} xe^{-3t} \sin nx \, dx = \frac{2e^{-3t}}{\pi} \int_0^{\pi} x \sin nx \, dx = \frac{2e^{-3t}(-1)^{n+1}}{n}.$$

Then

$$xe^{-3t} = \sum_{n=1}^{\infty} \frac{2e^{-3t}(-1)^{n+1}}{n} \sin nx$$

and

$$u_t - u_{xx} = \sum_{n=1}^{\infty} [u'_n(t) + n^2 u_n(t)] \sin nx = xe^{-3t} = \sum_{n=1}^{\infty} \frac{2e^{-3t}(-1)^{n+1}}{n} \sin nx.$$

Equating coefficients we obtain

$$u'_n(t) + n^2 u_n(t) = \frac{2e^{-3t}(-1)^{n+1}}{n}.$$

This is a linear first-order differential equation whose solution is

$$u_n(t) = \frac{2(-1)^{n+1}}{n(n^2 - 3)} e^{-3t} + C_n e^{-n^2 t}.$$

Thus

$$u(x, t) = \sum_{n=1}^{\infty} \frac{2(-1)^{n+1}}{n(n^2 - 3)} e^{-3t} \sin nx + \sum_{n=1}^{\infty} C_n e^{-n^2 t} \sin nx$$

and $u(x, 0) = 0$ implies

$$\sum_{n=1}^{\infty} \frac{2(-1)^{n+1}}{n(n^2 - 3)} \sin nx + \sum_{n=1}^{\infty} C_n \sin nx = 0$$

so that $C_n = 2(-1)^n / n(n^2 - 3)$. Therefore

$$u(x, t) = 2 \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n(n^2 - 3)} e^{-3t} \sin nx + 2 \sum_{n=1}^{\infty} \frac{(-1)^n}{n(n^2 - 3)} e^{-n^2 t} \sin nx.$$

14. Identifying $k = 1$ and $L = \pi$ we see that the eigenfunctions of $X'' + \lambda X = 0$, $X(0) = 0$, $X'(\pi) = 0$ are $1, \cos nx$, $n = 1, 2, 3, \dots$. Assuming that $u(x, t) = \frac{1}{2} u_0(t) + \sum_{n=1}^{\infty} u_n(t) \cos nx$, the formal partial derivatives of u are

$$\frac{\partial^2 u}{\partial x^2} = \sum_{n=1}^{\infty} u_n(t)(-n^2) \cos nx \quad \text{and} \quad \frac{\partial u}{\partial t} = \frac{1}{2} u'_0 + \sum_{n=1}^{\infty} u'_n(t) \cos nx.$$

Assuming that $xe^{-3t} = \frac{1}{2} F_0(t) + \sum_{n=1}^{\infty} F_n(t) \cos nx$ we have

$$F_0(t) = \frac{2e^{-3t}}{\pi} \int_0^{\pi} x \, dx = \pi e^{-3t}$$

and

$$F_n(t) = \frac{2e^{-3t}}{\pi} \int_0^{\pi} x \cos nx \, dx = \frac{2e^{-3t}[(-1)^n - 1]}{\pi n^2}.$$

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Then

$$xe^{-3t} = \frac{\pi}{2}e^{-3t} + \sum_{n=1}^{\infty} \frac{2e^{-3t}[(-1)^n - 1]}{\pi n^2} \cos nx$$

and

$$\begin{aligned} u_t - u_{xx} &= \frac{1}{2}u'_0(t) + \sum_{n=1}^{\infty} [u'_n(t) + n^2 u_n(t)] \cos nx \\ &= xe^{-3t} = \frac{\pi}{2}e^{-3t} + \sum_{n=1}^{\infty} \frac{2e^{-3t}[(-1)^n - 1]}{\pi n^2} \cos nx. \end{aligned}$$

Equating coefficients, we obtain

$$u'_0(t) = \pi e^{-3t} \quad \text{and} \quad u'_n(t) + n^2 u_n(t) = \frac{2e^{-3t}[(-1)^n - 1]}{\pi n^2} \cos nx.$$

The first equation yields $u_0(t) = -(\pi/3)e^{-3t} + C_0$ and the second equation, which is a linear first-order differential equation, yields

$$u_n(t) = \frac{2[(-1)^n - 1]}{\pi n^2(n^2 - 3)} e^{-3t} + C_n e^{-n^2 t}.$$

Thus

$$u(x, t) = -\frac{\pi}{3}e^{-3t} + C_0 + \sum_{n=1}^{\infty} \frac{2[(-1)^n - 1]}{\pi n^2(n^2 - 3)} e^{-3t} \cos nx + \sum_{n=1}^{\infty} C_n e^{-n^2 t} \cos nx$$

and $u(x, 0) = 0$ implies

$$-\frac{\pi}{3} + C_0 + \sum_{n=1}^{\infty} \frac{2[(-1)^n - 1]}{\pi n^2(n^2 - 3)} \cos nx + \sum_{n=1}^{\infty} C_n \cos nx = 0$$

so that $C_0 = \pi/3$ and $C_n = 2[(-1)^n - 1]/\pi n^2(n^2 - 3)$. Therefore

$$u(x, t) = \frac{\pi}{3}(1 - e^{-3t}) + \frac{2}{\pi} \sum_{n=1}^{\infty} \frac{(-1)^n - 1}{n^2(n^2 - 3)} e^{-3t} \cos nx + \frac{2}{\pi} \sum_{n=1}^{\infty} \frac{1 - (-1)^n}{n^2(n^2 - 3)} e^{-n^2 t} \cos nx.$$

15. Identifying $k = 1$ and $L = 1$ we see that the eigenfunctions of $X'' + \lambda X = 0$, $X(0) = 0$, $X(1) = 0$ are $\sin n\pi x$, $n = 1, 2, 3, \dots$. Assuming that $u(x, t) = \sum_{n=1}^{\infty} u_n(t) \sin n\pi x$, the formal partial derivatives of u are

$$\frac{\partial^2 u}{\partial x^2} = \sum_{n=1}^{\infty} u_n(t)(-n^2 \pi^2) \sin n\pi x \quad \text{and} \quad \frac{\partial u}{\partial t} = \sum_{n=1}^{\infty} u'_n(t) \sin n\pi x.$$

Assuming that $-1 + x - x \cos t = \sum_{n=1}^{\infty} F_n(t) \sin n\pi x$ we have

$$F_n(t) = \frac{2}{1} \int_0^1 (-1 + x - x \cos t) \sin n\pi x \, dx = \frac{2[-1 + (-1)^n \cos t]}{n\pi}.$$

Then

$$-1 + x - x \cos t = \frac{2}{\pi} \sum_{n=1}^{\infty} \frac{-1 + (-1)^n \cos t}{n} \sin n\pi x$$

and

$$\begin{aligned} u_t - u_{xx} &= \sum_{n=1}^{\infty} [u'_n(t) + n^2 \pi^2 u_n(t)] \sin n\pi x \\ &= -1 + x - x \cos t = \frac{2}{\pi} \sum_{n=1}^{\infty} \frac{-1 + (-1)^n \cos t}{n} \sin n\pi x. \end{aligned}$$

13.6 Nonhomogeneous Equations and Boundary Conditions

Equating coefficients we obtain

$$u'_n(t) + n^2\pi^2 u_n(t) = \frac{2[-1 + (-1)^n \cos t]}{n\pi}.$$

This is a linear first-order differential equation whose solution is

$$u_n(t) = \frac{2}{n\pi} \left[-\frac{1}{n^2\pi^2} + (-1)^n \frac{n^2\pi^2 \cos t + \sin t}{n^4\pi^4 + 1} \right] + C_n e^{-n^2\pi^2 t}.$$

Thus

$$u(x, t) = \sum_{n=1}^{\infty} \frac{2}{n\pi} \left[-\frac{1}{n^2\pi^2} + (-1)^n \frac{n^2\pi^2 \cos t + \sin t}{n^4\pi^4 + 1} \right] \sin n\pi x + \sum_{n=1}^{\infty} C_n e^{-n^2\pi^2 t} \sin n\pi x$$

and $u(x, 0) = x(1 - x)$ implies

$$\sum_{n=1}^{\infty} \frac{2}{n\pi} \left[-\frac{1}{n^2\pi^2} + (-1)^n \frac{n^2\pi^2}{n^4\pi^4 + 1} + C_n \right] \sin n\pi x = x(1 - x).$$

Hence

$$\frac{2}{n\pi} \left[-\frac{1}{n^2\pi^2} + (-1)^n \frac{n^2\pi^2}{n^4\pi^4 + 1} + C_n \right] = \frac{2}{1} \int_0^1 x(1 - x) \sin n\pi x \, dx = 2 \left[\frac{1 - (-1)^n}{n^3\pi^3} \right]$$

and

$$C_n = \frac{4 - 2(-1)^n}{n^3\pi^3} - (-1)^n \frac{2n\pi}{n^4\pi^4 + 1}.$$

Therefore

$$\begin{aligned} u(x, t) &= \sum_{n=1}^{\infty} \frac{2}{n\pi} \left[-\frac{1}{n^2\pi^2} + (-1)^n \frac{n^2\pi^2 \cos t + \sin t}{n^4\pi^4 + 1} \right] \sin n\pi x \\ &\quad + \sum_{n=1}^{\infty} \left[\frac{4 - 2(-1)^n}{n^3\pi^3} - (-1)^n \frac{2n\pi}{n^4\pi^4 + 1} \right] e^{-n^2\pi^2 t} \sin n\pi x. \end{aligned}$$

16. Identifying $k = 1$ and $L = \pi$ we see that the eigenfunctions of $X'' + \lambda X = 0$, $X(0) = 0$, $X(\pi) = 0$ are $\sin nx$, $n = 1, 2, 3, \dots$. Assuming that $u(x, t) = \sum_{n=1}^{\infty} u_n(t) \sin nx$, the formal partial derivatives of u are

$$\frac{\partial^2 u}{\partial x^2} = \sum_{n=1}^{\infty} u_n(t) (-n^2) \sin nx \quad \text{and} \quad \frac{\partial^2 u}{\partial t^2} = \sum_{n=1}^{\infty} u''_n(t) \sin nx.$$

Then

$$u_{tt} - u_{xx} = \sum_{n=1}^{\infty} [u''_n(t) + n^2 u_n(t)] \sin nx = \cos t \sin x.$$

Equating coefficients, we obtain $u''_1(t) + u_1(t) \cos t$ and $u''_n(t) + n^2 u_n(t) = 0$ for $n = 2, 3, 4, \dots$. Solving the first differential equation we obtain $u_1(t) = A_1 \cos t + B_1 \sin t + \frac{1}{2} t \sin t$. From the second differential equation we obtain $u_n(t) = A_n \cos nt + B_n \sin nt$ for $n = 2, 3, 4, \dots$. Thus

$$u(x, t) = \left(A_1 \cos t + B_1 \sin t + \frac{1}{2} t \sin t \right) \sin x + \sum_{n=2}^{\infty} (A_n \cos nt + B_n \sin nt) \sin nx.$$

From

$$u(x, 0) = A_1 \sin x + \sum_{n=2}^{\infty} A_n \sin nx = 0$$

we see that $A_n = 0$ for $n = 1, 2, 3, \dots$. Thus

$$u(x, t) = \left(B_1 \sin t + \frac{1}{2} t \sin t \right) \sin x + \sum_{n=2}^{\infty} B_n \sin nt \sin nx$$

and

$$\frac{\partial u}{\partial t} = \left(B_1 \cos t + \frac{1}{2}t \cos t + \frac{1}{2} \sin t \right) \sin x + \sum_{n=2}^{\infty} n B_n \cos nt \sin nx,$$

so

$$\left. \frac{\partial u}{\partial t} \right|_{t=0} = B_1 \sin x + \sum_{n=2}^{\infty} n B_n \sin nx = 0.$$

We see that $B_n = 0$ for all n so $u(x, t) = \frac{1}{2}t \sin t \sin x$.

17. This problem is very similar to Example 2 in the text. To match it to the boundary-value problem in (1) in the text we identify $k = 1$, $L = 1$, $F(x, t) = 0$, $u_0(t) = \sin t$, $u_1(t) = 0$, and $f(x) = 0$. To construct $\psi(x, t)$ we use

$$\psi(x, t) = u_0(x) + \frac{x}{L} [u_1(t) - u_0(t)] = \sin t + x[0 - \sin t] = (1 - x) \sin t,$$

so $G(x, t) = F(x, t) - \psi_t(x, t) = (x - 1) \cos t$. Then the substitution

$$u(x, t) = v(x, t) + \psi(x, t) = v(x, t) + (1 - x) \sin t$$

leads to the boundary-value problem

$$\begin{aligned} \frac{\partial^2 v}{\partial x^2} + (x - 1) \cos t &= \frac{\partial v}{\partial t}, \quad 0 < x < 1, \quad t > 0 \\ v(0, t) &= 0, \quad v(1, t) = 0, \quad t > 0 \\ v(x, 0) &= 0, \quad 0 < x < 1. \end{aligned}$$

The eigenvalues and eigenfunctions of the Sturm-Liouville problem

$$X'' + \lambda X = 0, \quad X(0) = 0, \quad X(1) = 0$$

are $\lambda_n = \alpha_n^2 = n^2 \pi^2$ and $\sin n\pi x$, $n = 1, 2, 3, \dots$. With $G(x, t) = (x - 1) \cos t$ we assume for fixed t that v and G can be written as Fourier sine series:

$$v(x, t) = \sum_{n=1}^{\infty} v_n(t) \sin n\pi x$$

and

$$G(x, t) = \sum_{n=1}^{\infty} G_n(t) \sin n\pi x.$$

By treating t as a parameter, the coefficients G_n can be computed:

$$G_n(t) = \frac{2}{1} \int_0^1 (x - 1) \cos t \sin n\pi x \, dx = 2 \cos t \int_0^1 (x - 1) \sin n\pi x \, dx = -\frac{2}{n\pi} \cos t.$$

Hence

$$(x - 1) \cos t = \sum_{n=1}^{\infty} \frac{-2 \cos t}{n\pi} \sin n\pi x.$$

Now, using the series representation for $v(x, t)$, we have

$$\frac{\partial^2 v}{\partial x^2} = \sum_{n=1}^{\infty} v_n(t) (-n^2 \pi^2) \sin n\pi x \quad \text{and} \quad \frac{\partial v}{\partial t} = \sum_{n=1}^{\infty} v'_n(t) \sin n\pi x.$$

Writing the partial differential equation as $v_t - v_{xx} = (x - 1) \cos t$ and using the above results we have

$$\sum_{n=1}^{\infty} [v'_n(t) + n^2 \pi^2 v_n(t)] \sin n\pi x = \sum_{n=1}^{\infty} \frac{-2 \cos t}{n\pi} \sin n\pi x.$$

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Equating coefficients we get

$$v'_n(t) + n^2\pi^2 v_n(t) = -\frac{2\cos t}{n\pi}.$$

For each n this is a linear first-order differential equation whose general solution is

$$v_n(t) = -\frac{2}{n\pi} \left[\frac{n^2\pi^2 \cos t + \sin t}{n^4\pi^4 + 1} \right] + C_n e^{-n^2\pi^2 t}.$$

Thus

$$v(x, t) = \sum_{n=1}^{\infty} \left[-\frac{2n^2\pi^2 \cos t + 2\sin t}{n\pi(n^4\pi^4 + 1)} + C_n e^{-n^2\pi^2 t} \right] \sin n\pi x.$$

The initial condition $v(x, 0) = 0$ implies

$$\sum_{n=1}^{\infty} \left[-\frac{2n\pi}{n^4\pi^4 + 1} + C_n \right] \sin n\pi x = 0$$

so that $C_n = 2n\pi/(n^4\pi^4 + 1)$. Therefore

$$\begin{aligned} v(x, t) &= \sum_{n=1}^{\infty} \left[-\frac{2n^2\pi^2 \cos t + 2\sin t}{n\pi(n^4\pi^4 + 1)} + \frac{2n\pi}{n^4\pi^4 + 1} e^{-n^2\pi^2 t} \right] \sin n\pi x \\ &= \frac{2}{\pi} \sum_{n=1}^{\infty} \left[\frac{n^2\pi^2 e^{-n^2\pi^2 t} - n^2\pi^2 \cos t - \sin t}{n(n^4\pi^4 + 1)} \right] \sin n\pi x \end{aligned}$$

and

$$u(x, t) = v(x, t) + \psi(x, t) = (1 - x) \sin t + \frac{2}{\pi} \sum_{n=1}^{\infty} \left[\frac{n^2\pi^2 e^{-n^2\pi^2 t} - n^2\pi^2 \cos t - \sin t}{n(n^4\pi^4 + 1)} \right] \sin n\pi x.$$

18. To match this problem to (1) in the text we identify $k = 1$, $L = 1$, $F(x, t) = 2t + 3tx$, $u_0(t) = t^2$, $u_1(t) = 1$, and $f(x) = x^2$. To construct $\psi(x, t)$ we use

$$\psi(x, t) = u_0(t) + \frac{x}{L} [u_1(t) - u_0(t)] = x + (1 - x)t^2,$$

so $G(x, t) = F(x, t) - \psi_t(x, t) = 2t + 3tx - 2(1 - x)t = 5tx$. Then the substitution

$$u(x, t) = v(x, t) + \psi(x, t) = v(x, t) + x + (1 - x)t^2$$

leads to the boundary-value problem

$$\begin{aligned} \frac{\partial^2 v}{\partial x^2} + 5tx &= \frac{\partial v}{\partial t}, \quad 0 < x < 1, \quad t > 0 \\ v(0, t) &= 0, \quad v(1, t) = 0, \quad t > 0 \\ v(x, 0) &= x^2 - x, \quad 0 < x < 1. \end{aligned}$$

The eigenvalues and eigenfunctions of the Sturm-Liouville problem

$$X'' + \lambda X = 0, \quad X(0) = 0, \quad X(1) = 0$$

are $\lambda_n = \alpha_n^2 = n^2\pi^2$ and $\sin n\pi x$, $n = 1, 2, 3, \dots$. With $G(x, t) = 5tx$ we assume for fixed t that v and G can be written as Fourier sine series:

$$v(x, t) = \sum_{n=1}^{\infty} v_n(t) \sin n\pi x$$

and

$$G(x, t) = \sum_{n=1}^{\infty} G_n(t) \sin n\pi x.$$

13.6 Nonhomogeneous Equations and Boundary Conditions

By treating t as a parameter, the coefficients G_n can be computed:

$$G_n(t) = \frac{2}{1} \int_0^1 5tx \sin n\pi x \, dx = 10t \int_0^1 x \sin n\pi x \, dx = \frac{10t}{n\pi} (-1)^{n+1}.$$

Hence

$$5tx = \sum_{n=1}^{\infty} (-1)^{n+1} \frac{10t}{n\pi} \sin n\pi x.$$

Now, using the series representation for $v(x, t)$, we have

$$\frac{\partial^2 v}{\partial x^2} = \sum_{n=1}^{\infty} v_n(t) (-n^2 \pi^2) \sin n\pi x \quad \text{and} \quad \frac{\partial v}{\partial t} = \sum_{n=1}^{\infty} v'_n(t) \sin n\pi x.$$

Writing the partial differential equation as $v_t - v_{xx} = 5tx$ and using the above results we have

$$\sum_{n=1}^{\infty} [v'_n(t) + n^2 \pi^2 v_n(t)] \sin n\pi x = \sum_{n=1}^{\infty} (-1)^{n+1} \frac{10t}{n\pi} \sin n\pi x.$$

Equating coefficients we get

$$v'_n(t) + n^2 \pi^2 v_n(t) = (-1)^{n+1} \frac{10t}{n\pi}.$$

For each n this is a linear first-order differential equation whose general solution is

$$v_n(t) = 10(-1)^{n+1} \frac{n^2 \pi^2 t - 1}{n^5 \pi^5} + C_n e^{-n^2 \pi^2 t}.$$

Thus

$$v(x, t) = \sum_{n=1}^{\infty} \left[10(-1)^{n+1} \frac{n^2 \pi^2 t - 1}{n^5 \pi^5} + C_n e^{-n^2 \pi^2 t} \right] \sin n\pi x.$$

The initial condition $v(x, 0) = x^2 - x$ implies

$$\sum_{n=1}^{\infty} \left[10(-1)^{n+1} \frac{-1}{n^5 \pi^5} + C_n \right] \sin n\pi x = x^2 - x.$$

Thinking of $x^2 - x$ as a Fourier sine series with coefficients $2 \int_0^1 (x^2 - x) \sin n\pi x \, dx = [4(-1)^n - 4]/n^3 \pi^3$ we equate coefficients to obtain

$$\frac{10(-1)^n}{n^5 \pi^5} + C_n = \frac{4(-1)^n - 4}{n^3 \pi^3}$$

so

$$C_n = \frac{4(-1)^n - 4}{n^3 \pi^3} - \frac{10(-1)^n}{n^5 \pi^5}.$$

Therefore

$$v(x, t) = \sum_{n=1}^{\infty} \left[10(-1)^{n+1} \frac{n^2 \pi^2 t - 1}{n^5 \pi^5} + \left(\frac{4(-1)^n - 4}{n^3 \pi^3} - \frac{10(-1)^n}{n^5 \pi^5} \right) e^{-n^2 \pi^2 t} \right] \sin n\pi x$$

and

$$\begin{aligned} u(x, t) &= v(x, t) + \psi(x, t) \\ &= x + (1-x)t^2 + \sum_{n=1}^{\infty} \left[10(-1)^{n+1} \frac{n^2 \pi^2 t - 1}{n^5 \pi^5} + \left(\frac{4(-1)^n - 4}{n^3 \pi^3} - \frac{10(-1)^n}{n^5 \pi^5} \right) e^{-n^2 \pi^2 t} \right] \sin n\pi x. \end{aligned}$$

19. After a long period of time we would intuitively expect the temperature at the center of the rod to be approximately equal to the average value of the temperatures at the ends of the rod. To prove this we note that as

13.6 Nonhomogeneous Equations and Boundary Conditions

t becomes large, $u(x, t)$ approaches $\psi(x)$, where $\psi(x) = u_0 + (x/L)[u_1 - u_0]$ (this is (12) in this section of the text). The result follows from

$$\psi\left(\frac{L}{2}\right) = u_0 + \frac{1}{2}[u_1 - u_0] = \frac{1}{2}(u_0 + u_1).$$

20. In the general case the associated Sturm-Liouville problem is

$$X'' + \lambda X = 0, \quad X'(0) = 0, \quad X'(L) = 0$$

with eigenvalues and eigenfunctions $\lambda_0 = 0$, $X_0 = 1$, and $\lambda_n = n^2\pi^2/L^2$, $X_n = \cos n\pi x/L$, $n = 1, 2, 3, \dots$. The entire set of eigenfunctions can be written as $X_n = \cos n\pi x/L$, $n = 0, 1, 2, \dots$, which serves as the basis for the Fourier cosine series. Hence, we assume in this problem that

$$u(x, t) = \frac{1}{2}u_0(t) + \sum_{n=1}^{\infty} u_n(t) \cos \frac{n\pi}{L}x$$

and

$$F(x, t) = \frac{1}{2}F_0(t) + \sum_{n=1}^{\infty} F_n(t) \cos \frac{n\pi}{L}x.$$

Taking $k = 1$, $L = 1$, $F(x, t) = tx$, and $f(x) = 0$ we have

$$u(x, t) = \frac{1}{2}u_0(t) + \sum_{n=1}^{\infty} u_n(t) \cos n\pi x$$

and

$$F(x, t) = \frac{1}{2}F_0(t) + \sum_{n=1}^{\infty} F_n(t) \cos n\pi x.$$

By treating t as a parameter, the coefficients F_n can be computed:

$$\begin{aligned} F_0(t) &= 2 \int_0^1 tx \, dx = t \\ F_n(t) &= 2 \int_0^1 tx \cos n\pi x \, dx = 2t \int_0^1 x \cos n\pi x \, dx = 2t \frac{(-1)^n - 1}{n^2\pi^2}. \end{aligned}$$

Hence

$$tx = \frac{1}{2}t + 2t \sum_{n=1}^{\infty} \frac{(-1)^n - 1}{n^2\pi^2} \cos n\pi x.$$

Now, using the series representation for $u(x, t)$, we have

$$\frac{\partial^2 u}{\partial x^2} = \sum_{n=1}^{\infty} u_n(t)(-n^2\pi^2) \cos n\pi x \quad \text{and} \quad \frac{\partial u}{\partial t} = \frac{1}{2}u'_0(t) + \sum_{n=1}^{\infty} u'_n(t) \cos n\pi x.$$

Writing the partial differential equation as $u_t - u_{xx} = tx$ and using the above results we have

$$\frac{1}{2}u'_0(t) + \sum_{n=1}^{\infty} [u'_n(t) + n^2\pi^2 u_n(t)] \cos n\pi x = \frac{1}{2}t + 2t \sum_{n=1}^{\infty} \frac{(-1)^n - 1}{n^2\pi^2} \cos n\pi x.$$

Equating coefficients we get

$$u'_0(t) = t \quad \text{and} \quad u'_n(t) + n^2\pi^2 u_n(t) = 2t \frac{(-1)^n - 1}{n^2\pi^2}.$$

From the first equation we obtain $u_0(t) = \frac{1}{2}t^2 + C_0$. The second equation is a linear, first-order differential equation whose general solution is

$$u_n(t) = 2 \left(\frac{(-1)^n - 1}{n^2\pi^2} \right) \left(\frac{n^2\pi^2 t - 1}{n^4\pi^4} \right) + C_n e^{-n^2\pi^2 t}.$$

Thus

$$u(x, t) = \frac{1}{4}t^2 + \frac{1}{2}C_0 + \sum_{n=1}^{\infty} \left[2 \left(\frac{(-1)^n - 1}{n^2\pi^2} \right) \left(\frac{n^2\pi^2 t - 1}{n^4\pi^4} \right) + C_n e^{-n^2\pi^2 t} \right] \cos n\pi x.$$

The initial condition $u(x, 0) = 0$ implies

$$\frac{1}{2}C_0 + \sum_{n=1}^{\infty} \left[2 \left(\frac{(-1)^n - 1}{n^2\pi^2} \right) \left(\frac{-1}{n^4\pi^4} \right) + C_n \right] \cos n\pi x = \frac{1}{2}C_0 + \sum_{n=1}^{\infty} \left[2 \frac{1 - (-1)^n}{n^6\pi^6} + C_n \right] \cos n\pi x = 0$$

so that

$$C_0 = 0 \quad \text{and} \quad C_n = 2 \frac{(-1)^n - 1}{n^6\pi^6}.$$

Therefore

$$\begin{aligned} u(x, t) &= \frac{1}{4}t^2 + \sum_{n=1}^{\infty} \left[2 \left(\frac{(-1)^n - 1}{n^2\pi^2} \right) \left(\frac{n^2\pi^2 t - 1}{n^4\pi^4} \right) + 2 \frac{(-1)^n - 1}{n^6\pi^6} e^{-n^2\pi^2 t} \right] \cos n\pi x \\ &= \frac{1}{4}t^2 + \frac{2}{\pi^6} \sum_{n=1}^{\infty} \left[\frac{[(-1)^n - 1][n^2\pi^2 t - 1 + e^{-n^2\pi^2 t}]}{n^6} \right] \cos n\pi x. \end{aligned}$$

EXERCISES 13.7

Orthogonal Series Expansions

- Referring to Example 1 in the text we have

$$X(x) = c_1 \cos \alpha x + c_2 \sin \alpha x$$

and

$$T(t) = c_3 e^{-k\alpha^2 t}.$$

From $X'(0) = 0$ (since the left end of the rod is insulated), we find $c_2 = 0$. Then $X(x) = c_1 \cos \alpha x$ and the other boundary condition $X'(1) = -hX(1)$ implies

$$-\alpha \sin \alpha + h \cos \alpha = 0 \quad \text{or} \quad \cot \alpha = \frac{\alpha}{h}.$$

Denoting the consecutive positive roots of this latter equation by α_n for $n = 1, 2, 3, \dots$, we have

$$u(x, t) = \sum_{n=1}^{\infty} A_n e^{-k\alpha_n^2 t} \cos \alpha_n x.$$

From the initial condition $u(x, 0) = 1$ we obtain

$$1 = \sum_{n=1}^{\infty} A_n \cos \alpha_n x$$

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and

$$\begin{aligned}
 A_n &= \frac{\int_0^1 \cos \alpha_n x \, dx}{\int_0^1 \cos^2 \alpha_n x \, dx} = \frac{\sin \alpha_n / \alpha_n}{\frac{1}{2} \left[1 + \frac{1}{2\alpha_n} \sin 2\alpha_n \right]} \\
 &= \frac{2 \sin \alpha_n}{\alpha_n \left[1 + \frac{1}{\alpha_n} \sin \alpha_n \cos \alpha_n \right]} = \frac{2 \sin \alpha_n}{\alpha_n \left[1 + \frac{1}{h\alpha_n} \sin \alpha_n (\alpha_n \sin \alpha_n) \right]} \\
 &= \frac{2h \sin \alpha_n}{\alpha_n [h + \sin^2 \alpha_n]}.
 \end{aligned}$$

The solution is

$$u(x, t) = 2h \sum_{n=1}^{\infty} \frac{\sin \alpha_n}{\alpha_n (h + \sin^2 \alpha_n)} e^{-k\alpha_n^2 t} \cos \alpha_n x.$$

2. Substituting $u(x, t) = v(x, t) + \psi(x)$ into the partial differential equation gives

$$k \frac{\partial^2 v}{\partial x^2} + k\psi'' = \frac{\partial v}{\partial t}.$$

This equation will be homogeneous if $\psi''(x) = 0$ or $\psi(x) = c_1 x + c_2$. The boundary condition $u(0, t) = 0$ implies $\psi(0) = 0$ which implies $c_2 = 0$. Thus $\psi(x) = c_1 x$. Using the second boundary condition we obtain

$$-\left(\frac{\partial v}{\partial x} + \psi' \right) \Big|_{x=1} = -h[v(1, t) + \psi(1) - u_0],$$

which will be homogeneous when

$$-\psi'(1) = -h\psi(1) + hu_0.$$

Since $\psi(1) = \psi'(1) = c_1$ we have $-c_1 = -hc_1 + hu_0$ and $c_1 = hu_0/(h-1)$. Thus

$$\psi(x) = \frac{hu_0}{h-1} x.$$

The new boundary-value problem is

$$\begin{aligned}
 k \frac{\partial^2 v}{\partial x^2} &= \frac{\partial v}{\partial t}, \quad 0 < x < 1, \quad t > 0 \\
 v(0, t) &= 0, \quad \frac{\partial v}{\partial x} \Big|_{x=1} = -hv(1, t), \quad h > 0, \quad t > 0 \\
 v(x, 0) &= f(x) - \frac{hu_0}{h-1} x, \quad 0 < x < 1.
 \end{aligned}$$

Referring to Example 1 in the text we see that

$$v(x, t) = \sum_{n=1}^{\infty} A_n e^{-k\alpha_n^2 t} \sin \alpha_n x$$

and

$$u(x, t) = v(x, t) + \psi(x) = \frac{hu_0}{h-1} x + \sum_{n=1}^{\infty} A_n e^{-k\alpha_n^2 t} \sin \alpha_n x$$

where

$$f(x) - \frac{hu_0}{h-1} x = \sum_{n=1}^{\infty} A_n \sin \alpha_n x$$

and α_n is a solution of $\alpha_n \cos \alpha_n = -h \sin \alpha_n$. The coefficients are

$$\begin{aligned} A_n &= \frac{\int_0^1 [f(x) - hu_0x/(h-1)] \sin \alpha_n x \, dx}{\int_0^1 \sin^2 \alpha_n x \, dx} = \frac{\int_0^1 [f(x) - hu_0x/(h-1)] \sin \alpha_n x \, dx}{\frac{1}{2} \left[1 - \frac{1}{2\alpha_n} \sin 2\alpha_n \right]} \\ &= \frac{2 \int_0^1 [f(x) - hu_0x/(h-1)] \sin \alpha_n x \, dx}{1 - \frac{1}{\alpha_n} \sin \alpha_n \cos \alpha_n} = \frac{2 \int_0^1 [f(x) - hu_0x/(h-1)] \sin \alpha_n x \, dx}{1 - \frac{1}{h\alpha_n} (h \sin \alpha_n) \cos \alpha_n} \\ &= \frac{2 \int_0^1 [f(x) - hu_0x/(h-1)] \sin \alpha_n x \, dx}{1 - \frac{1}{h\alpha_n} (-\alpha_n \cos \alpha_n) \cos \alpha_n} = \frac{2h}{h + \cos^2 \alpha_n} \int_0^1 \left[f(x) - \frac{hu_0}{h-1} x \right] \sin \alpha_n x \, dx. \end{aligned}$$

3. Separating variables in Laplace's equation gives

$$X'' + \alpha^2 X = 0$$

$$Y'' - \alpha^2 Y = 0$$

and

$$X(x) = c_1 \cos \alpha x + c_2 \sin \alpha x$$

$$Y(y) = c_3 \cosh \alpha y + c_4 \sinh \alpha y.$$

From $u(0, y) = 0$ we obtain $X(0) = 0$ and $c_1 = 0$. From $u_x(a, y) = -hu(a, y)$ we obtain $X'(a) = -hX(a)$ and

$$\alpha \cos \alpha a = -h \sin \alpha a \quad \text{or} \quad \tan \alpha a = -\frac{\alpha}{h}.$$

Let α_n , where $n = 1, 2, 3, \dots$, be the consecutive positive roots of this equation. From $u(x, 0) = 0$ we obtain $Y(0) = 0$ and $c_3 = 0$. Thus

$$u(x, y) = \sum_{n=1}^{\infty} A_n \sinh \alpha_n y \sin \alpha_n x.$$

Now

$$f(x) = \sum_{n=1}^{\infty} A_n \sinh \alpha_n b \sin \alpha_n x$$

and

$$A_n \sinh \alpha_n b = \frac{\int_0^a f(x) \sin \alpha_n x \, dx}{\int_0^a \sin^2 \alpha_n x \, dx}.$$

Since

$$\begin{aligned} \int_0^a \sin^2 \alpha_n x \, dx &= \frac{1}{2} \left[a - \frac{1}{2\alpha_n} \sin 2\alpha_n a \right] = \frac{1}{2} \left[a - \frac{1}{\alpha_n} \sin \alpha_n a \cos \alpha_n a \right] \\ &= \frac{1}{2} \left[a - \frac{1}{h\alpha_n} (h \sin \alpha_n a) \cos \alpha_n a \right] \\ &= \frac{1}{2} \left[a - \frac{1}{h\alpha_n} (-\alpha_n \cos \alpha_n a) \cos \alpha_n a \right] = \frac{1}{2h} [ah + \cos^2 \alpha_n a], \end{aligned}$$

we have

$$A_n = \frac{2h}{\sinh \alpha_n b [ah + \cos^2 \alpha_n a]} \int_0^a f(x) \sin \alpha_n x \, dx.$$

4. Letting $u(x, y) = X(x)Y(y)$ and separating variables gives

$$X''Y + XY'' = 0.$$

The boundary conditions

$$\left. \frac{\partial u}{\partial y} \right|_{y=0} = 0 \quad \text{and} \quad \left. \frac{\partial u}{\partial y} \right|_{y=1} = -hu(x, 1)$$

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correspond to

$$X(x)Y'(0) = 0 \quad \text{and} \quad X(x)Y'(1) = -hX(x)Y(1)$$

or

$$Y'(0) = 0 \quad \text{and} \quad Y'(1) = -hY(1).$$

Since these homogeneous boundary conditions are in terms of Y , we separate the differential equation as

$$\frac{X''}{X} = -\frac{Y''}{Y} = \alpha^2.$$

Then

$$Y'' + \alpha^2 Y = 0$$

and

$$X'' - \alpha^2 X = 0$$

have solutions

$$Y(y) = c_1 \cos \alpha y + c_2 \sin \alpha y$$

and

$$X(x) = c_3 e^{-\alpha x} + c_4 e^{\alpha x}.$$

We use exponential functions in the solution of $X(x)$ since the interval over which X is defined is infinite. (See the informal rule given in Section 12.5 of the text that discusses when to use the exponential form and when to use the hyperbolic form of the solution of $y'' - \alpha^2 y = 0$.) Now, $Y'(0) = 0$ implies $c_2 = 0$, so $Y(y) = c_1 \cos \alpha y$. Since $Y'(y) = -c_1 \alpha \sin \alpha y$, the boundary condition $Y'(1) = -hY(1)$ implies

$$-c_1 \alpha \sin \alpha = -h c_1 \cos \alpha \quad \text{or} \quad \cot \alpha = \frac{\alpha}{h}.$$

Consideration of the graphs of $f(\alpha) = \cot \alpha$ and $g(\alpha) = \alpha/h$ show that $\cot \alpha = \alpha/h$ has an infinite number of roots. The consecutive positive roots α_n for $n = 1, 2, 3, \dots$, are the eigenvalues of the problem. The corresponding eigenfunctions are $Y_n(y) = c_1 \cos \alpha_n y$. The condition $\lim_{x \rightarrow \infty} u(x, y) = 0$ is equivalent to $\lim_{x \rightarrow \infty} X(x) = 0$. Thus $c_4 = 0$ and $X(x) = c_3 e^{-\alpha x}$. Therefore

$$u_n(x, y) = X_n(x)Y_n(y) = A_n e^{-\alpha_n x} \cos \alpha_n y$$

and by the superposition principle

$$u(x, y) = \sum_{n=1}^{\infty} A_n e^{-\alpha_n x} \cos \alpha_n y.$$

[It is easily shown that there are no eigenvalues corresponding to $\alpha = 0$.] Finally, the condition $u(0, y) = u_0$ implies

$$u_0 = \sum_{n=1}^{\infty} A_n \cos \alpha_n y.$$

This is not a Fourier cosine series since the coefficients α_n of y are not integer multiples of π/p , where $p = 1$ in this problem. The functions $\cos \alpha_n y$ are however orthogonal since they are eigenfunctions of the Sturm-Liouville problem

$$Y'' + \alpha^2 Y = 0,$$

$$Y'(0) = 0$$

$$Y'(1) + hY(1) = 0,$$

with weight function $p(x) = 1$. Thus we find

$$A_n = \frac{\int_0^1 u_0 \cos \alpha_n y \, dy}{\int_0^1 \cos^2 \alpha_n y \, dy}.$$

Now

$$\int_0^1 u_0 \cos \alpha_n y \, dy = \frac{u_0}{\alpha_n} \sin \alpha_n y \Big|_0^1 = \frac{u_0}{\alpha_n} \sin \alpha_n$$

and

$$\begin{aligned} \int_0^1 \cos^2 \alpha_n y \, dy &= \frac{1}{2} \int_0^1 (1 + \cos 2\alpha_n y) \, dy = \frac{1}{2} \left[y + \frac{1}{2\alpha_n} \sin 2\alpha_n y \right]_0^1 \\ &= \frac{1}{2} \left[1 + \frac{1}{2\alpha_n} \sin 2\alpha_n \right] = \frac{1}{2} \left[1 + \frac{1}{\alpha_n} \sin \alpha_n \cos \alpha_n \right]. \end{aligned}$$

Since $\cot \alpha = \alpha/h$,

$$\frac{\cos \alpha}{\alpha} = \frac{\sin \alpha}{h}$$

and

$$\int_0^1 \cos^2 \alpha_n y \, dy = \frac{1}{2} \left[1 + \frac{\sin^2 \alpha_n}{h} \right].$$

Then

$$A_n = \frac{\frac{u_0}{\alpha_n} \sin \alpha_n}{\frac{1}{2} \left[1 + \frac{1}{h} \sin^2 \alpha_n \right]} = \frac{2hu_0 \sin \alpha_n}{\alpha_n (h + \sin^2 \alpha_n)}$$

and

$$u(x, y) = 2hu_0 \sum_{n=1}^{\infty} \frac{\sin \alpha_n}{\alpha_n (h + \sin^2 \alpha_n)} e^{-\alpha_n x} \cos \alpha_n y$$

where α_n for $n = 1, 2, 3, \dots$ are the consecutive positive roots of $\cot \alpha = \alpha/h$.

5. The boundary-value problem is

$$\begin{aligned} k \frac{\partial^2 u}{\partial x^2} &= \frac{\partial u}{\partial t}, \quad 0 < x < L, \quad t > 0, \\ u(0, t) &= 0, \quad \frac{\partial u}{\partial x} \Big|_{x=L} = 0, \quad t > 0, \\ u(x, 0) &= f(x), \quad 0 < x < L. \end{aligned}$$

Separation of variables leads to

$$\begin{aligned} X'' + \alpha^2 X &= 0 \\ T' + k\alpha^2 T &= 0 \end{aligned}$$

and

$$\begin{aligned} X(x) &= c_1 \cos \alpha x + c_2 \sin \alpha x \\ T(t) &= c_3 e^{-k\alpha^2 t}. \end{aligned}$$

From $X(0) = 0$ we find $c_1 = 0$. From $X'(L) = 0$ we obtain $\cos \alpha L = 0$ and

$$\alpha = \frac{\pi(2n-1)}{2L}, \quad n = 1, 2, 3, \dots$$

Thus

$$u(x, t) = \sum_{n=1}^{\infty} A_n e^{-k(2n-1)^2 \pi^2 t / 4L^2} \sin \left(\frac{2n-1}{2L} \right) \pi x$$

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where

$$A_n = \frac{\int_0^L f(x) \sin\left(\frac{2n-1}{2L}\right) \pi x dx}{\int_0^L \sin^2\left(\frac{2n-1}{2L}\right) \pi x dx} = \frac{2}{L} \int_0^L f(x) \sin\left(\frac{2n-1}{2L}\right) \pi x dx.$$

6. Substituting $u(x, t) = v(x, t) + \psi(x)$ into the partial differential equation gives

$$a^2 \frac{\partial^2 v}{\partial x^2} + \psi''(x) = \frac{\partial^2 v}{\partial t^2}.$$

This equation will be homogeneous if $\psi''(x) = 0$ or $\psi(x) = c_1 x + c_2$. The boundary condition $u(0, t) = 0$ implies $\psi(0) = 0$ which implies $c_2 = 0$. Thus $\psi(x) = c_1 x$. Using the second boundary condition, we obtain

$$E \left(\frac{\partial v}{\partial x} + \psi' \right) \Big|_{x=L} = F_0,$$

which will be homogeneous when

$$E\psi'(L) = F_0.$$

Since $\psi'(x) = c_1$ we conclude that $c_1 = F_0/E$ and

$$\psi(x) = \frac{F_0}{E} x.$$

The new boundary-value problem is

$$a^2 \frac{\partial^2 v}{\partial x^2} = \frac{\partial^2 v}{\partial t^2}, \quad 0 < x < L, \quad t > 0$$

$$v(0, t) = 0, \quad \frac{\partial v}{\partial x} \Big|_{x=L} = 0, \quad t > 0,$$

$$v(x, 0) = -\frac{F_0}{E} x, \quad \frac{\partial v}{\partial t} \Big|_{t=0} = 0, \quad 0 < x < L.$$

Referring to Example 2 in the text we see that

$$v(x, t) = \sum_{n=1}^{\infty} A_n \cos a \left(\frac{2n-1}{2L} \right) \pi t \sin \left(\frac{2n-1}{2L} \right) \pi x$$

where

$$-\frac{F_0}{E} x = \sum_{n=1}^{\infty} A_n \sin \left(\frac{2n-1}{2L} \right) \pi x$$

and

$$A_n = \frac{-F_0 \int_0^L x \sin \left(\frac{2n-1}{2L} \right) \pi x dx}{E \int_0^L \sin^2 \left(\frac{2n-1}{2L} \right) \pi x dx} = \frac{8F_0 L (-1)^n}{E \pi^2 (2n-1)^2}.$$

Thus

$$u(x, t) = v(x, t) + \psi(x) = \frac{F_0}{E} x + \frac{8F_0 L}{E \pi^2} \sum_{n=1}^{\infty} \frac{(-1)^n}{(2n-1)^2} \cos a \left(\frac{2n-1}{2L} \right) \pi t \sin \left(\frac{2n-1}{2L} \right) \pi x.$$

7. Separation of variables leads to

$$Y'' + \alpha^2 Y = 0$$

$$X'' - \alpha^2 X = 0$$

and

$$Y(y) = c_1 \cos \alpha y + c_2 \sin \alpha y$$

$$X(x) = c_3 \cosh \alpha x + c_4 \sinh \alpha x.$$

From $Y(0) = 0$ we find $c_1 = 0$. From $Y'(1) = 0$ we obtain $\cos \alpha = 0$ and

$$\alpha = \frac{\pi(2n-1)}{2}, \quad n = 1, 2, 3, \dots$$

Thus

$$Y(y) = c_2 \sin\left(\frac{2n-1}{2}\pi y\right).$$

From $X'(0) = 0$ we find $c_4 = 0$. Then

$$u(x, y) = \sum_{n=1}^{\infty} A_n \cosh\left(\frac{2n-1}{2}\pi x\right) \sin\left(\frac{2n-1}{2}\pi y\right)$$

where

$$u_0 = u(1, y) = \sum_{n=1}^{\infty} A_n \cosh\left(\frac{2n-1}{2}\pi\right) \sin\left(\frac{2n-1}{2}\pi y\right)$$

and

$$A_n \cosh\left(\frac{2n-1}{2}\pi\right) \pi = \frac{\int_0^1 u_0 \sin\left(\frac{2n-1}{2}\pi y\right) \pi y dy}{\int_0^1 \sin^2\left(\frac{2n-1}{2}\pi y\right) \pi y dy} = \frac{4u_0}{(2n-1)\pi}.$$

Thus

$$u(x, y) = \frac{4u_0}{\pi} \sum_{n=1}^{\infty} \frac{1}{(2n-1) \cosh\left(\frac{2n-1}{2}\pi\right)} \cosh\left(\frac{2n-1}{2}\pi x\right) \sin\left(\frac{2n-1}{2}\pi y\right).$$

8. The boundary-value problem is

$$\begin{aligned} k \frac{\partial^2 u}{\partial x^2} &= \frac{\partial u}{\partial t}, \quad 0 < x < 1, \quad t > 0 \\ \frac{\partial u}{\partial x} \Big|_{x=0} &= hu(0, t), \quad \frac{\partial u}{\partial x} \Big|_{x=1} = -hu(1, t), \quad h > 0, \quad t > 0, \\ u(x, 0) &= f(x), \quad 0 < x < 1. \end{aligned}$$

Referring to Example 1 in the text we have

$$X(x) = c_1 \cos \alpha x + c_2 \sin \alpha x \quad \text{and} \quad T(t) = c_3 e^{-k\alpha^2 t}.$$

Applying the boundary conditions, we obtain

$$X'(0) = hX(0)$$

$$X'(1) = -hX(1)$$

or

$$\alpha c_2 = hc_1$$

$$-\alpha c_1 \sin \alpha + \alpha c_2 \cos \alpha = -hc_1 \cos \alpha - hc_2 \sin \alpha.$$

Choosing $c_1 = \alpha$ and $c_2 = h$ (to satisfy the first equation above) we obtain

$$-\alpha^2 \sin \alpha + h\alpha \cos \alpha = -h\alpha \cos \alpha - h^2 \sin \alpha$$

$$2h\alpha \cos \alpha = (\alpha^2 - h^2) \sin \alpha.$$

The eigenvalues α_n are the consecutive positive roots of

$$\tan \alpha = \frac{2h\alpha}{\alpha^2 - h^2}.$$

Then

$$u(x, t) = \sum_{n=1}^{\infty} A_n e^{-k\alpha_n^2 t} (\alpha_n \cos \alpha_n x + h \sin \alpha_n x)$$

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where

$$f(x) = u(x, 0) = \sum_{n=1}^{\infty} A_n(\alpha_n \cos \alpha_n x + h \sin \alpha_n x)$$

and

$$\begin{aligned} A_n &= \frac{\int_0^1 f(x)(\alpha_n \cos \alpha_n x + h \sin \alpha_n x) dx}{\int_0^1 (\alpha_n \cos \alpha_n x + h \sin \alpha_n x)^2 dx} \\ &= \frac{2}{\alpha_n^2 + 2h + h^2} \int_0^1 f(x)(\alpha_n \cos \alpha_n x + h \sin \alpha_n x) dx. \end{aligned}$$

[Note: the evaluation and simplification of the integral in the denominator requires the use of the relationship $(\alpha^2 - h^2) \sin \alpha = 2h\alpha \cos \alpha$.]

9. The eigenfunctions of the associated homogeneous boundary-value problem are $\sin \alpha_n x$, $n = 1, 2, 3, \dots$, where the α_n are the consecutive positive roots of $\tan \alpha = -\alpha$. We assume that

$$u(x, t) = \sum_{n=1}^{\infty} u_n(t) \sin \alpha_n x \quad \text{and} \quad xe^{-2t} = \sum_{n=1}^{\infty} F_n(t) \sin \alpha_n x.$$

Then

$$F_n(t) = \frac{e^{-2t} \int_0^1 x \sin \alpha_n x dx}{\int_0^1 \sin^2 \alpha_n x dx}.$$

Since $\alpha_n \cos \alpha_n = -\sin \alpha_n$ and

$$\int_0^1 \sin^2 \alpha_n x dx = \frac{1}{2} \left[1 - \frac{1}{2\alpha_n} \sin 2\alpha_n \right],$$

we have

$$\begin{aligned} e^{-2t} \int_0^1 x \sin \alpha_n x dx &= e^{-2t} \left(\frac{\sin \alpha_n - \alpha_n \cos \alpha_n}{\alpha_n^2} \right) = \frac{2 \sin \alpha_n}{\alpha_n^2} e^{-2t} \\ \int_0^1 \sin^2 \alpha_n x dx &= \frac{1}{2} [1 + \cos^2 \alpha_n] \end{aligned}$$

and so

$$F_n(t) = \frac{4 \sin \alpha_n}{\alpha_n^2 (1 + \cos^2 \alpha_n)} e^{-2t}.$$

Substituting the assumptions into $u_t - ku_{xx} = xe^{-2t}$ and equating coefficients leads to the linear first-order differential equation

$$u'_n(t) + k\alpha_n^2 u(t) = \frac{4 \sin \alpha_n}{\alpha_n^2 (1 + \cos^2 \alpha_n)} e^{-2t}$$

whose solution is

$$u_n(t) = \frac{4 \sin \alpha_n}{\alpha_n^2 (1 + \cos^2 \alpha_n) (k\alpha_n^2 - 2)} e^{-2t} + C_n e^{-k\alpha_n^2 t}.$$

From

$$u(x, t) = \sum_{n=1}^{\infty} \left[\frac{4 \sin \alpha_n}{\alpha_n^2 (1 + \cos^2 \alpha_n) (k\alpha_n^2 - 2)} e^{-2t} + C_n e^{-k\alpha_n^2 t} \right] \sin \alpha_n x$$

and the initial condition $u(x, 0) = 0$ we see

$$C_n = -\frac{4 \sin \alpha_n}{\alpha_n^2 (1 + \cos^2 \alpha_n) (k\alpha_n^2 - 2)}.$$

The formal solution of the original problem is then

$$u(x, t) = \sum_{n=1}^{\infty} \frac{4 \sin \alpha_n}{\alpha_n^2 (1 + \cos^2 \alpha_n) (k\alpha_n^2 - 2)} (e^{-2t} - e^{-k\alpha_n^2 t}) \sin \alpha_n x.$$

10. (a) Using $u = XT$ and separation constant $-\lambda = \alpha^4$ we find

$$X^{(4)} - \alpha^4 X = 0 \quad \text{and} \quad X(x) = c_1 \cos \alpha x + c_2 \sin \alpha x + c_3 \cosh \alpha x + c_4 \sinh \alpha x.$$

Since $u = XT$ the boundary conditions become

$$X(0) = 0, \quad X'(0) = 0, \quad X''(1) = 0, \quad X'''(1) = 0.$$

Now $X(0) = 0$ implies $c_1 + c_3 = 0$, while $X'(0) = 0$ implies $c_2 + c_4 = 0$. Thus

$$X(x) = c_1 \cos \alpha x + c_2 \sin \alpha x - c_1 \cosh \alpha x - c_2 \sinh \alpha x.$$

The boundary condition $X''(1) = 0$ implies

$$-c_1 \cos \alpha - c_2 \sin \alpha - c_1 \cosh \alpha - c_2 \sinh \alpha = 0$$

while the boundary condition $X'''(1) = 0$ implies

$$c_1 \sin \alpha - c_2 \cos \alpha - c_1 \sinh \alpha - c_2 \cosh \alpha = 0.$$

We then have the system of two equations in two unknowns

$$(\cos \alpha + \cosh \alpha)c_1 + (\sin \alpha + \sinh \alpha)c_2 = 0$$

$$(\sin \alpha - \sinh \alpha)c_1 - (\cos \alpha + \cosh \alpha)c_2 = 0.$$

This homogeneous system will have nontrivial solutions for c_1 and c_2 provided

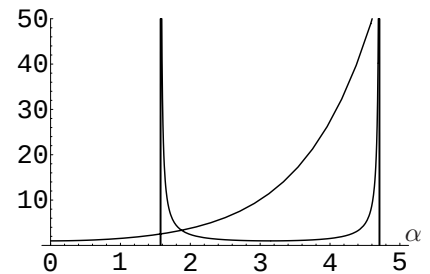
$$\begin{vmatrix} \cos \alpha + \cosh \alpha & \sin \alpha + \sinh \alpha \\ \sin \alpha - \sinh \alpha & -\cos \alpha - \cosh \alpha \end{vmatrix} = 0$$

or

$$-2 - 2 \cos \alpha \cosh \alpha = 0.$$

Thus, the eigenvalues are determined by the equation $\cos \alpha \cosh \alpha = -1$.

- (b) Using a computer to graph $\cosh \alpha$ and $-1/\cos \alpha = -\sec \alpha$ we see that the first two positive eigenvalues occur near 1.9 and 4.7. Applying Newton's method with these initial values we find that the eigenvalues are $\alpha_1 = 1.8751$ and $\alpha_2 = 4.6941$.



11. (a) In this case the boundary conditions are

$$\begin{aligned} u(0, t) &= 0, & \frac{\partial u}{\partial x} \Big|_{x=0} &= 0 \\ u(1, t) &= 0, & \frac{\partial u}{\partial x} \Big|_{x=1} &= 0. \end{aligned}$$

Separating variables leads to

$$X(x) = c_1 \cos \alpha x + c_2 \sin \alpha x + c_3 \cosh \alpha x + c_4 \sinh \alpha x$$

subject to

$$X(0) = 0, \quad X'(0) = 0, \quad X(1) = 0, \quad \text{and} \quad X'(1) = 0.$$

Now $X(0) = 0$ implies $c_1 + c_3 = 0$ while $X'(0) = 0$ implies $c_2 + c_4 = 0$. Thus

$$X(x) = c_1 \cos \alpha x + c_2 \sin \alpha x - c_1 \cosh \alpha x - c_2 \sinh \alpha x.$$

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The boundary condition $X(1) = 0$ implies

$$c_1 \cos \alpha + c_2 \sin \alpha - c_1 \cosh \alpha - c_2 \sinh \alpha = 0$$

while the boundary condition $X'(1) = 0$ implies

$$-c_1 \sin \alpha + c_2 \cos \alpha - c_1 \sinh \alpha - c_2 \cosh \alpha = 0.$$

We then have the system of two equations in two unknowns

$$(\cos \alpha - \cosh \alpha)c_1 + (\sin \alpha - \sinh \alpha)c_2 = 0$$

$$-(\sin \alpha + \sinh \alpha)c_1 + (\cos \alpha - \cosh \alpha)c_2 = 0.$$

This homogeneous system will have nontrivial solutions for c_1 and c_2 provided

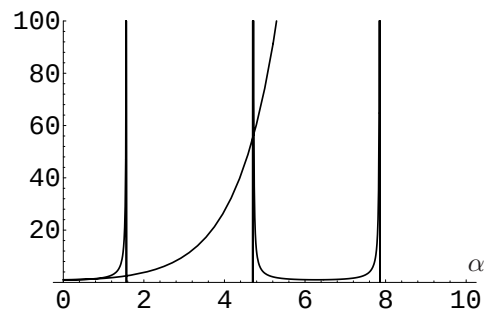
$$\begin{vmatrix} \cos \alpha - \cosh \alpha & \sin \alpha - \sinh \alpha \\ -\sin \alpha - \sinh \alpha & \cos \alpha - \cosh \alpha \end{vmatrix} = 0$$

or

$$2 - 2 \cos \alpha \cosh \alpha = 0.$$

Thus, the eigenvalues are determined by the equation $\cos \alpha \cosh \alpha = 1$.

- (b) Using a computer to graph $\cosh \alpha$ and $1/\cos \alpha = \sec \alpha$ we see that the first two positive eigenvalues occur near the vertical asymptotes of $\sec \alpha$, at $3\pi/2$ and $5\pi/2$. Applying Newton's method with these initial values we find that the eigenvalues are $\alpha_1 = 4.7300$ and $\alpha_2 = 7.8532$.



EXERCISES 13.8

Fourier Series in Two Variables

1. This boundary-value problem was solved in Example 1 in the text. Identifying $b = c = \pi$ and $f(x, y) = u_0$ we have

$$u(x, y, t) = \sum_{m=1}^{\infty} \sum_{n=1}^{\infty} A_{mn} e^{-k(m^2+n^2)t} \sin mx \sin ny$$

where

$$\begin{aligned} A_{mn} &= \frac{4}{\pi^2} \int_0^{\pi} \int_0^{\pi} u_0 \sin mx \sin ny \, dx \, dy \\ &= \frac{4u_0}{\pi^2} \int_0^{\pi} \sin mx \, dx \int_0^{\pi} \sin ny \, dy \\ &= \frac{4u_0}{mn\pi^2} [1 - (-1)^m][1 - (-1)^n]. \end{aligned}$$

2. As shown in Example 1 in the text, separation of variables leads to

$$X(x) = c_1 \cos \alpha x + c_2 \sin \alpha x$$

$$Y(y) = c_3 \cos \beta y + c_4 \sin \beta y$$

and

$$T(t) = c_5 e^{-k(\alpha^2 + \beta^2)t}.$$

The boundary conditions

$$\left. \begin{array}{l} u_x(0, y, t) = 0, \quad u_x(1, y, t) = 0 \\ u_y(x, 0, t) = 0, \quad u_y(x, 1, t) = 0 \end{array} \right\} \quad \text{imply} \quad \left\{ \begin{array}{l} X'(0) = 0, \quad X'(1) = 0 \\ Y'(0) = 0, \quad Y'(1) = 0. \end{array} \right.$$

Applying these conditions to

$$X'(x) = -\alpha c_1 \sin \alpha x + \alpha c_2 \cos \alpha x$$

and

$$Y'(y) = -\beta c_3 \sin \beta y + \beta c_4 \cos \beta y$$

gives $c_2 = c_4 = 0$ and $\sin \alpha = \sin \beta = 0$. Then

$$\alpha = m\pi, \quad m = 0, 1, 2, \dots \quad \text{and} \quad \beta = n\pi, \quad n = 0, 1, 2, \dots$$

By the superposition principle

$$\begin{aligned} u(x, y, t) = & A_{00} + \sum_{m=1}^{\infty} A_{m0} e^{-km^2\pi^2 t} \cos m\pi x + \sum_{n=1}^{\infty} A_{0n} e^{-kn^2\pi^2 t} \cos n\pi y \\ & + \sum_{m=1}^{\infty} \sum_{n=1}^{\infty} A_{mn} e^{-k(m^2+n^2)\pi^2 t} \cos m\pi x \cos n\pi y. \end{aligned}$$

We now compute the coefficients of the double cosine series: Identifying $b = c = 1$ and $f(x, y) = xy$ we have

$$A_{00} = \int_0^1 \int_0^1 xy \, dx \, dy = \int_0^1 \frac{1}{2} x^2 y \Big|_0^1 dy = \frac{1}{2} \int_0^1 y \, dy = \frac{1}{4},$$

$$\begin{aligned} A_{m0} &= 2 \int_0^1 \int_0^1 xy \cos m\pi x \, dx \, dy = 2 \int_0^1 \frac{1}{m^2\pi^2} (\cos m\pi x + m\pi x \sin m\pi x) \Big|_0^1 y \, dy \\ &= 2 \int_0^1 \frac{\cos m\pi - 1}{m^2\pi^2} y \, dy = \frac{\cos m\pi - 1}{m^2\pi^2} = \frac{(-1)^m - 1}{m^2\pi^2}, \end{aligned}$$

$$A_{0n} = 2 \int_0^1 \int_0^1 xy \cos n\pi y \, dx \, dy = \frac{(-1)^n - 1}{n^2\pi^2},$$

and

$$\begin{aligned} A_{mn} &= 4 \int_0^1 \int_0^1 xy \cos m\pi x \cos n\pi y \, dx \, dy = 4 \int_0^1 x \cos m\pi x \, dx \int_0^1 y \cos n\pi y \, dy \\ &= 4 \left(\frac{(-1)^m - 1}{m^2\pi^2} \right) \left(\frac{(-1)^n - 1}{n^2\pi^2} \right). \end{aligned}$$

In Problems 3 and 4 we need to solve the partial differential equation

$$a^2 \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} \right) = \frac{\partial^2 u}{\partial t^2}.$$

13.8 Fourier Series in Two Variables

To separate this equation we try $u(x, y, t) = X(x)Y(y)T(t)$:

$$a^2(X''YT + XY''T) = XYT''$$

$$\frac{X''}{X} = -\frac{Y''}{Y} + \frac{T''}{a^2T} = -\alpha^2.$$

Then

$$X'' + \alpha^2 X = 0 \tag{1}$$

$$\frac{Y''}{Y} = \frac{T''}{a^2T} + \alpha^2 = -\beta^2$$

$$Y'' + \beta^2 Y = 0 \tag{2}$$

$$T'' + a^2(\alpha^2 + \beta^2)T = 0. \tag{3}$$

The general solutions of equations (1), (2), and (3) are, respectively,

$$X(x) = c_1 \cos \alpha x + c_2 \sin \alpha x$$

$$Y(y) = c_3 \cos \beta y + c_4 \sin \beta y$$

$$T(t) = c_5 \cos a\sqrt{\alpha^2 + \beta^2}t + c_6 \sin a\sqrt{\alpha^2 + \beta^2}t.$$

3. The conditions $X(0) = 0$ and $Y(0) = 0$ give $c_1 = 0$ and $c_3 = 0$. The conditions $X(\pi) = 0$ and $Y(\pi) = 0$ yield two sets of eigenvalues:

$$\alpha = m, \quad m = 1, 2, 3, \dots \quad \text{and} \quad \beta = n, \quad n = 1, 2, 3, \dots$$

A product solution of the partial differential equation that satisfies the boundary conditions is

$$u_{mn}(x, y, t) = \left(A_{mn} \cos a\sqrt{m^2 + n^2}t + B_{mn} \sin a\sqrt{m^2 + n^2}t \right) \sin mx \sin ny.$$

To satisfy the initial conditions we use the superposition principle:

$$u(x, y, t) = \sum_{m=1}^{\infty} \sum_{n=1}^{\infty} \left(A_{mn} \cos a\sqrt{m^2 + n^2}t + B_{mn} \sin a\sqrt{m^2 + n^2}t \right) \sin mx \sin ny.$$

The initial condition $u_t(x, y, 0) = 0$ implies $B_{mn} = 0$ and

$$u(x, y, t) = \sum_{m=1}^{\infty} \sum_{n=1}^{\infty} A_{mn} \cos a\sqrt{m^2 + n^2}t \sin mx \sin ny.$$

At $t = 0$ we have

$$xy(x - \pi)(y - \pi) = \sum_{m=1}^{\infty} \sum_{n=1}^{\infty} A_{mn} \sin mx \sin ny.$$

Using (12) and (13) in the text, it follows that

$$\begin{aligned} A_{mn} &= \frac{4}{\pi^2} \int_0^\pi \int_0^\pi xy(x - \pi)(y - \pi) \sin mx \sin ny \, dx \, dy \\ &= \frac{4}{\pi^2} \int_0^\pi x(x - \pi) \sin mx \, dx \int_0^\pi y(y - \pi) \sin ny \, dy \\ &= \frac{16}{m^3 n^3 \pi^2} [(-1)^m - 1][(-1)^n - 1]. \end{aligned}$$

4. The conditions $X(0) = 0$ and $Y(0) = 0$ give $c_1 = 0$ and $c_3 = 0$. The conditions $X(b) = 0$ and $Y(c) = 0$ yield two sets of eigenvalues

$$\alpha = m\pi/b, \quad m = 1, 2, 3, \dots \quad \text{and} \quad \beta = n\pi/c, \quad n = 1, 2, 3, \dots$$

A product solution of the partial differential equation that satisfies the boundary conditions is

$$u_{mn}(x, y, t) = (A_{mn} \cos a\omega_{mn}t + B_{mn} \sin a\omega_{mn}t) \sin\left(\frac{m\pi}{b}x\right) \sin\left(\frac{n\pi}{c}y\right),$$

where $\omega_{mn} = \sqrt{(m\pi/b)^2 + (n\pi/c)^2}$. To satisfy the initial conditions we use the superposition principle:

$$u(x, y, t) = \sum_{m=1}^{\infty} \sum_{n=1}^{\infty} (A_{mn} \cos a\omega_{mn}t + B_{mn} \sin a\omega_{mn}t) \sin\left(\frac{m\pi}{b}x\right) \sin\left(\frac{n\pi}{c}y\right).$$

At $t = 0$ we have

$$f(x, y) = \sum_{m=1}^{\infty} \sum_{n=1}^{\infty} A_{mn} \sin\left(\frac{m\pi}{b}x\right) \sin\left(\frac{n\pi}{c}y\right)$$

and

$$g(x, y) = \sum_{m=1}^{\infty} \sum_{n=1}^{\infty} B_{mn} a\omega_{mn} \sin\left(\frac{m\pi}{b}x\right) \sin\left(\frac{n\pi}{c}y\right).$$

Using (12) and (13) in the text, it follows that

$$A_{mn} = \frac{4}{bc} \int_0^c \int_0^b f(x, y) \sin\left(\frac{m\pi}{b}x\right) \sin\left(\frac{n\pi}{c}y\right) dx dy$$

$$B_{mn} = \frac{4}{abc\omega_{mn}} \int_0^c \int_0^b g(x, y) \sin\left(\frac{m\pi}{b}x\right) \sin\left(\frac{n\pi}{c}y\right) dx dy.$$

In Problems 5 and 6 we try $u(x, y, z) = X(x)Y(y)Z(z)$ to separate Laplace's equation in three dimensions:

$$X''YZ + XY''Z + XYZ'' = 0$$

$$\frac{X''}{X} = -\frac{Y''}{Y} - \frac{Z''}{Z} = -\alpha^2.$$

Then

$$X'' + \alpha^2 X = 0 \tag{4}$$

$$\frac{Y''}{Y} = -\frac{Z''}{Z} + \alpha^2 = -\beta^2$$

$$Y'' + \beta^2 Y = 0 \tag{5}$$

$$Z'' - (\alpha^2 + \beta^2)Z = 0. \tag{6}$$

The general solutions of equations (4), (5), and (6) are, respectively

$$X(x) = c_1 \cos \alpha x + c_2 \sin \alpha x$$

$$Y(y) = c_3 \cos \beta y + c_4 \sin \beta y$$

$$Z(z) = c_5 \cosh \sqrt{\alpha^2 + \beta^2} z + c_6 \sinh \sqrt{\alpha^2 + \beta^2} z.$$

5. The boundary and initial conditions are

$$u(0, y, z) = 0, \quad u(a, y, z) = 0$$

$$u(x, 0, z) = 0, \quad u(x, b, z) = 0$$

$$u(x, y, 0) = 0, \quad u(x, y, c) = f(x, y).$$

13.8 Fourier Series in Two Variables

The conditions $X(0) = Y(0) = Z(0) = 0$ give $c_1 = c_3 = c_5 = 0$. The conditions $X(a) = 0$ and $Y(b) = 0$ yield two sets of eigenvalues:

$$\alpha = \frac{m\pi}{a}, \quad m = 1, 2, 3, \dots \quad \text{and} \quad \beta = \frac{n\pi}{b}, \quad n = 1, 2, 3, \dots$$

By the superposition principle

$$u(x, y, t) = \sum_{m=1}^{\infty} \sum_{n=1}^{\infty} A_{mn} \sinh \omega_{mn} z \sin \frac{m\pi}{a} x \sin \frac{n\pi}{b} y$$

where

$$\omega_{mn}^2 = \frac{m^2\pi^2}{a^2} + \frac{n^2\pi^2}{b^2}$$

and

$$A_{mn} = \frac{4}{ab \sinh \omega_{mn} c} \int_0^b \int_0^a f(x, y) \sin \frac{m\pi}{a} x \sin \frac{n\pi}{b} y \, dx \, dy.$$

6. The boundary and initial conditions are

$$\begin{aligned} u(0, y, z) &= 0, & u(a, y, z) &= 0, \\ u(x, 0, z) &= 0, & u(x, b, z) &= 0, \\ u(x, y, 0) &= f(x, y), & u(x, y, c) &= 0. \end{aligned}$$

The conditions $X(0) = Y(0) = 0$ give $c_1 = c_3 = 0$. The conditions $X(a) = Y(b) = 0$ yield two sets of eigenvalues:

$$\alpha = \frac{m\pi}{a}, \quad m = 1, 2, 3, \dots \quad \text{and} \quad \beta = \frac{n\pi}{b}, \quad n = 1, 2, 3, \dots$$

Let

$$\omega_{mn}^2 = \frac{m^2\pi^2}{a^2} + \frac{n^2\pi^2}{b^2}.$$

Then the boundary condition $Z(c) = 0$ gives

$$c_5 \cosh c\omega_{mn} + c_6 \sinh c\omega_{mn} = 0$$

from which we obtain

$$\begin{aligned} Z(z) &= c_5 \left(\cosh \omega_{mn} z - \frac{\cosh c\omega_{mn}}{\sinh c\omega_{mn}} \sinh \omega_{mn} z \right) \\ &= \frac{c_5}{\sinh c\omega_{mn}} (\sinh c\omega_{mn} \cosh \omega_{mn} z - \cosh c\omega_{mn} \sinh \omega_{mn} z) = c_{mn} \sinh \omega_{mn} (c - z). \end{aligned}$$

By the superposition principle

$$u(x, y, t) = \sum_{m=1}^{\infty} \sum_{n=1}^{\infty} A_{mn} \sinh \omega_{mn} (c - z) \sin \frac{m\pi}{a} x \sin \frac{n\pi}{b} y$$

where

$$A_{mn} = \frac{4}{ab \sinh c\omega_{mn}} \int_0^b \int_0^a f(x, y) \sin \frac{m\pi}{a} x \sin \frac{n\pi}{b} y \, dx \, dy.$$

7. The boundary and initial conditions are

$$\begin{aligned} u(0, y, z) &= 0, & u(1, y, z) &= 0, \\ u(x, 0, z) &= 0, & u(x, 1, z) &= 0, \\ u(x, y, 0) &= -u_0, & u(x, y, 1) &= u_0. \end{aligned}$$

Applying the superposition principle to the solutions in Problems 5 and 6, with $a = b = c = 1$ and $f(x, y) = u_0$ in Problem 5 and $f(x, y) = -u_0$ in Problem 6, we get

$$u(x, y, t) = \sum_{m=1}^{\infty} \sum_{n=1}^{\infty} A_{mn} [\sinh \omega_{mn} z - \sinh \omega_m (1 - z)] \sin m\pi x \sin n\pi y$$

where

$$\omega_{mn}^2 = (m^2 + n^2)\pi^2$$

and

$$\begin{aligned} A_{mn} &= \frac{4}{\sinh \omega_{mn}} \int_0^1 \int_0^1 u_0 \sin m\pi x \sin n\pi y \, dx \, dy = \frac{4u_0}{\sinh \omega_{mn}} \left(\int_0^1 \sin m\pi x \, dx \right) \left(\int_0^1 \sin n\pi y \, dy \right) \\ &= \frac{4u_0}{\sinh \omega_{mn}} \left(\frac{1}{m\pi} [1 - (-1)^n] \right) \left(\frac{1}{n\pi} [1 - (-1)^m] \right) = \frac{4u_0}{mn\pi^2 \sinh \omega_{mn}} [1 - (-1)^m] [1 - (-1)^n]. \end{aligned}$$

CHAPTER 13 REVIEW EXERCISES

1. Letting $u(x, y) = X(x) + Y(y)$ we have $X'Y' = XY$ and

$$\frac{X'}{X} = \frac{Y}{Y'} = -\lambda.$$

If $\lambda = 0$ then $X' = 0$ and $X(x) = c_1$. also $Y(y) = 0$ so $u = 0$.

If $\lambda \neq 0$ then $X' + \lambda X = 0$ and $Y + (1/\lambda)Y' = 0$. Thus $X(x) = c_1 e^{-\lambda x}$ and $Y(y) = c_2 e^{-y/\lambda}$ so

$$u(x, y) = A e^{(-\lambda x - y/\lambda)}.$$

2. Letting $u = XY$ we have $X''Y + XY'' + 2X'Y' + 2XY' = 0$ so that $(X'' + 2X')Y + X(Y'' + 2Y') = 0$. Separating variables and using the separation constant $-\lambda$ we obtain

$$\frac{X'' + 2X'}{-X} = \frac{Y'' + 2Y'}{Y} = -\lambda$$

so that

$$X'' + 2X' - \lambda X = 0 \quad \text{and} \quad Y'' + 2Y' + \lambda Y = 0.$$

The corresponding auxiliary equations are $m^2 + 2m - \lambda = 0$ and $m^2 + 2m + \lambda = 0$ with solutions $m = -1 \pm \sqrt{1 + \lambda}$ and $m = -1 \pm \sqrt{1 - \lambda}$, respectively. We consider five cases:

I. $\lambda = -1$: In this case $X = c_1 e^{-x} + c_2 x e^{-x}$ and $Y = c_3 e^{(-1+\sqrt{2})y} + c_4 e^{(-1-\sqrt{2})y}$ so that

$$u = (c_1 e^x + c_2 x e^{-x}) (c_3 e^{(-1+\sqrt{2})y} + c_4 e^{(-1-\sqrt{2})y}).$$

II. $\lambda = 1$: In this case $X = c_5 e^{(-1+\sqrt{2})x} + c_6 e^{(-1-\sqrt{2})x}$ and $Y = c_7 e^{-y} + c_8 y e^{-y}$ so that

$$u = (c_5 e^{(-1+\sqrt{2})x} + c_6 e^{(-1-\sqrt{2})x}) (c_7 e^{-y} + c_8 y e^{-y}).$$

CHAPTER 13 REVIEW EXERCISES

III. $-1 < \lambda < 1$: Here both $1 + \lambda$ and $1 - \lambda$ are positive so

$$u = \left(c_9 e^{(-1+\sqrt{1+\lambda})x} + c_{10} e^{(-1-\sqrt{1+\lambda})x} \right) \left(c_{11} e^{(-1+\sqrt{1-\lambda})y} + c_{12} e^{(-1-\sqrt{1-\lambda})y} \right).$$

IV. $\lambda < -1$: Here $1 + \lambda < 0$ and $1 - \lambda > 0$ so

$$u = e^{-x} (c_{13} \cos \sqrt{-1-\lambda} x + c_{14} \sin \sqrt{-1-\lambda} x) + \left(c_{15} e^{(-1+\sqrt{1-\lambda})y} + c_{16} e^{(-1-\sqrt{1-\lambda})y} \right).$$

V. $\lambda > 1$: Here $1 + \lambda > 0$ and $1 - \lambda < 0$ so

$$u = \left(c_{17} e^{(-1+\sqrt{1+\lambda})x} + c_{18} e^{(-1-\sqrt{1+\lambda})x} \right) + e^{-x} (c_{19} \cos \sqrt{\lambda-1} y + c_{20} \sin \sqrt{\lambda-1} y).$$

3. Substituting $u(x, t) = v(x, t) + \psi(x)$ into the partial differential equation we obtain

$$k \frac{\partial^2 v}{\partial x^2} + k \psi''(x) = \frac{\partial v}{\partial t}.$$

This equation will be homogeneous provided ψ satisfies

$$k\psi'' = 0 \quad \text{or} \quad \psi = c_1 x + c_2.$$

Considering

$$u(0, t) = v(0, t) + \psi(0) = u_0$$

we set $\psi(0) = u_0$ so that $\psi(x) = c_1 x + u_0$. Now

$$-\frac{\partial u}{\partial x} \Big|_{x=\pi} = -\frac{\partial v}{\partial x} \Big|_{x=\pi} - \psi'(x) = v(\pi, t) + \psi(\pi) - u_1$$

is equivalent to

$$\frac{\partial v}{\partial x} \Big|_{x=\pi} + v(\pi, t) = u_1 - \psi'(x) - \psi(\pi) = u_1 - c_1 - (c_1 \pi + u_0),$$

which will be homogeneous when

$$u_1 - c_1 - c_1 \pi - u_0 = 0 \quad \text{or} \quad c_1 = \frac{u_1 - u_0}{1 + \pi}.$$

The steady-state solution is

$$\psi(x) = \left(\frac{u_1 - u_0}{1 + \pi} \right) x + u_0.$$

4. The solution of the problem represents the heat of a thin rod of length π . The left boundary $x = 0$ is kept at constant temperature u_0 for $t > 0$. Heat is lost from the right end of the rod by being in contact with a medium that is held at constant temperature u_1 .

5. The boundary-value problem is

$$\begin{aligned} a^2 \frac{\partial^2 u}{\partial x^2} &= \frac{\partial^2 u}{\partial t^2}, \quad 0 < x < 1, \quad t > 0, \\ u(0, t) &= 0, \quad u(1, t) = 0, \quad t > 0, \\ u(x, 0) &= 0, \quad \frac{\partial u}{\partial t} \Big|_{t=0} = g(x), \quad 0 < x < 1. \end{aligned}$$

From Section 13.4 in the text we see that $A_n = 0$,

$$\begin{aligned} B_n &= \frac{2}{n\pi a} \int_0^1 g(x) \sin n\pi x \, dx = \frac{2}{n\pi a} \int_{1/4}^{3/4} h \sin n\pi x \, dx \\ &= \frac{2h}{n\pi a} \left(-\frac{1}{n\pi} \cos n\pi x \right) \Big|_{1/4}^{3/4} = \frac{2h}{n^2 \pi^2 a} \left(\cos \frac{n\pi}{4} - \cos \frac{3n\pi}{4} \right) \end{aligned}$$

and

$$u(x, t) = \sum_{n=1}^{\infty} B_n \sin n\pi at \sin n\pi x.$$

6. The boundary-value problem is

$$\begin{aligned} \frac{\partial^2 u}{\partial x^2} + x^2 &= \frac{\partial^2 u}{\partial t^2}, & 0 < x < 1, & \quad t > 0, \\ u(0, t) &= 1, & u(1, t) &= 0, & \quad t > 0, \\ u(x, 0) &= f(x), & u_t(x, 0) &= 0, & \quad 0 < x < 1. \end{aligned}$$

Substituting $u(x, t) = v(x, t) + \psi(x)$ into the partial differential equation gives

$$\frac{\partial^2 v}{\partial x^2} + \psi''(x) + x^2 = \frac{\partial^2 v}{\partial t^2}.$$

This equation will be homogeneous provided $\psi''(x) + x^2 = 0$ or

$$\psi(x) = -\frac{1}{12}x^4 + c_1x + c_2.$$

From $\psi(0) = 1$ and $\psi(1) = 0$ we obtain $c_1 = -11/12$ and $c_2 = 1$. The new problem is

$$\begin{aligned} \frac{\partial^2 v}{\partial x^2} &= \frac{\partial^2 v}{\partial t^2}, & 0 < x < 1, & \quad t > 0, \\ v(0, t) &= 0, & v(1, t) &= 0, & \quad t > 0, \\ v(x, 0) &= f(x) - \psi(x), & v_t(x, 0) &= 0, & \quad 0 < x < 1. \end{aligned}$$

From Section 13.4 in the text we see that $B_n = 0$,

$$A_n = 2 \int_0^1 [f(x) - \psi(x)] \sin n\pi x \, dx = 2 \int_0^1 \left[f(x) + \frac{1}{12}x^4 + \frac{11}{12}x - 1 \right] \sin n\pi x \, dx,$$

and

$$v(x, t) = \sum_{n=1}^{\infty} A_n \cos n\pi t \sin n\pi x.$$

Thus

$$u(x, t) = v(x, t) + \psi(x) = -\frac{1}{12}x^4 - \frac{11}{12}x + 1 + \sum_{n=1}^{\infty} A_n \cos n\pi t \sin n\pi x.$$

7. Using $u = XY$ and $-\lambda$ as a separation constant leads to

$$\begin{aligned} X'' - \lambda X &= 0, \\ X(0) &= 0, \end{aligned}$$

and

$$\begin{aligned} Y'' + \lambda Y &= 0, \\ Y(0) &= 0, \\ Y(\pi) &= 0. \end{aligned}$$

This leads to

$$Y = c_4 \sin ny \quad \text{and} \quad X = c_2 \sinh nx$$

for $n = 1, 2, 3, \dots$ so that

$$u = \sum_{n=1}^{\infty} A_n \sinh nx \sin ny.$$

CHAPTER 13 REVIEW EXERCISES

Imposing

$$u(\pi, y) = 50 = \sum_{n=1}^{\infty} A_n \sinh n\pi \sin ny$$

gives

$$A_n = \frac{100}{n\pi} \frac{1 - (-1)^n}{\sinh n\pi}$$

so that

$$u(x, y) = \frac{100}{\pi} \sum_{n=1}^{\infty} \frac{1 - (-1)^n}{n \sinh n\pi} \sinh nx \sin ny.$$

8. Using $u = XY$ and $-\lambda$ as a separation constant leads to

$$X'' - \lambda X = 0,$$

and

$$Y'' + \lambda Y = 0,$$

$$Y'(0) = 0,$$

$$Y'(\pi) = 0.$$

This leads to

$$Y = c_3 \cos ny \quad \text{and} \quad X = c_2 e^{-nx}$$

for $n = 1, 2, 3, \dots$. In this problem we also have $\lambda = 0$ is an eigenvalue with corresponding eigenfunctions 1 and 1. Thus

$$u = A_0 + \sum_{n=1}^{\infty} A_n e^{-nx} \cos ny.$$

Imposing

$$u(0, y) = 50 = A_0 + \sum_{n=1}^{\infty} A_n \cos ny$$

gives

$$A_0 = \frac{1}{\pi} \int_0^{\pi} 50 \, dy = 50$$

and

$$A_n = \frac{2}{\pi} \int_0^{\pi} 50 \cos ny \, dy = 0$$

for $n = 1, 2, 3, \dots$ so that

$$u(x, y) = 50.$$

9. Using $u = XY$ and $-\lambda$ as a separation constant leads to

$$X'' - \lambda X = 0,$$

and

$$Y'' + \lambda Y = 0,$$

$$Y(0) = 0,$$

$$Y(\pi) = 0.$$

Then

$$X = c_1 e^{nx} + c_2 e^{-nx} \quad \text{and} \quad Y = c_3 \cos ny + c_4 \sin ny$$

for $n = 1, 2, 3, \dots$. Since u must be bounded as $x \rightarrow \infty$, we define $c_1 = 0$. Also $Y(0) = 0$ implies $c_3 = 0$ so

$$u = \sum_{n=1}^{\infty} A_n e^{-nx} \sin ny.$$

Imposing

$$u(0, y) = 50 = \sum_{n=1}^{\infty} A_n \sin ny$$

gives

$$A_n = \frac{2}{\pi} \int_0^{\pi} 50 \sin ny \, dy = \frac{100}{n\pi} [1 - (-1)^n]$$

so that

$$u(x, y) = \sum_{n=1}^{\infty} \frac{100}{n\pi} [1 - (-1)^n] e^{-nx} \sin ny.$$

10. The boundary-value problem is

$$\begin{aligned} k \frac{\partial^2 u}{\partial x^2} &= \frac{\partial u}{\partial t}, & -L < x < L, & \quad t > 0, \\ u(-L, t) &= 0, & u(L, t) &= 0, & \quad t > 0, \\ u(x, 0) &= u_0, & -L < x < L. \end{aligned}$$

Referring to Section 13.3 in the text we have

$$X(x) = c_1 \cos \alpha x + c_2 \sin \alpha x$$

and

$$T(t) = c_3 e^{-k\alpha^2 t}.$$

Using the boundary conditions $u(-L, 0) = X(-L)T(0) = 0$ and $u(L, 0) = X(L)T(0) = 0$ we obtain $X(-L) = 0$ and $X(L) = 0$. Thus

$$\begin{aligned} c_1 \cos(-\alpha L) + c_2 \sin(-\alpha L) &= 0 \\ c_1 \cos \alpha L + c_2 \sin \alpha L &= 0 \end{aligned}$$

or

$$\begin{aligned} c_1 \cos \alpha L - c_2 \sin \alpha L &= 0 \\ c_1 \cos \alpha L + c_2 \sin \alpha L &= 0. \end{aligned}$$

Adding, we find $\cos \alpha L = 0$ which gives the eigenvalues

$$\alpha = \frac{2n-1}{2L} \pi, \quad n = 1, 2, 3, \dots$$

Thus

$$u(x, t) = \sum_{n=1}^{\infty} A_n e^{-[(2n-1)\pi/2L]^2 kt} \cos \left(\frac{2n-1}{2L} \pi x \right).$$

From

$$u(x, 0) = u_0 = \sum_{n=1}^{\infty} A_n \cos \left(\frac{2n-1}{2L} \pi x \right)$$

we find

$$A_n = \frac{2 \int_0^L u_0 \cos \left(\frac{2n-1}{2L} \pi x \right) \pi x \, dx}{2 \int_0^L \cos^2 \left(\frac{2n-1}{2L} \pi x \right) \pi x \, dx} = \frac{u_0 (-1)^{n+1} 2L / \pi (2n-1)}{L/2} = \frac{4u_0 (-1)^{n+1}}{\pi (2n-1)}.$$

CHAPTER 13 REVIEW EXERCISES

11. (a) The coefficients of the series

$$u(x, 0) = \sum_{n=1}^{\infty} B_n \sin nx$$

are

$$\begin{aligned} B_n &= \frac{2}{\pi} \int_0^{\pi} \sin x \sin nx \, dx = \frac{2}{\pi} \int_0^{\pi} \frac{1}{2} [\cos(1-n)x - \cos(1+n)x] \, dx \\ &= \frac{1}{\pi} \left[\frac{\sin(1-n)x}{1-n} \Big|_0^{\pi} - \frac{\sin(1+n)x}{1+n} \Big|_0^{\pi} \right] = 0 \text{ for } n \neq 1. \end{aligned}$$

For $n = 1$,

$$B_1 = \frac{2}{\pi} \int_0^{\pi} \sin^2 x \, dx = \frac{1}{\pi} \int_0^{\pi} (1 - \cos 2x) \, dx = 1.$$

Thus

$$u(x, t) = \sum_{n=1}^{\infty} B_n e^{-n^2 t} \sin nx$$

reduces to $u(x, t) = e^{-t} \sin x$ for $n = 1$.

- (b) This is like part (a), but in this case, for $n \neq 3$ and $n \neq 5$,

$$B_n = \frac{2}{\pi} \int_0^{\pi} (100 \sin 3x - 30 \sin 5x) \sin nx \, dx = 0;$$

while $B_3 = 100$ and $B_5 = -30$. Therefore

$$u(x, t) = 100e^{-9t} \sin 3x - 30e^{-25t} \sin 5x.$$

12. Substituting $u(x, t) = v(x, t) + \psi(x)$ into the partial differential equation results in $\psi'' = -\sin x$ and $\psi(x) = c_1 x + c_2 + \sin x$. The boundary conditions $\psi(0) = 400$ and $\psi(\pi) = 200$ imply $c_1 = -200/\pi$ and $c_2 = 400$ so

$$\psi(x) = -\frac{200}{\pi} x + 400 + \sin x.$$

Solving

$$\begin{aligned} \frac{\partial^2 v}{\partial x^2} &= \frac{\partial v}{\partial t}, & 0 < x < \pi, & \quad t > 0 \\ v(0, t) &= 0, & v(\pi, t) &= 0, & \quad t > 0 \\ u(x, 0) &= 400 + \sin x - \left(-\frac{200}{\pi} x + 400 + \sin x \right) = \frac{200}{\pi} x, & \quad 0 < x < \pi \end{aligned}$$

using separation of variables with separation constant $-\lambda$, where $\lambda = \alpha^2$, gives

$$X'' + \alpha^2 X = 0 \quad \text{and} \quad T' + \alpha^2 T = 0.$$

Using $X(0) = 0$ and $X(\pi) = 0$ we determine $\alpha^2 = n^2$, $X(x) = c_2 \sin nx$, and $T(t) = c_3 e^{-n^2 t}$. Then

$$v(x, t) = \sum_{n=1}^{\infty} A_n e^{-n^2 t} \sin nx$$

and

$$v(x, 0) = \frac{200}{\pi} x = \sum_{n=1}^{\infty} A_n \sin nx$$

so

$$A_n = \frac{400}{\pi^2} \int_0^{\pi} x \sin nx \, dx = \frac{400}{n\pi} (-1)^{n+1}.$$

Thus

$$u(x, t) = -\frac{200}{\pi}x + 400 + \sin x + \frac{400}{\pi} \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} e^{-n^2 t} \sin nx.$$

13. Using $u = XT$ and $-\lambda$, where $\lambda = \alpha^2$, as a separation constant we find

$$X'' + 2X' + \alpha^2 X = 0 \quad \text{and} \quad T'' + 2T' + (1 + \alpha^2)T = 0.$$

Thus for $\alpha > 1$

$$\begin{aligned} X &= c_1 e^{-x} \cos \sqrt{\alpha^2 - 1} x + c_2 e^{-x} \sin \sqrt{\alpha^2 - 1} x \\ T &= c_3 e^{-t} \cos \alpha t + c_4 e^{-t} \sin \alpha t. \end{aligned}$$

For $0 \leq \alpha \leq 1$ we only obtain $X = 0$. Now the boundary conditions $X(0) = 0$ and $X(\pi) = 0$ give, in turn, $c_1 = 0$ and $\sqrt{\alpha^2 - 1} \pi = n\pi$ or $\alpha^2 = n^2 + 1$, $n = 1, 2, 3, \dots$. The corresponding solutions are $X = c_2 e^{-x} \sin nx$. The initial condition $T'(0) = 0$ implies $c_3 = \alpha c_4$ and so

$$T = c_4 e^{-t} \left[\sqrt{n^2 + 1} \cos \sqrt{n^2 + 1} t + \sin \sqrt{n^2 + 1} t \right].$$

Using $u = XT$ and the superposition principle, a formal series solution is

$$u(x, t) = e^{-(x+t)} \sum_{n=1}^{\infty} A_n \left[\sqrt{n^2 + 1} \cos \sqrt{n^2 + 1} t + \sin \sqrt{n^2 + 1} t \right] \sin nx.$$

14. Letting $c = XT$ and separating variables we obtain

$$\frac{kX'' - hX'}{X} = \frac{T'}{T} \quad \text{or} \quad \frac{X'' - aX'}{X} = \frac{T'}{kT} = -\lambda$$

where $a = h/k$. Setting $\lambda = \alpha^2$ leads to the separated differential equations

$$X'' - aX' + \alpha^2 X = 0 \quad \text{and} \quad T' + k\alpha^2 T = 0.$$

The solution of the second equation is

$$T(t) = c_3 e^{-k\alpha^2 t}.$$

For the first equation we have $m = \frac{1}{2}(a \pm \sqrt{a^2 - 4\alpha^2})$, and we consider three cases using the boundary conditions $X(0) = X(1) = 0$:

$\boxed{a^2 > 4\alpha^2}$ The solution is $X = c_1 e^{m_1 x} + c_2 e^{m_2 x}$, where the boundary conditions imply $c_1 = c_2 = 0$, so $X = 0$. (Note in this case that if $\alpha = 0$, the solution is $X = c_1 + c_2 e^{ax}$ and the boundary conditions again imply $c_1 = c_2 = 0$, so $X = 0$.)

$\boxed{a^2 = 4\alpha^2}$ The solution is $X = c_1 e^{m_1 x} + c_2 x e^{m_1 x}$, where the boundary conditions imply $c_1 = c_2 = 0$, so $X = 0$.

$\boxed{a^2 < 4\alpha^2}$ The solution is

$$X(x) = c_1 e^{ax/2} \cos \frac{\sqrt{4\alpha^2 - a^2}}{2} x + c_2 e^{ax/2} \sin \frac{\sqrt{4\alpha^2 - a^2}}{2} x.$$

From $X(0) = 0$ we see that $c_1 = 0$. From $X(1) = 0$ we find

$$\frac{1}{2} \sqrt{4\alpha^2 - a^2} = n\pi \quad \text{or} \quad \alpha^2 = \frac{1}{4}(4n^2\pi^2 + a^2).$$

Thus

$$X(x) = c_2 e^{ax/2} \sin n\pi x,$$

and

$$c(x, t) = \sum_{n=1}^{\infty} A_n e^{ax/2} e^{-k(4n^2\pi^2 + a^2)t/4} \sin \pi x.$$

CHAPTER 13 REVIEW EXERCISES

The initial condition $c(x, 0) = c_0$ implies

$$c_0 = \sum_{n=1}^{\infty} A_n e^{ax/2} \sin n\pi x. \quad (1)$$

From the self-adjoint form

$$\frac{d}{dx} [e^{-ax} X'] + \alpha^2 e^{-ax} X = 0$$

the eigenfunctions are orthogonal on $[0, 1]$ with weight function e^{-ax} . That is

$$\int_0^1 e^{-ax} (e^{ax/2} \sin n\pi x) (e^{ax/2} \sin m\pi x) dx = 0, \quad n \neq m.$$

Multiplying (1) by $e^{-ax} e^{ax/2} \sin m\pi x$ and integrating we obtain

$$\int_0^1 c_0 e^{-ax} e^{ax/2} \sin m\pi x dx = \sum_{n=1}^{\infty} A_n \int_0^1 e^{-ax} e^{ax/2} (\sin m\pi x) e^{ax/2} \sin n\pi x dx$$

$$c_0 \int_0^1 e^{-ax/2} \sin n\pi x dx = A_n \int_0^1 \sin^2 n\pi x dx = \frac{1}{2} A_n$$

and

$$A_n = 2c_0 \int_0^1 e^{-ax/2} \sin n\pi x dx = \frac{4c_0[2e^{a/2}n\pi - 2n\pi(-1)^n]}{e^{a/2}(a^2 + 4n^2\pi^2)} = \frac{8n\pi c_0[e^{a/2} - (-1)^n]}{e^{a/2}(a^2 + 4n^2\pi^2)}.$$